

Lazarus - Mission Analysis Report

astroFUEL

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Mission Requirements Revision (version 2 - 23/10/2026)

Before entering into the MAR, it is necessary to present the revision made to the Mission Requirements, due to updates in the mission definition and analysis.

MIS_010
The satellite shall be capable of providing refuelling services for valuable GEO satellites weighing between roughly 1,000 kg and an estimated 6,500 kg at launch, provided that they have a compatible refuelling interface and that they use xenon for their station-keeping.

MIS_020
The satellite shall have a mechanical interface compatible with the OrbitFab standard refueling port RAFTI, allowing a safe and stable connection during xenon transfer, as well as a controlled and safe undocking once the operation is completed.

MIS_040
The satellite shall have a minimum operational lifetime of 8 years after insertion into its nominal orbit, during which it shall maintain the capabilities required to perform the planned refuelling services. After performing all of them and once the payload xenon is finished, the satellite will perform a transfer to a graveyard orbit of GEO + 300 km.

MIS_050
The satellite subsystems shall be designed and sized considering the new space assets that are being launched into GEO orbit in recent years and are planned to be launched in future years, and which use xenon for station-keeping. As a reference, the AIRBUS full-electric spacecraft Eurostar Neo should be considered.

MIS_060
The total mass of the satellite at launch (wet mass), including the structure, subsystems, xenon payload and propellant for its own manoeuvres, shall not exceed 4,500 kg.

MIS_065
The satellite's dimensions shall not exceed the following values: 5.2 m for the diameter and 11.8 m for the height.

MIS_070
The satellite shall have a minimum payload capacity of 500 kg of xenon to supply different client satellites.

MIS_080
The satellite's tank shall ensure the safe storage of the xenon payload throughout the mission's operational lifetime, maintaining its structural integrity and preventing leaks under the conditions of the space environment.

MIS_091
The spacecraft shall be capable of autonomously executing rendezvous and proximity operations with a cooperative client spacecraft during predefined servicing windows and within the operational GEO region.

MIS_092
During critical rendezvous and proximity-operation phases, telemetry and command capability with the AstroFUEL ground segment and coordination data exchange with the client operator shall be available under nominal conditions.

MIS_093
Loss of ground or client communication shall not result in loss of collision safety; the spacecraft shall autonomously transition to a predefined safe RPO state.

MIS_100
The spacecraft shall be capable of approaching a cooperative client spacecraft from the initial relative-navigation acquisition point to the docking interface while remaining inside the predefined approach corridor and outside all defined keep-out zones.

MIS_101
A nominal hold point shall be established at a relative distance of approximately 500 m from the client, at which relative navigation shall be validated before continuing the approach.

Continuation from each hold point shall only be permitted when predefined limits on relative position, relative velocity, attitude error, navigation uncertainty and available abort capability are satisfied.

The corresponding numerical limits shall be defined by the RPO GNC and docking requirements.

MIS_110
The spacecraft shall autonomously detect unsafe conditions during rendezvous, proximity operations and docking and shall initiate the corresponding safety response without requiring ground intervention.

Abort or retreat shall be triggered when at least one predefined safety limit is exceeded, including:

- loss or unacceptable degradation of the relative-navigation solution;
- violation of the approach corridor or keep-out-zone constraints;
- relative closing velocity above the permitted limit;
- loss of sufficient translational or attitude-control authority;
- critical propulsion or GNC failure;
- unexpected client motion or attitude;

- violation of the minimum collision-avoidance capability.

The safe state following an abort shall correspond to a collision-free relative trajectory with increasing or non-decreasing separation from the client and sufficient distance to allow mission reassessment before any further approach attempt.

MIS_120
The satellite shall have a system capable of transferring xenon to the client satellite under vacuum and microgravity conditions, using the necessary connection, pressure regulation and flow control mechanisms to ensure a safe and controlled transfer, while respecting the operating pressure and temperature limits of both spacecraft.

MIS_130
The satellite shall maintain continuous communication with the ground, enabling operators to monitor all the mission phases and to command an abort in case of safety-critical concerns.

Acronyms

C_R Solar Radiation Pressure Coefficient.

I_{sp} Specific Impulse.

AOCS Attitude and Orbit Control System.

CAM Collision Avoidance and Abort.

CP Commissioning Phase.

DKM Docking Mode.

DP Disposal Phase.

DPU Data Processing Unit.

EMC Inertial Reference Frame.

EP Electric Propulsion.

GEO Geostationary Earth Orbit.

GEO-P Geostationary Earth Parking Orbit.

GTO Geostationary Transfer Orbit.

IADC Inter-Agency Space Debris Coordination Committee.

LEOP Launch and Early Orbit Phase.

MTM Mated Mode.

OP Operational Phase.

RPO Rendezvous and Proximity Operations.

RVM Rendezvous Mode.

S/C Spacecraft.

SRP Solar Radiation Pressure.

TBD To Be Determined.

TFM Transfer Mode.

1 Introduction

Geostationary orbit is critical for global telecommunications, direct broadcasting, and sovereign defense space assets. Until recently, the lifetime of a satellite placed within the geostationary orbit has generally been dictated by station-keeping propellant and not by the electronic or structural capabilities of the asset. Upon being depleted of internal propellant reserves, fully functioning satellites (in terms of electronics and structure) are retired and placed into graveyard orbits, resulting in expensive loss of investment and high cost of new launches to replace them. With the proliferation of geostationary assets and active space debris mitigation regulations, which promote zero debris accumulation, the space industry is turning to in-orbit servicing to create reusable infrastructure. Within this emerging domain, the AstroFUEL initiative proposes the use of the Lazarus technology demonstrator to provide standardized in-orbit refuelling service to cooperative geostationary orbit satellites. Instead of trying to capture uncooperative spacecraft, the Lazarus demonstrator will target next-generation satellites with cooperative standard mechanical and fluid interfaces. Through the in-orbit refuelling of Xenon propellant within the microgravity environment by means of an autonomous rendezvous, docking, and regulated fluid transfer, Lazarus will be able to provide a cost-effective life extension to cooperative target satellites which will maximize return on investment while promoting orbital sustainability. This Mission Analysis Report outlines in detail the Lazarus Concept of Operations and provides quantitative mission analysis in support of the spacecraft design and selected mission architecture. In particular, Chapter 2 describes the general mission requirements and constraints dictating the payload, host platform, ground segment, and launch vehicle. Chapter 3 explores various mission concept options and their trade-offs, which leads to the development of the baseline mission architecture described in Chapter 4. The following chapters, 5 to 9, provide quantitative analysis of the Launch Window opportunities, Launch and Early Orbit Phase (LEOP), operational orbit flight dynamics, eclipse analysis, Ground Station visibility, and Delta-V / propellant budget to establish the full engineering baseline of the Lazarus refuelling mission.

2 Mission requirements and constraints

2.1 General Description

The Lazarus spacecraft is an in-orbit servicing technology demonstrator designed to perform successive propellant-transfer servicing operations with active geostationary spacecraft. The spacecraft is designed to support a range of client classes from SmallGEO class platforms to large and heavy assets. These spacecraft would have cooperated standardized interfaces and would use Xenon-based propulsion for station-keeping operations. The primary function of the Lazarus spacecraft is to provide multiple servicing operations to extend the lifetime of the target spacecraft. The spacecraft would carry a payload of Xenon propellant within high-pressure tanks and would be equipped with autonomous proximity navigation sensors and a standard docking interface. The operational concept of the spacecraft would involve autonomous rendezvous and proximity operations to the target spacecraft followed by capture and rigid mechanical docking. After the mechanical docking is achieved, the fluid lines would be connected to facilitate regulated Xenon transfer within the

microgravity environment of the geostationary orbit. After the Xenon transfer is completed and the fluid interface is confirmed to be isolated, the spacecraft will perform a departure maneuver to clear the client from its operational area and enter its GEO-P (Geostationary Parking orbit defined as 50 km above the GEO belt) to await the next servicing operation.

2.2 Description of Strategies of Payload Operations

The operational architecture of the Lazarus mission (WP-1300) is structured sequentially to ensure the safe, autonomous, and efficient refuelling of cooperative client satellites in GEO. The operational profile is divided into the following primary phases:

2.2.1 GEO-P

- **Insertion and Parking:** Following the GTO insertion, the Lazarus spacecraft autonomously transfers to its GEO-P, preliminarily defined as a designated waiting orbit situated safely above the nominal GEO belt and below the graveyard orbit to avoid interference with active orbital slots.
- **Standby Mode:** The spacecraft remains in a standby state, monitoring its subsystem health and maintaining continuous coverage with the Ground Segment for supervision, image processing support, and instruction updates.

2.2.2 Transfer and Approach Phase (TFM)

- **Target Acquisition:** Upon identification of a client satellite, the Ground Segment calculates the optimal transfer window and transmits time-tagged commands containing the transfer trajectory, delegating only the strictly necessary proximity tasks to the spacecraft.
- **Orbital Phasing and Rear Alignment:** Lazarus executes the necessary phasing manoeuvres to match the client's GEO slot. It is imperative that the transfer strategy is designed for Lazarus to join the client's orbit from behind, advancing towards the target while continuously pointing its docking port at the client spacecraft.
- **Far-Range Conventional Navigation:** During the far-approach phase, the spacecraft utilizes conventional navigation.
- **Safety Hold Point and Pre-Heating:** The TFM concludes when Lazarus reaches a designated safety hold point at a predefined relative distance with a high degree of positional certainty. At this point, the spacecraft halts its advance to verify alignment, while the isolated thermal plenum is pre-heated to prepare the propellant for transfer without penalizing the mated timeline.

2.2.3 Rendezvous and Docking Phase (RVM and DKM)

- **RVM Activation:** Once the safety hold point is reached and RVM is activated, the spacecraft's navigation switches from conventional GNSS tracking to relative Optical Navigation

with respect to the client satellite, utilizing the onboard proximity sensor suite.

- **Satellite Managed Corridors:** The autonomous final approach is strictly constrained within designated satellite managed corridors—defined as controlled orbital traffic routes that leverage satellite technology for continuous monitoring, real-time traffic management, and enhanced safety—recognizing cooperative optical markers on the client to maintain precise alignment.
- **Client Coordination and Port Geometry:** The operational activities and Earth-pointing communications of the client satellite shall not be disturbed or interrupted during the procedure. To ensure geometric compatibility during the refuelling timeline, it is imperative that the client's docking port is situated by design on the left side of its body with respect to its Earth-pointing antenna. Beyond this structural layout, the only required client intervention is the temporary deactivation of its own Safe Mode (SFM) to permit physical mating.
- **Capture and Mating (Docking Mode (DKM)/Mated Mode (MTM)):** Once the relative distance and velocity reach predefined critical thresholds, the final docking sequence engages, capturing and rigidly latching onto the client satellite before taking over the combined attitude control.
- **Controlled Failure and Abort (Collision Avoidance and Abort (CAM)):** In the event of an anomaly, such as mechanical bouncing during contact or an aborted docking when relative state thresholds are violated, the system executes a defined controlled failure protocol. This ensures that the spacecraft safely withdraws via a predetermined Collision Avoidance trajectory without entering an unpredictable emergency mode or performing sudden, hazardous manoeuvres near the client.

2.2.4 Refuelling Operations

- **Interface Integrity Verification:** Following mechanical docking, the fluidic connection and seal integrity at the transfer interface are verified via pressure-decay leak testing prior to opening the primary isolation valves.
- **Passive Equalization Transfer:** Refuelling initiates in a passive blowdown mode, leveraging the initial pressure gradient between the storage assembly and the client tank. Regulated mass flow and active in-line heating compensate for expansion-induced temperature drops, transferring the initial propellant batch without mechanical disturbance.
- **Active Thermal Compression Transfer:** To overcome adverse pressure gradients and minimize residual propellant in the mother tanks, the active branch engages cyclic thermal compression. Alternating thermal condensation and heating cycles drive the remaining propellant into the client tank while dedicated thermal control regulates the injected fluid to prevent client overheating.
- **Depressurization and Separation:** Upon achieving target fill levels, transfer lines are isolated and depressurized to recover trapped propellant. The docking mechanism disengages,

nominal attitude control is restored to the client, and the servicer executes a safe departure manoeuvre to return to its parking orbit.

2.3 Payload Requirements on the Operational Orbit

The Lazarus payload is dedicated to the safe approach, capture, and refuelling of cooperative client satellites in GEO. The payload architecture is composed of three main sub-assemblies: the Docking Mechanism, the Fluid Transfer System, and the Proximity Sensors. The specific system-level requirements for the payload are defined as follows:

2.3.1 Docking Mechanism

- **Standard Interface & Port Geometry:** The docking mechanism shall feature a universal docking port fully compatible with factory-standardized service interfaces integrated into next-generation GEO platforms (identified in the OrbitFab RAFTI model), designed to mate from a rear approach vector with a client port situated on the left side of the client's body relative to its Earth-pointing antenna.
- **Capture and Latching:** The system shall provide secure capture, alignment, and rigid structural latching via mechanical clamps to ensure no unintended physical perturbations affect the client's orbit during the refuelling process.
- **Mated Mode Control:** The system shall safely take over the client spacecraft's attitude control during the Mated Mode (MTM), maintaining the client's Earth-pointing orientation to fulfil operational requirements without interrupting its communication links.
- **Docking Reliability & Corridor Compliance:** The mechanism shall execute the mating and capture operations strictly within the designated satellite managed corridors once relative distance and velocity reach their critical engagement thresholds, ensuring a docking success probability of at least 75% prior to the execution of any automated abort sequence.
- **Safe Separation & Controlled Failure:** The mechanism shall guarantee a safe demating and separation sequence upon nominal completion of the fluid transfer, and shall implement a defined controlled failure protocol in the event of mechanical bouncing or an emergency abort to prevent sudden, uncontrolled manoeuvres.

2.3.2 Fluid Transfer System

- **Propellant Storage & Transfer Manifold:** The primary storage tank shall safely accommodate up to 500 kg of Xenon gas at high pressure (MEOP $\sim$ 187 bar). This shall be paired with a dual-branch hybrid transfer architecture, consisting of a passive blowdown manifold and an active, low-volume (1.5 L) cyclic thermal compressor [55].
- **Hybrid Transfer Operations:** The system shall execute a static, vibration-free transfer of 100 kg of Xenon into a depleted client tank ($\sim$ 10 bar) within 2.0 hours without causing

damages to it. This shall be achieved by transferring approximately 65% of the mass via initial pressure-equalization (passive blowdown) and the remaining 35% via active thermal compression cycles, ensuring Xenon utilization from the mother tank down to a residual pressure of 10 bar ($> 95\%$ efficiency) [55].

- **Thermal Conditioning & Protection:** The system shall integrate high-power in-line Joule-Thomson heaters (610 W peak) downstream of the passive flow restrictions to prevent localized freezing, and a dedicated Thermal Fluid Loop (TFL) with radiator to manage compressor cycling. The injection temperature at the client interface shall be strictly regulated to 20 °C (safely above the 16.6 °C critical point), preventing terminal adiabatic compression overpressure inside the client's tank and ensuring the client tank temperature does not exceed 50 °C during the operation [53].
- **Mass Flow Control:** The Flow Control System shall integrate a proportional throttling valve, a Coriolis mass flow meter, and redundant isolation valves to regulate the nominal mass flow rate (averaging 24.1 g/s during the passive phase and 7.8 g/s during the active phase) and continuously monitor telemetry to strictly prevent any propellant leakage [54].
- **Cleanliness and Contamination Prevention:** The system shall implement high-purity filtration (for particles < 2–$5\,\mu$m) and adhere to stringent cleanliness procedures on the subsystem level to prevent contamination of the transferred Xenon, ensuring compatibility with sensitive electric propulsion cathode elements on the target spacecraft [54].
- **Environmental Resilience & Qualification:** All fluid transfer components (heaters, valves, seals, and the thermal compressor) shall be designed to withstand recurrent cyclic pressure and thermal stresses across multiple refuelling cycles, verified via rigorous Thermal Vacuum (TVAC) ground testing in Phase D [54].

2.3.3 Proximity Sensors

- **Relative Navigation:** The proximity sensor suite—comprising 3D LiDAR, optical cameras, and short-range radars—shall provide high-accuracy relative position, attitude, and velocity data to the AOCS/GNC subsystem to maintain the spacecraft within the satellite managed corridors.
- **Target Illumination & Ranging:** The payload shall incorporate a Target Illumination System (TIS) and redundant laser sensors on the capture tool to ensure accurate ranging and image acquisition independently of external Sun illumination conditions during the final docking phase.
- **Hold Point Accuracy:** The sensor suite shall possess the range and resolution required to enable the spacecraft to accurately halt and loiter at a designated safety hold point located 500 m from the target satellite with a 95% positional certainty.
- **Autonomous Approach:** Upon activation of the RVM at the 500 m hold point, the sensors shall support fully autonomous optical/relative navigation to recognize cooperative optical markers on the client satellite.

- **Data Processing & FDIR:** A dedicated payload Data Processing Unit (DPU) shall process raw sensor data in real-time to monitor the critical relative distance and velocity engagement thresholds and support autonomous FDIR (Fault Detection, Isolation and Recovery) contingency modes, executing immediate Collision Avoidance (CAM) and controlled abort sequences during both solo and mated configurations if predefined tolerances are exceeded.

2.4 Ground Segment Requirements

To ensure the success of telemetry, tracking, and command (TT&C) operations, as well as high-capacity data downlink, the ground segment must comply with the following requirements:

- **Frequency Compatibility and Allocation:** The ground segment shall operate under a compatible dual-band architecture.
  - Low-gain network: TT&C and emergencies. Capability to transmit (uplink) and receive (downlink) in S-band.
  - Must be capable of receiving (downlink) in K-band the videos and images transmitted from the satellite, combined with uplink in X-band.
- **G/T Sensitivity:** It must meet minimum requirements to ensure video reception, being G/T $>$ 55.8 dB/K for high-gain stations and G/T $>$ 22 dB/K for the low-gain network.
- **EIRP:** To guarantee uplink closure at the maximum orbit distance, ground stations shall provide an EIRP $>$ 107 dBW in high gain and $>$ 68 dBm (34dBW) in low gain, ensuring command reception even if the satellite loses its nominal attitude.
- **Elevation:** Operations will be performed with an elevation angle defined by the ground stations of 10° for high-gain cases and 5° for low-gain stations.
- **Pointing Accuracy and Polarization:** Ground antennas shall operate in circular polarization (RHCP / LHCP). The maximum mechanical pointing error of the stations must be $<$ 0.006° for 35m antennas and $<$ 0.1° for 15m antennas in S-band.
- **Modulation and Coding Standards:** The entire ground segment must comply with CCSDS interoperability standards to allow cross-support between ESTRACK (ESA), DSN (NASA), and SSC networks. QPSK / OQPSK modulation with LDPC coding will be used for the K-band video link, and BPSK / GMSK modulation for the S-band TT&C link.
- **Global Availability and Continuous Handover:** Since Lazarus will move along the 360° longitude of the geostationary belt to service multiple client satellites, the ground segment must provide longitudinal coverage without blind spots. The network must execute smooth and synchronized signal handovers between adjacent stations within their geometric overlapping zones, avoiding interruptions in telemetry or video during critical approach phases.

2.5 Launcher Requirements

To increase mission flexibility and reduce dependence on a single launch provider, the Lazarus spacecraft shall satisfy the following preliminary launcher requirements:

- **Launch Vehicle Compatibility:** The spacecraft shall be designed to remain compatible with Ariane 62, Ariane 64 and Falcon 9, minimising the modifications required for integration with each launch vehicle.
- **European Launch Preference:** European launch vehicles shall be prioritised, with Ariane 62 and Ariane 64 considered the primary launch options. Falcon 9 shall be retained as a backup alternative in case of delays, unavailability or changes in mission planning.
- **Payload Mass Constraints:** The spacecraft launch mass shall remain within the payload capacity available for the selected GTO injection conditions. The most restrictive preliminary mass limit among the retained launch configurations shall be considered during spacecraft sizing.
- **Payload Fairing and Mechanical Compatibility:** The spacecraft, including its folded solar arrays and external equipment, shall fit within the usable payload fairing envelopes of the selected launch vehicles. The design shall also comply with the corresponding mechanical interfaces, structural loads and integration requirements.
- **Injection Orbit Compatibility:** The spacecraft shall be capable of operating from the GTO injection conditions provided by the selected launcher and performing the subsequent transfer to GEO-P. Differences in initial orbital inclination shall be considered when determining the propulsion requirements, propellant budget and transfer duration.

2.6 Satellite Requirements

The top-level system and performance requirements for the Lazarus satellite platform (bus) are defined in this section. They derive from the broader mission requirements and from the constraints imposed by the launch vehicle, and establish the design requirements and the operating boundaries for the platform's subsystems. Specifically, this section covers Propulsion, Attitude and Orbit Control (AOCS), Guidance, Navigation and Control (GNC), Fault Detection, Isolation, and Recovery (FDIR), Electrical Power (EPS), Communications, and End-of-Life Disposal. Following established space systems engineering standards, these requirements define exactly what the spacecraft must accomplish during every mission phase, providing the definitive baseline for all subsystem sizing and design.

2.6.1 System Reference, Mass and Volume Constraints

- **SAT-SYS-010** [Platform Reference]: The satellite platform shall be designed as a highly reliable bus capable of supporting every operational phase of the Lazarus mission in the geostationary environment. Its structural, thermal, and electrical architectures shall be

designed and sized taking as a reference new-generation GEO platforms utilizing xenon-based electric propulsion for station-keeping, specifically the Airbus Eurostar Neo classes.

- **SAT-SYS-020** [Mass Constraint]: The total wet mass of the satellite at launch, including the primary structure, platform subsystems, servicing payload, and all onboard Xenon propellant reserves, shall not exceed 4,500 kg.
- **SAT-SYS-030** [Dimensional Envelope]: The stowed satellite launch configuration shall fit strictly within a cylindrical static envelope of 5.2 m in diameter and 11.8 m in height, ensuring physical compatibility with the payload fairings of Ariane 62, Ariane 64, and Falcon 9.
- **SAT-SYS-040** [Subsystem Accommodation]: Within the specified mass and volume envelopes, the satellite bus shall provide internal volume, mounting surfaces, and thermal interfaces to accommodate all platform subsystems (Structure, Propulsion, AOCS, EPS, OBDH, TT&C, TCS) and payload elements while maintaining a minimum mass growth margin of 20% at the current design maturity level.

2.6.2 Propulsion and Attitude Control Subsystem (AOCS)

- **SAT-PROP-010** [All-Electric Architecture]: The satellite shall utilize a 100% xenon-based electric propulsion architecture for all orbital and attitude maneuvers throughout its operational lifetime, including GTO-to-GEO-P orbit raising, orbital phasing, Rendezvous and Proximity Operations (RPO), and reaction-wheel momentum desaturation.
- **SAT-AOCS-020** [Standby Pointing]: While in standby mode within the Geostationary Parking Orbit (GEO-P), the Attitude and Orbit Control System (AOCS) shall maintain continuous Nadir-pointing orientation to preserve thermal equilibrium, power generation, and ground communications.
- **SAT-AOCS-020** [Momentum Desaturation]: The AOCS shall perform three-axis reaction-wheel angular momentum unloading exclusively using the onboard distributed auxiliary electric thrusters without inducing net translational perturbations that exceed the GEO-P orbital tolerance.

2.6.3 Autonomy, GNC and Proximity Safety (FDIR)

- **SAT-GNC-010** [Supervised Autonomy]: The satellite shall execute Rendezvous and Proximity Operations (RPO) autonomously under ground supervision, providing the onboard computational capability required to process relative navigation sensor data and execute guidance and control algorithms in real time.
- **SAT-FDIR-010** [Proximity FDIR and CAM]: To prevent physical contact during close approaches, the platform On-Board Computer (OBC) shall integrate proximity-operations-specific Fault Detection, Isolation, and Recovery (FDIR) logic. It shall autonomously detect system anomalies and trajectory corridor violations, executing an unprompted Collision Avoidance Maneuver (CAM) and abort sequence without ground intervention.

- **SAT-FDIR-020** [Communication Loss Contingency]: In the event of a total loss of telemetry/command link or client communication during RPO, collision safety shall not be compromised; the satellite shall autonomously transition to a predefined safe RPO loitering state along a collision-free relative trajectory where separation from the client does not decrease.
- **SAT-COM-010** [Continuous Ground Communication]: The satellite shall maintain continuous communication with the ground, enabling operators to monitor all the mission phases and to command an abort in case of safety-critical concerns.

2.6.4 Electrical Power System, Lifetime and Disposal

- **SAT-EPS-010** [Power System Sizing]: The Electrical Power System (EPS) shall supply the continuous high-power demand required by the main electric propulsion thruster during orbit-raising and phasing maneuvers, while simultaneously powering all platform subsystems and payload thermal pre-heating functions.
- **SAT-EPS-020** [Eclipse Operations]: The secondary energy storage subsystem shall sustain the most demanding power condition, which corresponds to the refuelling operations, during the maximum continuous Earth eclipse duration of 72 minutes in GEO-P.
- **SAT-SYS-050** [Operational Lifetime]: The satellite bus and its constituent subsystems shall be designed and qualified for a minimum operational lifetime of 8 years in orbit, providing the structural, thermal, and electrical robustness required to complete all the planned refuelling operations.
- **SAT-EOL-010** [Graveyard Disposal and Passivation]: Upon completion of all servicing operations or fuel depletion, the satellite shall retain sufficient propellant reserve and control authority to execute an autonomous transfer to a graveyard orbit at GEO + 300 km, followed by complete system passivation (tank depressurization, battery isolation, and reaction-wheel spin-down) in compliance with space debris mitigation standards.

2.7 Main System Drivers

The main design constraints of Lazarus are determined by the mission objectives, particularly by the requirements associated with orbital refuelling operations. The principal system drivers are identified below:

- **Docking and Xenon Transfer Operations:** The spacecraft shall accommodate a docking and refuelling interface on its upper face to enable xenon transfer to client satellites. Its location will influence the external configuration of the spacecraft and the structural elements required to support the loads generated during docking operations.
- **Xenon Storage and Mass Distribution:** The spacecraft will incorporate two independent xenon storage systems, each consisting of three tanks: one dedicated to the refuelling payload and the other reserved for spacecraft propulsion. Their arrangement shall minimise variations in the spacecraft centre of mass as xenon is consumed or transferred throughout the mission.

- **Attitude and Orbit Control System (AOCS):** The spacecraft shall accommodate four reaction wheels arranged in a pyramidal configuration to provide three-axis attitude control and redundancy. Their integration, together with the auxiliary thrusters and navigation sensors, will influence the internal equipment distribution and the structural mounting requirements.
- **Electrical Power System (EPS):** The high electrical power demand associated with electric propulsion requires large deployable solar arrays. Their dimensions, folded configuration and deployment mechanisms will influence the spacecraft external geometry and available volume during launch. The batteries and power management equipment will also contribute significantly to the spacecraft mass and internal distribution.
- **Launch Vehicle Compatibility:** The spacecraft configuration shall satisfy the geometric, mass and mechanical constraints of the retained launch vehicles. This requirement will influence the overall spacecraft dimensions, structural design and integration of deployable equipment.

3 Mission Concepts

3.1 Alternative Scenarios

The Lazarus mission can be implemented through different architectures depending on the launch strategy, operational orbit, propulsion system, degree of cooperation of the client spacecraft, rendezvous and docking strategy, relative navigation architecture, angular momentum management, ground operations concept, and level of onboard autonomy.

This section identifies the main technically feasible alternatives capable of satisfying the mission requirements. The reference architecture is subsequently selected in Section 3.2 through the corresponding trade-off.

3.1.1 Launch and Initial Injection Strategy

The launch and initial injection strategy directly affects the mass that can be placed into orbit, the propulsion capability required from the spacecraft, the duration of the transfer phase, and the overall mission architecture.

Three main alternatives are considered.

L1 — Direct Injection into GEO The launch vehicle performs practically the entire energy transfer required and injects Lazarus directly into the GEO region.

This option minimises the ΔV that must be provided by the spacecraft itself and considerably reduces the time required to reach the operational region.

However, direct GEO injection requires greater performance from the launch vehicle and generally reduces the payload mass that can be delivered compared with a GTO injection. It also reduces

the number of launch services potentially compatible with the mission.

L2 — Direct Injection into the Operational GEO+50 km Orbit A second alternative consists of injecting Lazarus directly into, or very close to, the GEO+50 km operational parking orbit.

This configuration practically eliminates the initial orbit-raising manoeuvre and therefore minimises the spacecraft propellant required before the beginning of nominal operations.

However, it requires a high-energy and relatively specific injection from the launch vehicle. Consequently, the available payload capability is reduced and the mission becomes more dependent on launch services capable of providing the required final orbital conditions.

L3 — GTO Injection and Autonomous Transfer The third alternative consists of injecting Lazarus into an elliptical and inclined Geostationary Transfer Orbit (GTO), from which the spacecraft autonomously performs the transfer to the GEO+50 km operational orbit.

This alternative provides a significantly greater payload capability than direct injection into the final operational orbit and is compatible with conventional launch profiles used for missions targeting the geostationary region.

The main disadvantage is that the spacecraft propulsion system must provide the complete orbit-raising capability. This increases transfer duration, propulsion-system utilisation, electrical-energy demand and exposure to the radiation environment.

The required transfer effort also depends on the orbital conditions provided by the selected launch vehicle. In particular, the initial inclination has a significant impact on the propulsion required to reach the final equatorial orbit.

Preliminary mission analyses therefore considered two reference GTO injection conditions:

$$i_0 = 6^\circ$$

for a representative low-inclination European launch scenario, and

$$i_0 = 27^\circ$$

for a more demanding reference injection scenario.

These cases are used to assess the sensitivity of the GTO-to-GEO+50 km transfer to the selected launch vehicle.

3.1.2 Launch Vehicle Alternatives

Once the possible orbital injection strategies had been identified, the launch vehicles capable of providing the required injection conditions were analysed.

The initial study focused on European launch vehicles. Ariane 62, Ariane 64 and Vega-C were

considered as candidate launchers. Falcon 9 was subsequently introduced as an additional non-European alternative in order to increase mission flexibility and reduce dependence on a single launch vehicle family.

Four alternatives are therefore considered.

LV1 — Vega-C Vega-C represents the lightest European launch alternative considered for Lazarus.

Its use would preserve a fully European launch architecture. However, its payload capability becomes restrictive for a spacecraft with the expected mass and operational requirements of Lazarus, particularly when injection into a high-energy orbit such as GTO is required.

The limited available mass would place additional constraints on the spacecraft mass budget and on the amount of xenon that could be carried.

LV2 — Ariane 62 Ariane 62 represents the two-booster configuration of the Ariane 6 family.

Its performance provides a significant improvement over Vega-C and makes it a potentially compatible solution for GTO injection while maintaining the advantages associated with a European launch architecture.

Its lower payload capability compared with Ariane 64 nevertheless provides less margin for future spacecraft mass growth.

LV3 — Ariane 64 Ariane 64 represents the highest-performance configuration of the Ariane 6 family considered for the mission.

Its greater payload capability provides a larger mass margin for Lazarus, including the spacecraft subsystems and the large xenon inventory required for servicing operations.

This additional margin also reduces the sensitivity of the mission to spacecraft mass growth during later design phases.

LV4 — Falcon 9 Falcon 9 is considered as an additional non-European launch alternative.

Its inclusion increases launcher availability and reduces the dependence of the programme on a single launcher family.

However, the orbital conditions associated with the Falcon 9 reference scenario can be more demanding from the spacecraft propulsion point of view.

In the preliminary mission analysis, a reference initial inclination of approximately 27° was considered for the higher-inclination injection case, compared with approximately 6° for the low-inclination European reference case.

A higher initial inclination increases the propulsive effort required to reach the final equatorial GEO+50 km orbit and therefore affects xenon consumption and transfer duration.

Consequently, launcher compatibility must be evaluated not only in terms of payload capability

but also in terms of the orbital conditions delivered at separation.

Maintaining several launch alternatives allows a common Lazarus spacecraft configuration to remain compatible with different launch opportunities and increases overall programme robustness.

3.1.3 Parking Orbit

Between two servicing operations, Lazarus requires an orbit in which it can remain while waiting for the assignment of a new client.

Three alternatives are considered.

O1 — Fixed GEO Slot Lazarus occupies a defined geostationary position and performs the required maintenance manoeuvres to preserve its orbital longitude.

This solution simplifies operations conceptually and provides a permanent and known position, but requires the allocation of a GEO *slot* and continuous *station keeping* manoeuvres.

O2 — GEO+50 km Drift Orbit Lazarus is placed approximately 50 km above geostationary altitude.

Since its orbital period does not exactly match the Earth's rotation period, the spacecraft slowly drifts with respect to the GEO belt without maintaining a fixed longitude.

In this case, the mission does not require permanent allocation of a GEO *slot*. Navigation and orbit maintenance focus on keeping the spacecraft within a defined semi-major axis, eccentricity, and inclination *envelope*, while allowing longitudinal drift.

O3 — GEO-50 km Drift Orbit A conceptually equivalent alternative consists of placing Lazarus approximately 50 km below the GEO.

As in the GEO+50 km case, the vehicle would experience longitudinal drift with respect to the spacecraft located in GEO, and maintaining a fixed orbital longitude would not be required.

However, because Lazarus would be located below the GEO belt, it could periodically be positioned between client spacecraft and the Earth when crossing longitudes close to those occupied by the clients.

This geometry is less favourable from an operational point of view, as it increases the possibility of the vehicle crossing regions close to the lines of sight of antennas, cameras, or other Earth-pointing sensors on GEO spacecraft.

Although these situations could be managed through orbital planning and conjunction restrictions, an orbit above the GEO belt naturally reduces this interaction with the fields of view and Earth-pointing directions of operational satellites.

3.1.4 Propulsion Architecture

The propulsion architecture constitutes one of the main system drivers of Lazarus.

The spacecraft must perform manoeuvres with very different requirements, ranging from the high-ΔV transfer from GTO to GEO+50 km to precise low-acceleration rendezvous, docking and attitude-control operations.

Three propulsion architectures are considered.

P1 — Fully Chemical Propulsion In this architecture, chemical propulsion is used for the initial orbit transfer, operational transfers, rendezvous, docking, abort manoeuvres and reaction-wheel unloading.

The main advantage is the high thrust available, which considerably reduces manoeuvre duration and provides high acceleration for emergency operations.

However, chemical propulsion has a significantly lower specific impulse than electric propulsion.

For a mission requiring a large GTO-to-GEO transfer, this results in a substantial propellant mass requirement and therefore significantly increases the spacecraft launch mass.

P2 — Hybrid Chemical-Electric Architecture A hybrid architecture uses electric propulsion for major orbital transfers and a chemical Reaction Control System (RCS) for proximity operations, docking, abort manoeuvres and reaction-wheel unloading.

This concept combines the high propellant efficiency of electric propulsion with the high acceleration capability of chemical propulsion during safety-critical proximity operations.

However, it requires two independent propulsion technologies, including separate propellant storage and feed systems, valves, control electronics and associated hardware.

This increases spacecraft mass, system complexity and the number of potential failure modes.

P3 — Fully Electric Propulsion In this architecture, all propulsion functions are performed using electric propulsion and xenon.

The propulsion system is divided functionally into two levels.

A main electric propulsion system is used for major orbital manoeuvres, including:

- GTO-to-GEO+50 km orbit raising;
- transfers between the parking orbit and client orbital regions;
- orbital phasing;
- return to the GEO+50 km parking orbit.

A distributed auxiliary electric propulsion system is used for:

- rendezvous and proximity operations;
- final approach and docking;

- undocking and retreat;
- reaction-wheel momentum unloading.

The principal advantages are the high specific impulse, reduced propellant mass, elimination of a separate chemical propulsion system, and the possibility of using xenon as the common spacecraft propulsion resource.

The main disadvantages are the higher electrical-power demand, longer manoeuvre durations and reduced instantaneous acceleration, particularly during close-range abort manoeuvres.

3.1.5 Degree of Client Cooperation

The servicing concept depends significantly on the capabilities and level of participation required from the client spacecraft.

Three possibilities are considered.

C1 — Active Client Both Lazarus and the client actively participate in the rendezvous manoeuvre. The client may modify its orbit, attitude, or relative motion to facilitate the approach.

This solution can reduce the performance required from the servicer, but considerably increases coordination between both vehicles and requires the client to have GNC capabilities compatible with the operation.

C2 — Passive but Cooperative Client Lazarus actively performs the rendezvous and approach while the client maintains a previously agreed attitude and configuration.

Although the client does not actively perform the approach, it is considered cooperative because it provides orbital and attitude information, has a known servicing interface, and may incorporate elements specifically designed to facilitate the operation, such as navigation fiducials and a standardised docking and propellant-transfer interface.

C3 — Non-Cooperative Client Lazarus must approach and capture a vehicle that was not specifically designed to receive servicing.

This alternative significantly increases the complexity of navigation, target identification, attitude estimation, mechanical capture, and operational safety.

3.1.6 Rendezvous and Approach Strategy

Two main concepts are considered.

R1 — Continuous Approach Lazarus follows a continuous controlled approach trajectory from the rendezvous region until contact.

This alternative reduces the total operation time but limits the opportunities to stop the sequence and verify the status of both vehicles.

R2 — Hold-Point Approach The trajectory is divided into different phases separated by hold points.

A possible sequence is:

$$500 \text{ m} \rightarrow 100 \text{ m} \rightarrow 20 \text{ m} \rightarrow 5 \text{ m} \rightarrow \text{Docking}.$$

At each *hold point*, the navigation status, alignment, actuator availability, client status, and conditions required to continue can be verified.

3.1.7 Relative Navigation Architecture

During proximity operations, absolute navigation alone does not provide the accuracy required to perform docking.

Three alternatives are considered.

N1 — Optical-Only Navigation Relative position and attitude are estimated using cameras and image processing.

This architecture has relatively low power consumption and provides a large amount of geometrical information, although it depends more strongly on illumination conditions and correct target identification.

N2 — Navigation Using Active Sensors Sensors such as LiDAR or laser rangefinders are used to obtain direct measurements of relative distance and geometry.

These sensors reduce dependence on illumination conditions, although they increase power demand and system complexity.

N3 — Multi-Sensor Architecture Cameras and active sensors are combined, using different technologies depending on the distance to the client.

A possible architecture includes two relative-navigation cameras, one three-dimensional LiDAR, and one laser rangefinder. Their measurements can be fused through a relative-navigation filter.

3.1.8 Reaction-Wheel Angular Momentum Management

External disturbances produce a progressive accumulation of angular momentum in the reaction wheels, making a desaturation mechanism necessary.

Three options are considered.

M1 — Magnetorquers Magnetorquers generate an external torque through interaction with the Earth's magnetic field.

Their main advantage is that they do not consume propellant. However, their authority depends directly on the magnitude and direction of the available magnetic field.

M2 — Chemical RCS Chemical thrusters allow the reaction wheels to be unloaded rapidly and provide high control authority, but require a dedicated chemical propulsion system.

M3 — Distributed Electric Propulsion The same electric thrusters used for proximity operations can be combined to generate external torques with low net translational force.

This makes it possible to transfer the angular momentum stored in the reaction wheels out of the spacecraft without incorporating an additional propulsion system.

3.1.9 Coordination Between Ground Segments

At least two operators are involved during servicing: the Lazarus operator and the client spacecraft operator.

Two main concepts are considered.

G1 — Single or Joint Control Centre During servicing, both spacecraft are coordinated from the same facility or operations team.

This alternative facilitates immediate coordination but requires the integration of procedures, personnel, infrastructure, and responsibilities potentially belonging to different organisations.

G2 — Coordinated Independent Control Centres Each operator retains authority over its own vehicle.

During rendezvous and servicing, a specific coordination protocol is established to exchange status information, GO/NO-GO decisions, planning information, and contingency procedures.

A joint servicing cell may also be established exclusively to synchronise operations without requiring physical integration of both control centres.

3.1.10 Level of Autonomy During RPO

Different levels of ground-segment participation are considered.

A1 — Direct Ground Control Navigation and guidance commands are continuously generated from the ground during the approach.

This solution maintains direct operator involvement but makes safety functions dependent on link availability and the ground communication cycle.

A2 — Fully Autonomous Operation After receiving the mission objective, the spacecraft autonomously makes decisions related to rendezvous, phase transitions, and docking.

This option reduces dependence on the ground segment but requires a high level of onboard autonomy.

A3 — Supervised Autonomy The ground segment plans and authorises the operation and transmits command sequences or authorises specific phases.

Once each sequence has been initiated, Lazarus autonomously executes the required navigation, guidance, and control functions while the ground segment monitors its progress.

Safety actions requiring rapid response always remain onboard.

3.1.11 FDIR Strategy During Proximity Operations

Proximity operations represent a special situation for the Failure Detection, Isolation and Recovery system.

Two alternatives are considered.

F1 — Conventional FDIR The same automatic responses used during nominal spacecraft operations are maintained throughout the rendezvous.

This solution reduces the amount of specific logic required, but certain *safe mode* responses may be incompatible with the presence of another vehicle at short range.

F2 — Proximity-Operations-Specific FDIR During rendezvous and docking, a dedicated set of safety responses is activated.

For certain failures, the priority becomes stabilising the relative motion and avoiding a collision through actions such as:

$$\text{Hold} \rightarrow \text{Retreat} \rightarrow \text{Abort}.$$

Once a safe distance has been reached, the conventional FDIR logic can be restored.

3.1.12 Fleet Architecture

Finally, the number of Lazarus vehicles deployed in the GEO region is considered.

S1 — Single Lazarus Vehicle A single vehicle provides the service and enables an initial complete demonstration of the technology with the minimum number of spacecraft.

S2 — Lazarus Fleet Several vehicles operate simultaneously in the GEO region.

This solution increases service availability, reduces response times, and provides mission-level redundancy, although it increases cost and fleet-coordination complexity.

3.2 Scenario Trade-off

Based on the alternatives described in the previous section, a preliminary trade-off has been performed in order to select the Lazarus reference architecture.

Since the different architecture decisions affect very different aspects of the mission, a single global matrix has not been used. Instead, independent trade-offs have been performed for the main design decisions.

For each criterion, a relative weight w_i is assigned, and each alternative receives a score s_i between 1 and 5, where:

- 1: very unfavourable performance;
- 2: unfavourable performance;
- 3: acceptable performance;
- 4: favourable performance;
- 5: very favourable performance.

The weighted score for each alternative is calculated as:

$$S = \sum_{i=1}^{N} w_i s_i, \tag{1}$$

where the weights for each trade-off satisfy:

$$\sum_{i=1}^{N} w_i = 1. \tag{2}$$

The resulting scores should be interpreted as a decision-support tool during the preliminary design phase. The final selection also takes into account the results of the analyses and simulations specifically developed for Lazarus.

3.2.1 Launch and Initial Injection Strategy

The three alternatives defined in the previous section are compared:

- L1: direct injection into GEO;
- L2: direct injection into the GEO+50 km operational orbit;
- L3: GTO injection followed by autonomous transfer.

The criteria considered are shown in Table 3.1.

Table 3.1: Trade-off of the launch and initial injection strategy.

Criterion	**Weight**	**L1**	**L2**	**L3**
Launch service cost	25%	2	1	5
Launcher availability/flexibility	20%	3	1	5
Deliverable spacecraft mass	20%	2	1	4
Spacecraft propulsion demand	15%	5	5	3
Time to operational orbit	10%	5	5	2
Mission flexibility	10%	3	1	5
Weighted score		**3.05**	**2.00**	**4.20**

Alternative **L3, GTO injection followed by autonomous transfer**, is selected.

Direct injection into GEO or GEO+50 km considerably reduces the propulsion required from Lazarus and allows the operational orbit to be reached rapidly. However, these alternatives require greater launch vehicle performance and significantly reduce the payload mass available for the spacecraft.

GTO injection provides a more favourable launch-mass capability and is compatible with a larger number of conventional missions towards the geostationary region.

The principal drawback is that Lazarus must provide the complete orbit-raising capability.

A preliminary analysis initially considered performing this transfer using chemical propulsion. The resulting propellant requirement was of the order of:

$$m_{\text{prop,chem}} \approx 2000 \text{ kg}. \tag{3}$$

Such a propellant requirement represents an excessive mass penalty for Lazarus and would substantially reduce the mass available for the spacecraft systems and the xenon intended for client refuelling.

For this reason, the GTO transfer was subsequently analysed using electric propulsion.

Two initial inclination cases and two thrust strategies were studied. The resulting preliminary transfer performance is summarised in Table 3.2.

Table 3.2: Preliminary electric-propulsion GTO-to-GEO+50 km transfer results.

Initial inclination	**Thrust strategy**	**Transfer time**	ΔV	**Xe mass**
6°	Continuous	72.12 days	2.2393 km/s	423.5 kg
27°	Continuous	85.84 days	2.7065 km/s	504.0 kg
6°	Non-continuous	115.66 days	1.7222 km/s	331.3 kg
27°	Non-continuous	149.18 days	2.1395 km/s	406.0 kg

The results demonstrate that the GTO-to-GEO+50 km transfer is feasible for both initial inclination cases considered.

The injection inclination has a significant impact on transfer performance. For the continuous-thrust case, increasing the initial inclination from 6° to 27° increases the required ΔV by approximately:

$$\Delta(\Delta V) \simeq 0.47 \text{ km/s}, \tag{4}$$

increases xenon consumption by approximately:

$$\Delta m_{\text{Xe}} \simeq 80.5 \text{ kg}, \tag{5}$$

and increases the transfer duration by approximately:

$$\Delta t \simeq 13.7 \text{ days}. \tag{6}$$

The comparison between continuous and non-continuous thrust also shows a clear trade-off between propellant consumption and transfer duration.

For the 6° case, non-continuous thrust reduces the xenon requirement by approximately 92 kg but increases the total flight time from approximately 72 to 116 days.

For the 27° case, the xenon saving is approximately 98 kg, while the total transfer duration increases from approximately 86 to 149 days.

A preliminary Cowell-based mission analysis estimated a transfer duration of approximately 95 days. Under this reference, the continuous-thrust solutions remain within the same operational timescale, whereas non-continuous thrust provides improved propellant efficiency at the expense of a significantly longer transfer.

The detailed thrust-coast strategy will therefore remain subject to further optimisation considering power availability, thruster lifetime, transfer-time requirements and xenon consumption.

The selected mission-level architecture is consequently:

$$\text{GTO Injection} + \text{Electric Orbit Raising} \tag{7}$$

This solution combines the greater launch mass available in GTO with the high propellant efficiency of electric propulsion.

3.2.2 Launch Vehicle Selection

The four launch vehicle alternatives identified in Section 3.1 are compared:

- LV1: Vega-C;
- LV2: Ariane 62;

- LV3: Ariane 64;
- LV4: Falcon 9.

The comparison considers both launcher performance and the effect of the injection conditions on the subsequent electric transfer.

Table 3.3: Trade-off of the launch vehicle alternatives.

Criterion	**Weight**	**LV1**	**LV2**	**LV3**	**LV4**
Payload capability	25%	1	4	5	5
Compatibility with GTO architecture	20%	2	5	5	5
Transfer penalty from injection orbit	20%	4	5	5	2
Launch availability/flexibility	15%	3	4	4	5
European strategic preference	10%	5	5	5	2
Mass-growth margin	10%	1	3	5	5
Weighted score		**2.50**	**4.40**	**4.85**	**4.10**

Vega-C is not retained as a baseline launcher because its payload capability provides insufficient margin for the expected Lazarus mass under the required injection conditions.

The three launch vehicles retained as technically relevant alternatives are therefore:

$$\text{Ariane 62}, \qquad \text{Ariane 64}, \qquad \text{Falcon 9} \tag{8}$$

The Ariane 6 family is preferred because it maintains a European launch architecture while providing suitable performance for the selected GTO injection strategy.

Ariane 64 obtains the highest preliminary score because its larger payload capability provides the greatest margin for the Lazarus launch mass and future spacecraft mass growth.

Ariane 62 nevertheless remains a viable alternative if the final spacecraft mass is compatible with its available performance.

Falcon 9 is retained as a backup option in order to increase mission flexibility and reduce dependence on the availability of Ariane 6.

The transfer analyses also demonstrate that the launch vehicle cannot be selected based exclusively on payload capability.

The initial GTO inclination directly affects the xenon required to reach the final equatorial GEO+50 km orbit.

The preliminary analysis indicates that the higher-inclination reference case requires approximately 80 kg more xenon than the low-inclination case when continuous thrust is used.

Therefore, the propulsion system and spacecraft propellant budget shall be capable of supporting the most demanding retained injection scenario.

This approach allows a common Lazarus spacecraft configuration to remain compatible with different launch vehicles without requiring a redesign of the propulsion architecture.

The preliminary launcher strategy is therefore:

$$\text{Ariane 6 preferred} \qquad \text{Falcon 9 backup} \tag{9}$$

with the final selection between Ariane 62 and Ariane 64 dependent on the consolidated Lazarus launch mass and the performance available for the specific launch opportunity.

3.2.3 Parking Orbit Selection

The following alternatives are compared:

- O1: fixed GEO slot;
- O2: GEO+50 km drift orbit;
- O3: GEO-50 km drift orbit.

Table 3.4: Trade-off of the parking orbit.

Criterion	**Weight**	**O1**	**O2**	**O3**
Need for a GEO slot	25%	1	5	5
Interaction with GEO spacecraft	20%	5	5	2
Orbit-maintenance demand	20%	2	4	4
Scalability to multiple servicers	20%	1	5	5
Operational simplicity	15%	5	4	4
Weighted score		**2.60**	**4.65**	**4.05**

Alternative **O2, GEO+50 km drift orbit**, is selected.

The use of a fixed GEO slot is not particularly appropriate for a vehicle whose main function is to travel between different clients. Geostationary positions are also a highly restricted resource, and operation in a defined slot would require the corresponding *station keeping* manoeuvres throughout the operational lifetime.

The GEO+50 km and GEO-50 km alternatives eliminate the need to maintain a fixed longitude and allow the use of a drift orbit.

However, the GEO-50 km option presents a less favourable geometry. Since Lazarus would be located closer to the Earth than the spacecraft in GEO, it could temporarily be positioned between them and the Earth during its drift.

Since many communication antennas, cameras, and other sensors on GEO spacecraft have Earth-pointing fields of view, this configuration increases the possibility of the servicer crossing regions close to their lines of sight or pointing directions.

This does not necessarily imply direct interference, but would require additional operational constraints and analyses.

The GEO+50 km orbit retains the advantages of longitudinal drift without routinely placing the servicer on the Earth-facing side of the geostationary belt.

Furthermore, this solution facilitates a future expansion of the architecture to several AstroFUEL vehicles without requiring the allocation of a conventional GEO slot to each one.

3.2.4 Propulsion Architecture Selection

The following alternatives are compared:

- P1: fully chemical propulsion;
- P2: hybrid chemical-electric architecture;
- P3: fully electric architecture.

Table 3.5: Trade-off of the propulsion architecture.

Criterion	**Weight**	**P1**	**P2**	**P3**
Efficiency / propellant mass	25%	1	4	5
System mass and complexity	20%	3	2	5
Sustainability and common xenon use	15%	1	2	5
RPO performance and safety	15%	5	5	3
Power demand	10%	5	3	2
Suitability for major transfers	15%	1	5	5
Weighted score		**2.40**	**3.50**	**4.40**

Alternative **P3, fully electric propulsion**, is selected.

The propulsion trade is strongly coupled with the selected GTO injection strategy.

The preliminary chemical transfer analysis resulted in a propellant requirement of approximately:

$$m_{\text{prop,chem}} \approx 2000 \text{ kg}, \tag{10}$$

which represents an excessive mass penalty for the expected Lazarus architecture.

In comparison, the preliminary electric-transfer analyses resulted in xenon requirements between approximately:

$$331 \text{ kg} \lesssim m_{\text{Xe,GTO}-\text{GEO+50}} \lesssim 504 \text{ kg}, \tag{11}$$

depending on the initial inclination and selected thrust strategy.

Although the chemical and electric analyses use different transfer formulations and therefore do not constitute a strict one-to-one comparison, the results clearly show the significant reduction in propulsion mass achievable through electric orbit raising.

This is particularly important for Lazarus because every kilogram allocated to the spacecraft propulsion system reduces the mass available for client xenon or other spacecraft subsystems.

A hybrid architecture would retain the advantages of electric propulsion for the large orbital transfers while using chemical propulsion during RPO. However, this would require two independent propulsion technologies and their associated tanks, feed systems, valves and control hardware.

The fully electric architecture avoids this additional complexity.

The selected architecture therefore consists of:

- one main electric propulsion system for GTO orbit raising and major orbital transfers;
- one distributed system of twelve auxiliary electric thrusters for rendezvous, docking, undocking, retreat and reaction-wheel unloading.

Preliminary 6DoF analyses have demonstrated the feasibility of performing the complete proximity-operation sequence using the distributed electric propulsion system.

The main penalty is the increase in manoeuvre duration due to the low available acceleration. However, since Lazarus is an uncrewed servicing vehicle, this increase in duration is considered acceptable compared with the reductions in propellant mass and system complexity.

Very-close-range abort capability remains one of the principal design drivers of the auxiliary propulsion architecture. This risk is mitigated through reduced closing velocities, hold points, strict GO/NO-GO criteria and dedicated onboard abort logic.

The selected propulsion architecture is therefore:

$$\text{100\% Electric Propulsion} \tag{12}$$

3.2.5 Client Cooperation Selection

The following alternatives are compared:

- C1: active client;
- C2: passive but cooperative client;
- C3: non-cooperative client.

Table 3.6: Trade-off of the degree of client cooperation.

Criterion	Weight	C1	C2	C3
Required client modifications	20%	2	4	5
Safety and predictability	25%	4	5	1
Servicing compatibility	20%	4	5	2
Operational complexity	15%	2	4	1
Commercial scalability	20%	2	4	5
Weighted score		**2.90**	**4.45**	**2.80**

A **passive but cooperative client** is selected.

Lazarus will be the active vehicle during rendezvous, avoiding simultaneous translational manoeuvres by both vehicles at short range.

However, the client shall cooperate with the operation by:

- maintaining a known servicing attitude;
- transmitting orbital and attitude information;
- providing a compatible docking and xenon-transfer interface;
- providing cooperative navigation elements, such as fiducials;
- coordinating operations with the Lazarus ground segment.

This solution concentrates GNC complexity in the servicer and reduces the modifications required on client spacecraft.

Furthermore, it is consistent with the AstroFUEL philosophy of developing a refuelling system based on a standardised interface for future GEO Spacecraft (S/C).

3.2.6 Rendezvous Strategy Selection

The following alternatives are compared:

- R1: continuous approach;
- R2: hold-point approach.

Table 3.7: Trade-off of the rendezvous strategy.

Criterion	**Weight**	**R1**	**R2**
Safety	30%	2	5
Abort capability	25%	2	5
Operational simplicity	15%	4	3
Operation duration	15%	5	2
Intermediate state verification	15%	2	5
Weighted score		**2.75**	**4.25**

The **hold-point approach** is selected.

The preliminary sequence is:

$$500 \text{ m} \rightarrow 100 \text{ m} \rightarrow 20 \text{ m} \rightarrow 5 \text{ m} \rightarrow \text{Docking}. \tag{13}$$

At each hold point, the main conditions required to continue are verified:

- relative-navigation solution;
- client attitude and status;
- alignment;
- actuator availability;
- power availability;
- capability to execute an abort;
- GO/NO-GO criteria.

This strategy is particularly suitable for a low-acceleration architecture based on electric propulsion, as it allows the closing velocity to be limited before entering higher-risk regions.

3.2.7 Relative Navigation Architecture Selection

The following alternatives are compared:

- N1: optical-only navigation;
- N2: navigation based mainly on active sensors;
- N3: multi-sensor architecture.

Table 3.8: Trade-off of the relative-navigation architecture.

Criterion	**Weight**	**N1**	**N2**	**N3**
Robustness	30%	2	3	5
Accuracy	25%	4	4	5
Independence from illumination	15%	1	5	5
Redundancy	20%	2	2	5
Power and complexity	10%	5	3	2
Weighted score		**2.65**	**3.35**	**4.70**

A **multi-sensor architecture** is selected, preliminarily consisting of:

- two relative-navigation cameras;
- one three-dimensional LiDAR;
- one laser rangefinder.

The cameras allow line-of-sight and relative pose estimation using the known geometry of the client and cooperative elements.

The LiDAR provides active range and geometry information with reduced dependence on illumination conditions.

The laser rangefinder provides a simple and independent measurement of range and closing velocity.

The use of different sensing technologies provides heterogeneous redundancy. Degradation of an optical sensor does not necessarily affect an active sensor in the same way, allowing inconsistencies to be detected and improving FDIR functions.

3.2.8 Reaction-Wheel Desaturation System Selection

The following alternatives are compared:

- M1: magnetorquers;
- M2: chemical RCS;
- M3: distributed electric propulsion.

Table 3.9: Trade-off of the reaction-wheel desaturation system.

Criterion	Weight	M1	M2	M3
Control authority at GEO	30%	1	5	5
Added mass and complexity	25%	2	1	5
Propellant consumption	20%	5	2	4
Pointing compatibility	15%	2	5	5
Reuse of existing hardware	10%	1	1	5
Weighted score		**2.20**	**3.00**	**4.80**

Distributed electric propulsion is selected.

At GEO altitude, the available terrestrial magnetic field does not provide sufficiently robust control authority for magnetorquers to be used as the primary desaturation system.

A chemical RCS would allow the wheels to be unloaded rapidly, but would require the introduction of a second propulsion system solely for this and other auxiliary functions.

The selected solution reuses the same twelve electric thrusters employed during RPO.

Preliminary simulations have demonstrated:

- three-axis unloading capability;
- continuous nadir-pointing maintenance;
- tolerance to a single thruster failure;
- low xenon consumption.

Desaturation is not normally an urgent manoeuvre and can therefore be performed slowly and scheduled during favourable solar-power-generation windows in order to reduce peak power demand.

3.2.9 Ground Coordination Concept Selection

The following alternatives are compared:

- G1: single or joint control centre;
- G2: coordinated independent control centres.

Table 3.10: Trade-off of the ground coordination concept.

Criterion	Weight	G1	G2
Clarity of responsibilities	25%	3	5
Interoperability between operators	25%	4	5
Commercial scalability	20%	2	5
Coordination speed	20%	5	4
Implementation complexity	10%	2	4
Weighted score		**3.35**	**4.70**

Coordinated independent control centres are selected.

Each operator retains authority over its own spacecraft, avoiding the need to physically integrate control centres belonging to different organisations.

During rendezvous, docking, and refuelling, reinforced operational coordination is established, potentially including a joint servicing cell responsible for:

- synchronising the operational timeline;
- exchanging relevant information;
- coordinating GO/NO-GO decisions;
- verifying the configuration of both vehicles;
- managing contingencies.

It is not necessary for both spacecraft to use the same ground station or to be operated from the same physical control room.

3.2.10 Level of Autonomy Selection

The following alternatives are compared:

- A1: direct ground control;
- A2: fully autonomous operation;
- A3: supervised autonomy.

Table 3.11: Trade-off of the level of autonomy during RPO.

Criterion	Weight	A1	A2	A3
Safety / response time	30%	1	5	5
Ground operational workload	15%	2	5	4
Development complexity	20%	4	1	4
Operational robustness	20%	2	4	5
Human supervision	15%	5	1	5
Weighted score		**2.55**	**3.40**	**4.65**

Ground-supervised autonomy is selected.

The ground segment:

- plans the operation;
- transmits command sequences;
- monitors execution;
- authorises the main operational transitions when required.

Once a sequence has been initiated, AstroFUEL autonomously performs:

- relative navigation;
- guidance;
- translational control;
- attitude control;
- actuator allocation;
- safety-limit monitoring.

Functions requiring immediate response cannot depend on the ground link.

3.2.11 FDIR Strategy Selection

The following alternatives are compared:

- F1: conventional FDIR;
- F2: proximity-operations-specific FDIR.

Table 3.12: Trade-off of the FDIR strategy during RPO.

Criterion	Weight	F1	F2
Collision safety	40%	1	5
Failure response	25%	2	5
Implementation simplicity	15%	5	3
Operational robustness	20%	2	5
Weighted score		**2.05**	**4.70**

A **proximity-operations-specific FDIR** is selected.

During nominal operation, certain failures may automatically trigger a transition to *safe mode*. However, this response may be dangerous when AstroFUEL is located only a few metres from the client.

A sudden attitude change, uncontrolled loss of specific actuators, or an automatic manoeuvre not designed for RPO could increase the risk of collision.

During proximity operations, the priority of the FDIR system will therefore be to maintain a safe relative trajectory.

Detection of a situation requiring an abort shall trigger an autonomous onboard response without waiting for ground authorisation.

Once a safe distance has been reached, Lazarus may leave the proximity-operations FDIR mode and restore the conventional *safe mode* logic.

3.2.12 Fleet Architecture Selection

For the initial demonstration, the following alternatives are compared:

- S1: a single AstroFUEL vehicle;
- S2: initial deployment of a fleet of vehicles.

Table 3.13: Trade-off of the architecture for the initial demonstration.

Criterion	Weight	S1	S2
Initial investment	35%	5	1
Demonstration simplicity	25%	5	2
Initial redundancy	15%	1	5
Future scalability	15%	3	5
Commercial response time	10%	2	5
Weighted score		**3.80**	**2.85**

For the first mission, **a single AstroFUEL vehicle** is selected, with the objective of demonstrating the complete servicing concept.

The mission will allow validation of orbital transfer using electric propulsion, navigation and rendezvous, docking, xenon transfer, undocking, return to the parking orbit and vehicle reuse for multiple servicing operations.

Once the technology has been demonstrated, the commercial architecture will evolve towards the deployment of **at least two AstroFUEL vehicles**.

The use of multiple servicers provides:

- mission-level redundancy;
- increased service availability;
- reduced average response time;

The use of GEO+50 km as the parking orbit facilitates this evolution by avoiding the need to allocate an independent conventional GEO slot to each vehicle.

3.2.13 Selected Mission Architecture

As a result of the previous trade-offs, the AstroFUEL baseline architecture is defined as:

Table 3.14: Selected mission architecture.

Element	Selection
Initial injection	GTO + autonomous electric transfer
Parking orbit	GEO+50 km drift orbit
Propulsion	100% electric
Client	Passive but cooperative
Rendezvous	Hold-point approach
Relative navigation	2 cameras + LiDAR + laser rangefinder
Reaction-wheel desaturation	Distributed electric propulsion
Ground coordination	Coordinated independent control centres
Autonomy	Ground-supervised, onboard-executed
FDIR	Proximity-operations-specific FDIR
Initial demonstrator	1 AstroFUEL
Commercial architecture	$\geq$ 2 AstroFUEL vehicles

4 Mission Overview

4.1 Mission Baseline Description

The mission baseline starts from the client chart (Table 4.1): the orbit of each client and the expected end of its propellant.

Table 4.1: Client chart. All clients are on a circular equatorial GEO at 36 000 km of altitude.

Client	Slot [° E]	Expected end of propellant
A	19.0	Jun 2035
B	-43.6	Jul 2036
C	170.6	Nov 2037
D	113.6	Sep 2038
E	129.3	Feb 2040

To serve every client in time, the mission has to manage the following:

- **Refuelling windows.** The end of propellant is an estimate, so each window is placed 104 d to 127 d before it to cover its uncertainty. The end of the window is the latest service date.
- **Launch date.** The low-thrust transfer takes 370 d to 505 d, so the launch must be early enough to reach GEO-P before the first window, leave time for commissioning and place the first service, which is the demonstration, outside an eclipse season.
- **Sequence.** The services are performed one after another from GEO-P, so the availability of Lazarus after one client fixes the time available for the next.
- **Relative position.** Lazarus can only depart towards a client with the optimal relative angle. It either waits for the natural drift to bring it (Case 1) or phases during the last 100 d (Case 2).
- **Station keeping** of GEO-P during every wait.
- **Propellant**: transfer, services, station keeping and disposal must fit in the 500 kg tank.
- **Disposal** to a graveyard orbit 250 km above GEO-P after the last client.

4.2 Launcher Description

Based on the analysis of the performance of the launch vehicles considered, and taking a preliminary spacecraft mass of approximately 3,000 kg as a reference, the most restrictive constraints have been identified to ensure the compatibility of Lazarus with the different launch alternatives.

The most restrictive payload capacity corresponds to Ariane 62 , with a maximum mass of 4,500 kg for the considered injection orbit. Regarding geometric limitations, Falcon 9 establishes the reference maximum diameter of 5.3 m, while Ariane 62 presents the most restrictive maximum height of 11.8 m.

Launcher	Configuración	Volume [m^3]	Mass [kg]	Price [M$]
Ariane 62	Single launch – Short fairing	196,104	4500	77
Ariane 62	Single launch – Long fairing	299,142	4500	77
Ariane 64	Single launch	299,142	11500	126
Falcon 9	Single launch	280,3306	4800	61

Table 4.2: Comparison of launcher volume, mass, and price.

The spacecraft dimensions are constrained by the usable payload volumes defined in the Ariane 6 User's Manual [28] and the Falcon User's Guide [29].

Based on the initial spacecraft mass estimate, only a dedicated launch configuration is currently considered, in which Lazarus is the only primary payload of the launch vehicle.

However, if the spacecraft mass can be significantly reduced during the design and sizing process, the possibility of a shared launch may be evaluated. In this scenario, the mass and volume constraints would need to be reviewed, as they would depend on the distribution of the payloads, the adapters used and the integration conditions established for the shared mission.

Therefore, the identified constraints will be used as a reference for the preliminary sizing, although they may be modified as the design progresses and the available launch opportunities become more clearly defined.

4.3 Satellite Description

The Lazarus servicer spacecraft main body is a rectangular prismatic geometry with baseline dimensions of 2.9 m × 2.9 m × 5.0 m, expanding to a stowed launch envelope of 3.4 m × 2.9 m × 5.0 m with folded solar array wings. The spacecraft is sized for a nominal launch wet mass of 3,000 kg. This mass budget provides a generous unallocated margin of 1,500 kg against the 4,500 kg maximum capacity of the Ariane 62 launcher, ensuring high system flexibility and growth capability. Primary structural loads are transmitted through a central Carbon Fiber Reinforced Polymer (CFRP) cylindrical tube measuring 1.2 m in diameter and 5.0 m in height, which serves as the primary load path between the launcher payload adapter interface and the upper servicing plane. The core is enclosed by an outer structural framework of Aluminium 7075-T6 longerons and transverse beams supporting CFRP/Aluminium honeycomb sandwich panels, reinforced internally by five stiffening rings and twelve shear walls. The spacecraft utilizes a 100% Xenon-based electric propulsion architecture for all mission phases, stored in 3 tanks of 90 liters, and has a payload capacity of 500 kg of Xenon, stored in 3 separate tanks of 120 liters each. Main orbit raising from Geostationary Transfer Orbit (GTO) to the Geostationary Earth Parking Orbit (GEO-P) and long-range orbital phasing are executed by four Safran PPS5000 Hall Effect thrusters (delivering 300 mN nominal thrust at a specific impulse Isp of 2,000 s) arranged on 2-axis pointing mechanisms at the lower deck. Rendezvous and Proximity Operations (RPO), fine relative maneuvering, docking/undocking, and three-axis reaction-wheel momentum desaturation are performed by twelve auxiliary SITAEL HT400 electric thrusters arranged in four canted corner clusters of three thrusters each to generate the required forces and torques while minimizing unwanted perturbations. The avionics,

power, and servicing payload architectures are optimized for autonomous proximity operations and continuous high power demand. Attitude control is managed by a three-axis Attitude and Orbit Control System (AOCS) utilizing four 12.0 Nms pyramidal reaction wheels, desaturated exclusively via the auxiliary electric thrusters without chemical RCS or magnetorquers. The GNC navigation suite combines Sodern HORUS star trackers, fine/coarse Sun sensors, a Safran STIM300 IMU, and a Beyond Gravity GEO GNSS receiver. The upper-deck relative navigation sensor suite comprises dual Infinity Avionics cameras (including an infrared camera), a Jena-Optronik 3D LiDAR, a Jenoptik laser rangefinder, and an active Target Illumination System. Primary power is generated by two deployable solar array wings providing 30 m^2 of total solar cell area, supported by batteries sized to sustain the most demanding power condition, which corresponds to the refuelling operations, during the maximum continuous Earth eclipse duration of 72 minutes in GEO-P. Servicing operations are executed via an upper-deck OrbitFab RAFTI universal docking interface and a static hybrid thermal-compression Fluid Transfer System, supported by S-band low-gain antennas for TT&C and a steerable K-band high-gain antenna for high-rate proximity telemetry and video downlink.

4.4 Ground Stations

High-gain network (K-band): We will use the 35 m antennas from ESA's deep space network. The Cebreros ground station (DSA-2) will be the main ground station during orbital movement in GEO (it receives in X, Ka, or K band, and transmits in X band). However, for orbital maneuvers, we will use the rest of the ESTRACK antennas, such as New Norcia in Australia and Malargüe in Argentina. As the coverage is not sufficient to cover the entire Earth, time will be leased on NASA's DSN network in Goldstone, California.

We define the characteristics of our main ground station below, with the characteristics of the other ESTRACK antennas being identical.

Parameter	**Specification**
Frequency	K band (25.5 - 27.0 GHz) downlink X band (7145 - 7235 MHz) uplink
G/T Sensitivity	$>$ 55.8 dB/K
EIRP Emission Power	$>$ 107 dBW
Pointing Accuracy	$<$ 0.006°
Modulation	QPSK, OQPSK, CCSDS

Table 4.3: Cebreros ESTRACK Station Characteristics

Low-gain network (S-band): This will be the network we use for telemetry and telecommand communications. Additionally, it will be used in emergency situations when the pointing of our HGA antenna cannot be guaranteed. Our main station will be the Kourou Station (15 m). This station operates in S-band for uplink and downlink, and X-band for downlink. We will also use

the Malindi station in Kenya and subcontract the services of the Dongara (Australia) and South Point (Hawaii) ground stations, both belonging to the SSC (Swedish Space Corporation).

We define the characteristics of our main station below, which are identical for the rest of the ESA stations and similar for the other antennas.

Parameter	**Specification**
Frequency	S band (2025 - 2300 MHz) downlink and uplink X band (7145 - 8500 MHz) downlink
G/T Sensitivity (S)	22 dB/K
EIRP Emission Power	$>$ 68 dBm (38 dBW)
Pointing Accuracy	$< 0.1°$
Modulation	BPSK, GMSK, CCSDS

Table 4.4: Kourou Station Characteristics

4.5 Mission Operation Phases

The mission is divided into four phases: LEOP (transfer from GTO to GEO-P), Commissioning Phase (CP), OP (five services with station keeping between them) and Disposal Phase (DP) (transfer to the graveyard orbit). Lazarus serves five clients in GEO whose slots and expected end of propellant are known. All the results come from the Lazarus mission simulation, with the parameters of Table 4.5 and two launch cases: $i_0 = 6°$ (29 May 2033) and $i_0 = 27°$ (2 June 2033).

Table 4.5: Lazarus parameters used in the simulations.

Parameter	**Value**
Initial mass, m_0 [kg]	3000
Thrust, T [N]	0.3
I_{sp} [s]	2000
SRP area, A [m^2]	41.7
C_R [–]	1.30

4.5.1 Launch and Early Orbit Phase

The first phase of the mission, called (LEOP), comprises the launch of the satellite into space, its separation from the launch vehicle and its injection into a specific GTO. It is important to clarify that the GTO is this mission, does not have its apogee at 36,000 km (GEO) as usual, since Lazarus' nominal orbit is at 36,050 km (GEO + 50km). Therefore the apogee of the GTO in this mission has its perigee as usual at 250km, but its apogee at 36,050 km. During this phase of the mission, the satellite is switched off and all antennas and solar panels are folded.

4.5.2 Commissioning Phase

When the satellite separates from the launch vehicle, already in the above described GTO, it boots up its systems triggered by a separation switch. The initialization of the satellite starts immediately executing a pre-programmed autonomous boot sequence. This sequence includes the deployment of the solar panels and the high-gain directional antenna of the satellite. The solar panels are aligned towards the sun, ensuring the power supply of the satellite, and the antenna is aligned towards Earth for allowing communication with Earth.

As soon as the satellite establishes connection with Earth, it sends a first signal to the ground stations to confirm it is functioning. Then, an exhaustive verification of every subsystem onboard is performed, to ensure the correct functioning of every single of them after launch. This phase is called the CP of the mission.

4.5.3 Operational Phase

The OP begins once the CP has been successfully completed and Lazarus has been declared fully operational. During this phase, the vehicle provides xenon refueling services to client spacecraft located in GEO. The operation is based on a repetitive cycle that begins in GEO-P, continues with a transfer towards the client to rendezvous, the docking and refueling, and it ends with Lazarus returning to GEO-P.

First, Lazarus has to circularize its orbit to a circular orbit 50 km above GEO, while also reducing the inclination of its orbital plane to 0°. As stated before, this orbit is refered to as GEO-P and it is Lazarus' nominal orbit.

Once in GEO-P, the commercial activity of Lazarus can begin. The general sequence of a servicing mission is:

$$\text{Waiting} \rightarrow \text{Mission Preparation} \rightarrow \text{Transfer} \rightarrow \text{Rendezvous} \rightarrow \text{Proximity Operations}$$

$$\rightarrow \text{Docking} \rightarrow \text{Refueling} \rightarrow \text{Undocking} \rightarrow \text{Retreat} \rightarrow \text{Return to GEO} - \text{P}$$

Each of these operations is associated to a specific spacecraft mode. This allows the sensors, actuators, navigation algorithms, power-system configuration and FDIR functions required at each step to be appropriately configured. The main operational modes of the spacecraft are summarized below in Table 4.6.

Table 4.6: Main operational modes during OP.

Mode	Main function
Waiting Mode	Nominal waiting in GEO-P
Service Preparation Mode	Preparation of a new servicing mission
Transfer Mode	Transfer and phasing towards the client
Rendezvous Acquisition Mode	Initial acquisition of the client
Far-Range RPO Mode	Long-range relative approach
Close-Range RPO Mode	Precision relative approach
Docking Mode	Final approach and capture
Docked/Refueling Mode	Xenon transfer
Undocking Mode	Initial separation from the client
Retreat Mode	Safe departure from the client
Return Transfer Mode	Return to GEO-P
Abort Mode	Safe separation in case of contingency

In addition, some functions, such as reaction-wheel desaturation, may be executed as submodes within different operational modes whenever mission conditions allow it.

Between two servicing operations, Lazarus remains in GEO-P. During this phase, the spacecraft operates in **Waiting Mode**, which represents the nominal long-duration configuration of the vehicle. In this mode, Lazarus mainly performs:

- orbit determination and propagation;
- maintenance of the orbit within the operational envelope;
- maintenance of nominal attitude;
- communications with the ground segment;
- spacecraft health monitoring;
- thermal management;
- power management;
- monitoring of reaction-wheel angular momentum;
- reception of mission updates and potential client data.

The nominal attitude is assumed to be NADIR-pointing, compatible with communication, navigation, and thermal-management requirements.

The reaction wheels are the primary actuators for attitude control. When the accumulated angular momentum reaches a predefined threshold, Lazarus autonomously transitions from **Waiting Mode** to **Desaturation Mode**.

During Desaturation Mode, the auxiliary electric thrusters are used to unload the accumulated angular momentum from the reaction wheels while the spacecraft continues to maintain its nominal NADIR-pointing attitude. The reaction wheels remain active during the manoeuvre and compensate for attitude disturbances generated by the electric propulsion system.

Since reaction-wheel desaturation is normally not time-critical, the manoeuvre can be scheduled during favourable power-generation conditions. Once the reaction-wheel angular momentum has been reduced below the predefined recovery threshold and nominal attitude-control conditions have been restored, Lazarus autonomously returns to **Waiting Mode**.

During Waiting Mode, the spacecraft continuously monitors its subsystem status through the onboard FDIR functions. If an anomaly or failure is detected that prevents nominal operation or could compromise spacecraft safety, Lazarus autonomously transitions to **Safe Mode**.

In Safe Mode, non-essential and potentially hazardous systems are switched off or placed in a safe configuration, while the minimum functions required to preserve spacecraft safety are maintained. These include, as applicable, basic attitude control, power generation, thermal control, onboard health monitoring, and communications with the ground segment.

The spacecraft remains in a stable and power-positive configuration and waits for ground commands before returning to nominal operations or initiating the corresponding recovery procedure.

Under nominal conditions, Lazarus remains in Waiting Mode until the ground segment assigns a new client and authorises the start of a new servicing operation.

Service Preparation Mode

Once a client has been selected, Lazarus enters the **Service Preparation Mode**. The objective of this mode is to verify that both the servicer and the client are in suitable conditions before leaving the parking orbit.

The ground segment provides the spacecraft with the information required for the new operation including:

- client ephemerides;
- expected orbital state;
- client servicing attitude;
- docking-interface characteristics;
- xenon quantity to be transferred;
- operational timeline;
- transfer and phasing strategy;
- planned rendezvous sequence;
- safety constraints;

- GO/NO-GO criteria.

Before initiating the transfer, checks are performed on:

- propulsion system;
- navigation system;
- communications;
- available power;
- reaction-wheel status;
- relative-navigation sensors;
- docking system status;
- availability of the xenon required by the client.

Once all checks have been successfully completed, the ground segment authorises departure from the parking orbit.

Transfer Mode

After leaving GEO-P, Lazarus enters the **Transfer Mode**. During this phase, the main electric propulsion system is primarily used to progressively modify the orbit and perform the phasing required with the client spacecraft. The sequence is:

$$\text{GEO-P} \rightarrow \text{Phasing} \rightarrow \text{Client Orbital Region}.$$

During this phase, navigation continues to rely mainly on absolute navigation and orbit propagation. The GNC system maintains the attitude required by the main thruster, solar arrays, communications and thermal constraints.

Due to the low thrust of the electric propulsion system, the transfer is performed through long-duration manoeuvres.

When Lazarus reaches GEO 500m behind the client's satellite, the rendezvous between the two spacecraft can begin.

Rendezvous Acquisition Mode

Before initiating proximity operations, Lazarus enters the **Rendezvous Acquisition Mode**.

The main objective is to detect the client, initialize relative navigation and verify that its state is consistent with the information previously provided by the ground segment. The main electric thrusters are turned off and the smaller electric thrusters take over.

During this phase, the transition between absolute navigation and relative navigation is progressively performed. The relative-navigation sensors begin to be activated according to their operational range. The preliminary architecture of the relative-navigation system includes:

- two relative-navigation cameras;
- one three-dimensional LiDAR;
- one laser rangefinder.

Initially, the cameras provide target acquisition and line-of-sight estimation. As the distance decreases, the active sensors provide additional measurements of range, closing velocity and relative geometry. The information from the different sensors is fused by the relative-navigation system. The rendezvous-maneuver is divided into five "hold points". Before continuing towards the next hold point, the acquisition and navigation criteria defined at every hold point must be satisfied.

Far-Range RPO Mode

Once a sufficiently accurate relative-navigation solution has been established, Lazarus enters the **Far-Range Rendezvous and Proximity Operations (RPO) Mode**. During this phase, the first part of the controlled approach to the client is performed in five steps:

$$500\text{m} \rightarrow 100\text{m} \rightarrow 20\text{m} \rightarrow 5\text{m}.$$

The hold points allow the spacecraft status to be verified multiple times before getting so close to the client's satellite that the time available to react to an anomaly becomes very short. At each hold point, the following aspects are checked:

- accuracy and consistency of relative navigation;
- relative position and velocity;
- client attitude;
- alignment with the approach corridor;
- availability of the auxiliary thrusters;
- reaction-wheel status;
- available power;
- docking-system status;
- capability to execute a retreat or abort manoeuvre.

The ground segment monitors the operation and may authorise the main transitions between phases. However, trajectory execution and relative-motion control are performed autonomously onboard.

Close-Range RPO Mode

As Lazarus approaches the client, the system enters the **Close-Range RPO Mode**.

This mode mainly includes operations between the final two hold points (20m and 5m) and the region immediately prior to docking.

During this phase, increased importance is given to:

- LiDAR;
- laser rangefinder;
- close-range approach camera;
- fiducial markers installed on the client;
- precise relative-pose estimation.

The closing velocity is progressively reduced as the distance decreases. This is especially important because the architecture uses low-acceleration electric thrusters. Maintaining low approach velocities ensures that sufficient distance remains available for performing a separation maneuver.

Table 4.7: Velocities between hold points.

Distance Range	**Velocity [m/s]**
500m - 200m	0.22
200m - 100m	0.089
100m - 20m	0.01
20m - 5m	0.005
5m - dock	< 0.005

During this phase, the proximity-operations FDIR remains fully active.

In the event of navigation degradation, loss of control authority or any other unsafe condition, the spacecraft may autonomously take the following decision: hold, retreat, abort.

Docking Mode

After successfully completing the final hold point at 5m from the client, Lazarus enters the **Docking Mode**. During this phase, the final approach towards the servicing interface is performed. The GNC system simultaneously controls:

- relative position;
- relative velocity;
- angular alignment;
- relative angular velocity;

- lateral offset with respect to the docking port.

Navigation is mainly based on short-range sensors and fiducial markers installed around the cooperative servicing interface of the client.

The closing velocity is deliberately kept low (<0.005m/s) in order to reduce contact loads and preserve abort capability.

The sequence is:

$$5\text{m} \rightarrow \text{Final Approach} \rightarrow \text{Contact} \rightarrow \text{Capture} \rightarrow \text{Docked}.$$

Once contact occurs, the docking mechanism performs capture and rigidisation of the connection between both vehicles. Before refueling can begin, it must be verified that the mechanical connection and propellant-transfer interface have been correctly established.

Docked/Refueling Mode

Once docking has been completed, Lazarus enters the **Docked/Refueling Mode**. During this mode, both spacecraft are temporarily considered as one mechanically coupled system.

Before the xenon transfer can begin, the following aspects must be verified:

- docking mechanism status;
- connection of the refuelling interface;
- interface sealing;
- initial pressures;
- valve status;
- transfer-system configuration;
- available xenon quantity;
- amount of xenon requested by the client.

Once these checks have been successfully completed, the xenon transfer can begin. During refueling, the following parameters are continuously monitored:

- pressure;
- temperature;
- flow rate;
- transferred mass;
- valve status;

- possible leaks or anomalies.

The transfer ends when the programmed xenon quantity has been delivered or when one of the system limits is reached. The lines are subsequently isolated and the required checks are performed before undocking is authorized.

Undocking Mode
Once the transfer has been completed, Lazarus enters the **Undocking Mode**. The process begins with the safe configuration of the xenon interface and release of the docking mechanism. The initial separation is performed at very low velocity in order to avoid unintended contact between both spacecraft. During this phase, full relative navigation is re-established and positive physical separation is verified before the retreat manoeuvre begins.

Retreat Mode
After the initial separation, the **Retreat Mode** is activated, which follows the same steps as used during the approach:

$$\text{Docking} \rightarrow 5\text{m} \rightarrow 20\text{m} \rightarrow 100\text{m} \rightarrow 500\text{m}.$$

The auxiliary electric thrusters control the relative motion throughout the sequence. Relative navigation remains active until a safe separation distance has been reached. Once the proximity region has been cleared, the use of relative-navigation sensors can be progressively reduced and the spacecraft transitions back towards absolute orbital navigation.

Return Transfer Mode
Once proximity operations have been completed, Lazarus enters the **Return Transfer Mode**. The main electric propulsion system is used again to leave the client's orbital region and perform the phasing required to recover the nominal drift orbit. The sequence is:

$$\text{Client Region} \rightarrow \text{Transfer} \rightarrow \text{GEO-P}.$$

Once the orbital elements are again within the defined waiting-orbit envelope in GEO-P, the servicing operation is considered complete and Lazarus can return to Waiting Mode and becomes available for a new mission.

Abort Mode
The **Abort Mode** is a safety mode available during rendezvous, proximity operations and docking. Unlike the other modes, it does not form part of the nominal mission sequence and is only activated when a condition is detected that makes continuation of the approach unsafe. Possible causes of activation include:

- loss or severe degradation of relative navigation;

- loss of control authority;
- failure of one or more relevant actuators;
- unexpected client behaviour;
- excessive closing velocity;
- departure from the approach corridor;
- critical power loss;
- docking-system failure;
- violation of a safety limit.

The response depends on the distance to the client and the severity of the anomaly. The Abort Mode must be capable of being initiated autonomously onboard, since the time available to avoid a collision may be shorter than that required to receive a decision from the ground segment.

The objective of the abort manoeuvre is to increase the distance between both vehicles in a controlled and predictable manner and reach a safe region from which the mission status can be reassessed. Once a safe distance has been reached, the vehicle may remain on hold, return to the nominal parking orbit, or enter the conventional safe mode depending on the type of failure.

Operational Cycle Summary

The Operational Phase is repeated as long as none of the end-of-life criteria is reached. These are primarily four:

- Critical hardware or subsystem degradation that threatens the spacecraft's ability to maintain nominal station-keeping, attitude control, or communication.
- Lazarus does not carry sufficient xenon to be able to perform another servicing maneuver and reaching afterwards its graveyard orbit.
- Lazarus does not carry the demanded amount of payload (xenon) the clients demand.
- Completion of the maximum extended mission timeline.

Once any of the end-of-life criteria is reached, the spacecraft leaves the Operational Phase and enters the Disposal Phase.

4.5.4 Disposal Phase

After reaching any of the above mentioned end-of-life criteria, Lazarus enters automatically the DP. This is its final mission phase, in which the last operational procedures are executed to

safely decommission the spacecraft. This phase ensures compliance with international space debris mitigation guidelines by permanently removing the satellite from the protected GEO region.

Once an end-of-life criterion is validated, Lazarus transitions from nominal operations to the orbit-raising sequence. The S/C will execute a series of targeted delta-v maneuvers to raise its perigee and apogee to approximately 300 kilometers above the standard GEO altitude. This designated graveyard orbit ensures the satellite remains completely clear of active operational slots and prevents long-term orbital perturbations from bringing the spacecraft back into the protected GEO belt, (it must have its perigee above the IADC minimum and an eccentricity below 0.003, both conditions are met, see in Section 7.8.1).

Upon confirmation from ground control that the graveyard orbit has been successfully achieved and stabilized, Lazarus will undergo complete system passivation. This protocol mitigates the risk of future accidental fragmentation or radio frequency interference. First, a complete depressurization and depletion of all remaining chemical propellants and pressurants is accomplished. Afterwards follows the system shutdown, including the final deactivation of all active transponders, telemetry streams, and reaction wheels. Last, the electrical deactivation of the satellite is completed, by permanently disconnecting the battery charge regulators from the solar arrays and fully discharging the stored power.

5 Launch Window Analysis

The Launch Window Analysis aims to determine the permissible launch periods throughout the year and the daily time intervals during which the candidate launch vehicles can lift off to inject the Lazarus spacecraft into a valid GTO, while satisfying all orbital, power, thermal, attitude, and ground segment constraints.
As stated in section 3.2.2, the mission considers the Ariane 6 as the primary launcher, being more suitable for the injection maneuvers, while keeping the Falcon 9 as a backup option to increase the mission flexibility. Lazarus is dedicated to a single launch in GTO, followed by low thrust electric propulsion to reach his operational orbit GEO-P; the process takes a transfer time that spans from 370 to 505.1 days, based on the inclination of the GTO orbit (determined by the launcher selected) as shown in table 7.3.

5.1 Launch Window Calculation

To guaranty operational flexibility and reduce program dependence on a single launch provider, two launch vehicle families and spaceports are retained for the Lazarus mission (as established in the launch vehicle trade-off of Section 3.2.2):

- **Nominal Baseline (Ariane 62 / Ariane 64):** Launched from the ELA-4 pad at the Guiana Space Centre (CSG) in Kourou, French Guiana ($\phi_{\text{CSG}} = 5°15'50''$ N $\approx 5.26°$ N, $\lambda_{\text{CSG}} = -52.79°$ E). Employing a direct-ascent mono-boost profile of the Vinci upper stage, the spacecraft reaches Geostationary Transfer Orbit (GTO) injection approximately 30–60 minutes after lift-off at a low inclination ($i_0 = 6°$–$7°$).
- **Backup Option (Falcon 9):** Launched from the Cape Canaveral Space Force Station (SLC-40), Florida, USA ($\phi_{\text{CC}} = 28.56°$ N, $\lambda_{\text{CC}} = -80.58°$ E) into an equivalent GTO with a higher initial inclination dictated by the launch site latitude ($i_0 = 27°$–$28.5°$).

The methodology for calculating the Lazarus Launch Window is governed by two coupled temporal scales; the day of the year and the daily lift-off time. The GEO-P arrival target year is set to 2034 to align the spacecraft's operational readiness with the servicing demand of client platforms.

For a fixed eastward launch azimuth compliant with range safety corridors over the Atlantic Ocean, the geometric shape (a, e), inclination (i_0), and argument of perigee of the injected GTO remain constant with respect to the Earth. As the Earth rotates at its sidereal rate $\omega_\oplus \approx 15.04107°/\text{h}$, the launch pad sweeps through 360° of inertial space every sidereal day (23h 56m 04s). Consequently, the selected Universal Time of lift-off (t_{UTC}) uniquely determines the initial inertial RAAN (Ω_0) in the Earth-Centered Inertial (ECI / EME2000) frame.

Referenced to the UTC time of the first (fictitious) perigee passage t_p on calendar day $D \in [1, 365]$ of year 2033, the inertial RAAN (Ω_0) of the injected GTO is computed as:

$$\Omega_0(D, t_{\text{ref}}) = [\theta_{\text{GMST}}(D, 0\text{h UTC}) + \omega_\oplus \cdot t_{\text{ref}} + \lambda_{\text{site}} + \Delta\lambda_{\Omega,\text{LV}}] \pmod{360°} \tag{14}$$

where:

- $\theta_{\mathrm{GMST}}(D, 0\mathrm{h\ UTC})$ is the Greenwich Mean Sidereal Time at 00:00 UTC on calendar day D of year 2033;
- t_{ref} is the reference UTC epoch (defined as the first fictitious perigee passage t_p in the Arianespace convention, or the second-stage orbital insertion epoch SECO-2 in the SpaceX convention);
- λ_{site} is the East longitude of the active launch pad ($\lambda_{\mathrm{CSG}} = -52.79°$ for Ariane 6 at Kourou; $\lambda_{\mathrm{SLC40}} = -80.58°$ for Falcon 9 at Cape Canaveral);
- $\Delta\lambda_{\Omega,\mathrm{LV}}$ is the vehicle-specific longitudinal ascent arc from the launch meridian to the ascending node ($\Delta\lambda_{\Omega,\mathrm{A6}} = 120.0°$ for the Ariane 6 reference GTO at $H_0 - 3$ s; $\Delta\lambda_{\Omega,\mathrm{F9}} \approx 95.5°$ for the Falcon 9 two-burn GTO parking-to-apogee transfer profile).

Is possible to determine a target optimal RAAN interval $[\Omega_{\min}, \Omega_{\max}]$ based on particular conditions wanted to be achieved on the GTO (and GEO-P considering the transfer maneuver) like the solar-eclipse and orbital plane constraints on a certain day D, Equation (14) is inverted to solve directly for the permissible UTC injection time t_p:

$$t_p(D, \Omega_0) = \frac{[\Omega_0 - \theta_{\mathrm{GMST}}(D, 0\mathrm{h\ UTC}) - \lambda_{\mathrm{site}} - \Delta\lambda_{\Omega}] \pmod{360°}}{\omega_{\oplus}} \tag{15}$$

Because the Sun's right ascension advances by $\sim 0.9856°$/day (~ 3.94 min/day) while $\theta_{\mathrm{GMST}}(D, 0\mathrm{h\ UTC})$ advances by the same rate, maintaining a constant relative angle between the GTO line of nodes (Ω_0) and the Sun vector ($\alpha_{\odot}$) results in a nearly constant UTC launch hour throughout the year, modulated only by the Equation of Time and the seasonal solar declination $\delta_{\odot}(D)$.

To determine an overview of the valid Launch Window for Lazarus, a numerical algorithm is implemented covering all calendar days of the year (D) and daily injection times (t_p). For each grid point (D, t_p), the algorithm computes the corresponding $\Omega(D, t_p)$ and inertial Sun position vector to verify that the spacecraft is injected in sunlight at separation. The valid (D, t_p) intervals that satisfy all constraints define the daily Satellite Injection Window. Subtracting the launch vehicle's powered flight duration yields the final Daily Launch Window, which must guarantee a continuous duration of at least 45 minutes, in the case of Ariane 6, per day as required by the launch service provider.

The results of the algorithm are shown in the following Figure; It's possible to obeserve that around the summer and winter solstices (June/July and November/January), the solar declination shifts the Earth's shadow cone away from the low-inclination GTO plane, creating wide eclipse-free regions (dark blue domains with 0 min eclipse) during late-evening UTC hours.The green horizontal band represents the nominal daily Launch Window that differs from Ariane 6 and Falcon 9.

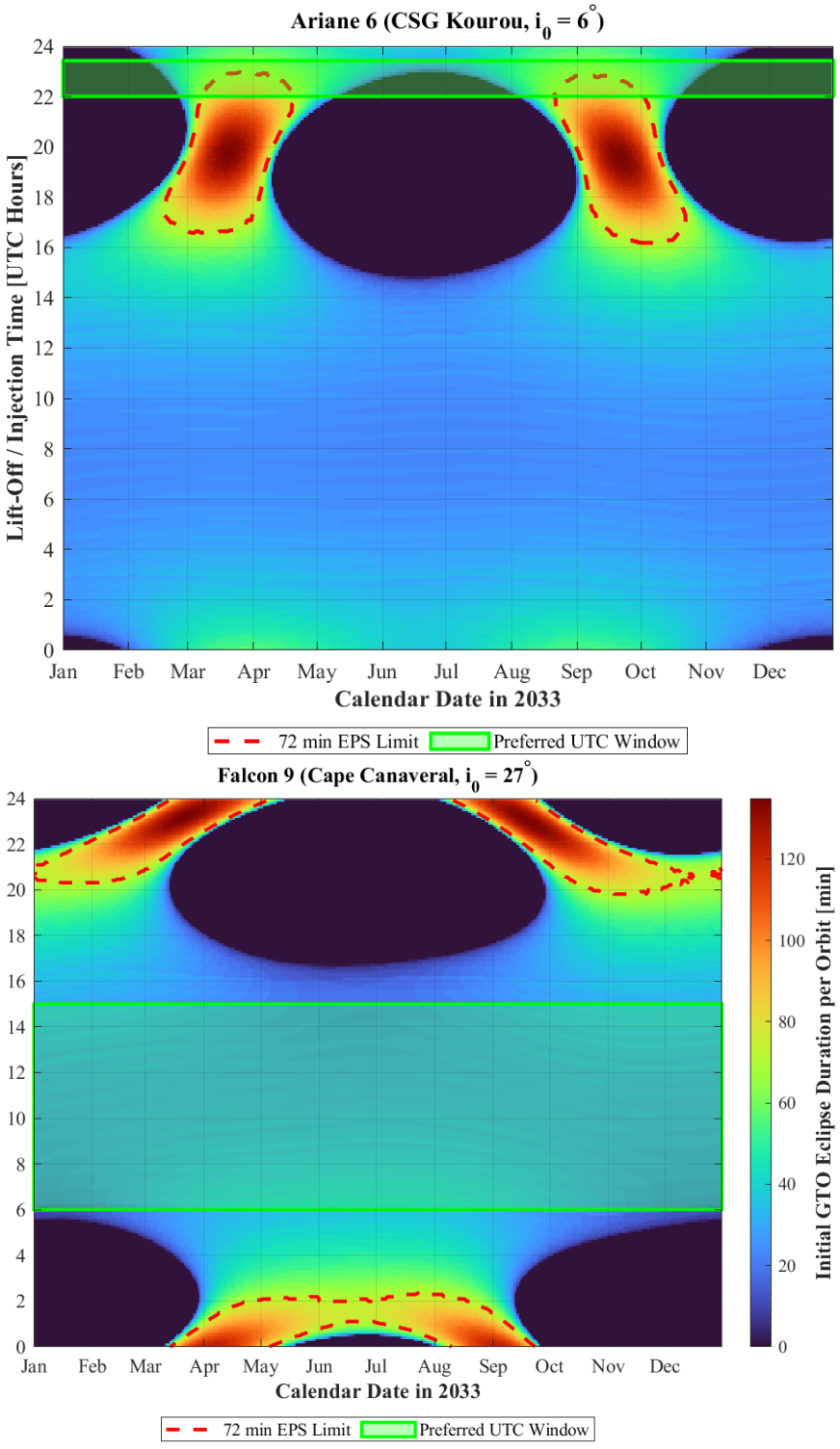


Figure 5.1: Annual initial GTO eclipse duration contour maps (in minutes per revolution) as a function of calendar date in 2033 and lift-off time (UTC) for Ariane 6 from Kourou ($i_0 = 6°$, left) and Falcon 9 from Cape Canaveral ($i_0 = 27°$, right). The dashed red contours denote the 72.0 min EPS maximum eclipse limit, the shaded green bands represent the preferred daily UTC launch windows

5.2 Definition of the Initial State

The Launch Window Analysis and the subsequent low-thrust trajectory propagation require a definition of the spacecraft's orbital, kinematic, mechanical, and physical state at the exact instant of separation from the launch vehicle upper stage ; this is determined by the initial boundary condition handed over from the launch service provider Arianespace or SpaceX. Following the mission architecture baseline (Section 3.2.13), Lazarus is injected into an elliptical Geostationary Transfer Orbit (GTO) that accommodates the specific ascent profiles and inclination capabilities of the two candidate launchers:

- **Ariane 62 / Ariane 64 (Preferred Injection):** In accordance with sections 2.4.1.1 and 2.7.2 the Ariane 6 User's Manual and the mission eclipse baseline (Section 7.3), the reference GTO features a perigee altitude of $h_p = 200$–250 km (250 km standard osculating perigee), an apogee altitude of $h_a = 35,836$–$36,000$ km ($35,943$ km standard Arianespace Launch Window reference apogee), an argument of perigee $\omega_p = 178.0°$, and a low inclination $i_0 = 6.0°$–$7.0°$
- **Falcon 9 (Backup Injection):** In accordance with Section 3.1 and Appendix H of the *Falcon User's Guide*, standard GTO missions from the Eastern Range feature a baseline perigee altitude of $h_p = 185$ km (100 nmi, extendable to 250 km upon customer request with a minor mass-to-orbit penalty), an apogee altitude matched to $h_a = 35,836$–$36,000$ km, an argument of perigee $\omega_p \approx 178.0°$, and an initial inclination of $i_0 = 27.0°$–$28.5°$.

In accordance with the launch vehicle performance baselines, Lazarus is injected into a standard Geostationary Transfer Orbit (GTO). To establish the initial boundary conditions for the Launch Window calculation, the nominal launcher capabilities are considered. Table 5.1 summarizes the pure nominal osculating Keplerian elements at injection for both candidate launch vehicles, exactly as specified in the Arianespace and SpaceX User's Manuals, alongside the 1σ injection accuracy dispersions.

Table 5.1: Nominal osculating Keplerian parameters at GTO injection for Ariane 6 and Falcon 9 standard missions.

Orbital Parameter	**Symbol**	**Ariane 6 (Nominal GTO)**	**Falcon 9 (Nominal GTO)**	**Dispersion**
Launch Site	—	CSG Kourou (ELA-4)	CCSFS / KSC	—
Ascent Flight Profile	—	Direct ascent (1 burn)	Two-burn	—
Injection Time	t_{inj}	$\sim$ 30.0–60.0 min	33.3 min (SECO-2 at 28.3 m)	—
Perigee Altitude	h_p	250 km	250 km	$\pm$1.3 km
Apogee Altitude	h_a	35,943 km	36000 km	$\pm$80.0 km
Perigee Radius	r_p	6,628.1 km	6,628.1 km	—
Apogee Radius	r_a	42,321.1 km	42,3211 km	—
Semi-Major Axis	a_0	24,474.6 km	24,363.6 km	$\pm$40.0 km
Eccentricity	e_0	0.7292	0.7307	$\pm 4.5 \times 10^{-4}$
Inclination	i_0	6.0°	27°	$\pm$0.02°
Argument of Perigee	ω_p	178.0°	178.0°	$\pm$0.2°
Initial RAAN	Ω_0	$f(D, t_{\text{UTC}}) \in [0°, 360°)$	$f(D, t_{\text{UTC}}) \in [0°, 360°)$	$\pm$0.2°

The σ injection accuracy dispersions ($\sigma_a = \pm 40.0$ km, $\sigma_e = \pm 4.5 \times 10^{-4}$, $\sigma_i = \pm 0.02°$, $\sigma_{\omega_p} = \pm 0.2°$, and $\sigma_\Omega = \pm 0.2°$, corresponding to $\sigma_{h_a} = \pm 80.0$ km and $\sigma_{h_p} = \pm 1.3$ km) are directly extracted from the official Arianespace specification for standard GTO missions.

5.3 Constraints

The determination of the permissible Launch Window for LAZARUS requires filtering the candidate calendar dates ($D \in [1, 365]$ in 2033) and daily lift-off times ($t_{\text{UTC}} \in [00{:}00, 24{:}00]$) against a set of coupled astrodynamic, subsystem, ground segment, and launch vehicle constraints. To maintain structural clarity across the Mission Analysis Report, a strict distinction is established between the **injection constraints** that directly bound the lift-off window (evaluated in this chapter) and the **long-term orbital and operational constraints** that govern the subsequent transfer and GEO-P servicing phases wich are analyzed with further details on the upcoming chapters.

The daily and seasonal launch opportunities are directly restricted by the following launch-phase constraints:

1. **GTO Apogee Shadow Avoidance:** As demonstrated in Section 5.1, the daily lift-off time t_{UTC} fixes the inertial Right Ascension of the Ascending Node (Ω_0) and consequently the angle between the GTO line of apsides and the Earth–Sun vector. Because orbital velocity at the GTO apogee is minimal ($v_a \approx 1.6$ km/s), any launch hour that aligns the apogee with the anti-Sun direction traps the spacecraft inside the Earth's shadow cylinder for extended periods. As quantified in the section 5.1 graphic the worst condition leaves the apogee in shadow with a eclipse duration that can surpass the 120 minutes. Such durations severely exceed the 72.0 min maximum design eclipse limit of the Electrical Power Subsystem and are critical for the active thermal conditioning required to maintain the 500 kg supercritical

Xenon payload above its storage conditions. Therefore, $\Omega_0(D, t_{\text{UTC}})$ is constrained to late-evening UTC hours (around 22:00 UTC) that guarantee zero apogee passes in shadow and restrict GTO eclipses to brief perigee passages (≤ 39.1 min).

2. **Seasonal Launch Period** On the seasonal timescale, the launch date D is constrained by the coupling between the transfer orbit reduced-power electric orbit-raising duration and the target arrival conditions in GEO-P. To ensure that Lazarus reaches GEO-P at the ending of the geostationary eclipse season—enabling complete in-orbit commissioning of the thermal, power, and relative navigation subsystems under eclipse conditions and aligning with client servicing schedules—the launch slot is constrained to a Solstice period. Departing at the onset of the solstice simultaneously maximizes the initial solar declination ($\delta_\odot \approx +23.44°$), keeping the Earth's shadow cone tilted away from the low-inclination GTO plane during the initial spiral revolutions.

3. **Launch Service Provider Operational, COLA, and Weather Constraints:** Finally, the daily Launch Window duration must satisfy the operational requirements of the launch authorities and spaceport ranges:

   - *Minimum Daily Window Duration:* For a single GTO launch on Ariane 6 from CSG, Arianespace mandates a daily Launch Window of at least 45 minutes (to permit a minimum of two launch attempts per day) and recommends a customer-specified window of up to 3 hours. Conversely, SpaceX requests customers to maintain a daily window of up to 4 hours where possible to maximize launch availability, with an absolute minimum window of 15 seconds to accommodate launch Collision Avoidance conjunction screening by the Combined Space Operations Center.
   - *Climatological and Weather Violation Risk:* While Kourou enjoys a stable climate year-round, launches on the Falcon 9 backup from Cape Canaveral during the June–July solstice period face an elevated statistical Probability of Violation (POV) for weather constraints during late afternoon/evening UTC hours (20.2%–29.9% POV between 18:00 UTC and 23:00 UTC due to Florida summer convective thunderstorms, compared to 5.4%–8.0% between 06:00 UTC and 15:00 UTC).

Several mission constraints are analyzed in full quantitative detail in the subsequent sections of this report:

- **Detailed transfer orbit Earth and Moon Eclipse Profiles (Section 7.3):** The conical shadow model (umbra and penumbra) across both the GTO transfer and the 8-year GEO-P operational lifetime—including the 72.0 min equinoctial worst-case Earth eclipse in GEO-P (Table 7.7) and the partial lunar eclipses (Table 7.8)—is thoroughly evaluated in Sections 7.3.1 and 7.3.2.

- **Sun–Spacecraft–Earth (SES) Angle Thresholds (Section 7.4):** The geometric evolution of the SES angle in GEO-P and the exact angular thresholds for penumbra, cylindrical shadow, and umbra entry are established in Section 7.4..

- **Global Ground Station Handovers and Communication Constraints (Section 7.5 and Chapter 8):** Longitudinal coverage ranges and signal handovers across the S-band network (Kourou, Malindi, Dongara, South Point) and the K-band deep-space network (Cebreros, Malargüe, New Norcia, Goldstone) are mapped in Sections 7.5 and 8.1, while operational link constraints are detailed in Section 8.2

5.4 Overall Launch Window

Building upon the analytical launch window formulation established in Section 5.1, the initial injection state defined in Section 5.2, and the multi-disciplinary constraints detailed in Section 5.3, it is possible to synthesize the optimal temporal interval—in terms of target year, calendar season, and daily Universal Time (UTC)—to execute the Lazarus launch. As demonstrated in the launch vehicle trade-off (Section 3.2.2) and transfer performance analysis (Table 3.2), Ariane 6 (Ariane 62/64) remains the preferred baseline launch vehicle over the Falcon 9 backup option, owing to the superior compatibility of its near-equatorial GTO injection ($i_0 = 6.0°$–$7.0°$ from Kourou versus $i_0 = 27.0°$–$28.5°$ from Cape Canaveral) with the final equatorial GEO-P operational parking orbit, resulting in a significantly shorter electric transfer time and lower Xenon propellant consumption.

5.4.1 Target Launch Year and Seasonal Launch Period Selection

First, on a programmatic level, the target launch year is constrained by system and commercial market readiness: the 2033 timeframe directly precedes the period in which the first target cooperative GEO client platforms will require in-orbit life-extension refueling services (beginning in early 2035).

Second, regarding the calendar launch date within the year, the spacecraft is governed by two primary astrodynamic and subsystem constraints linked to the initial GTO apogee geometry and the arrival conditions at GEO-P:

1. **GTO Apogee Illumination Condition:** As evidenced by the annual GTO eclipse contour map derived in Section 5.1, the most favorable departure periods for initial orbital insertion coincide with the onset of the astronomical solstices (represented by the dark-blue regions of the contour map), where the high solar declination tilts the Earth's shadow cone away from the orbital plane, yielding **zero minutes of apogee eclipse** and limiting shadow transits to brief perigee passages.

2. **GEO-P Arrival and Power Subsystem Sizing Condition:** The launch date must be backward-propagated across the low-thrust electric transfer duration so as (i) to arrive at GEO-P outside the geostationary eclipse seasons (which span 26 February to 12 April and 30 August to 15 October, Section 7.3.1), thereby guaranteeing at least the first week of operational orbit—and ideally the entire Commissioning Phase—at full solar power without eclipse interruptions, and (ii) to keep the longest eclipse of the whole mission at the value imposed by GEO-P (72 min), so that the transfer does not drive the sizing of the power subsystem.

5.4.2 Daily Lift-Off Time (UTC) Selection

With respect to the daily lift-off hour (t_{UTC}), the dominant constraints are dictated by the launch vehicle standard flight profiles and range availability:

- **Ariane 6 (CSG Kourou):** In full agreement with the annual eclipse contour map and the *Ariane 6 User's Manual*, lift-off is scheduled within the standard late-evening window around 23:00 **UTC** (opening daily between 22:00 UTC and 22:40 UTC and closing between 22:45 UTC and 23:25 UTC, ensuring a minimum continuous duration of 45 minutes). Launching at 22:00 UTC guarantees sunlit separation and projects the GTO apogee into the sunlit hemisphere with zero apogee shadow passes.
- **Falcon 9 (CCSFS / KSC):** Following the operational and climatological guidelines of the *Falcon User's Guide* the daily launch window is constrained within the 06:00 **UTC to** 15:00 **UTC** interval. This broad daytime/morning window satisfies SpaceX's requirement for a $\geq$ 4-hour daily window while avoiding the severe late-afternoon convective thunderstorm risk at the Eastern Range (reducing the statistical Weather Probability of Violation from $\sim$ 20.2–29.9% at 18:00–23:00 UTC down to 5.4–8.0% between 06:00 UTC and 15:00 UTC).

5.4.3 Transfer Eclipse Trade-Off and Final Launch Window Baseline

Based on these considerations a sweep of launch dates and hours has been carried out over the whole of 2033, with one launch date every 4 days. Foreach launch, the eclipses of the electric transfer and of the 8 years of the mission are computed.

Table 5.2 shows five representative options for each launcher, with the selected one in bold.

Launch (UTC)	Longest transfer eclipse [min]	Eclipse-free days after arrival	Total eclipses	Total time in shadow [h]
29 May 2033, 23:00	**67.5**	**88**	**973**	**822.0**
30 Jun 2033, 23:00	71.1	56	930	784.4
2 Sep 2033, 23:00	83.3	0	857	734.1
1 Nov 2033, 22:00	81.4	112	874	760.8
31 Dec 2033, 23:00	70.3	52	933	785.6

(a) Ariane 6.

Launch (UTC)	Longest transfer eclipse [min]	Eclipse-free days after arrival	Total eclipses	Total time in shadow [h]
18 Mar 2033, 10:00	77.8	26	790	693.2
1 May 2033, 12:00	71.0	0	785	699.9
2 Jun 2033, 11:00	**71.0**	**129**	**796**	**703.3**
4 Jul 2033, 09:00	69.7	97	818	710.6
29 Nov 2033, 20:00	90.2	135	792	717.8

(b) Falcon 9.

Table 5.2: Launch options from the 2033 launch date sweep (selected option in bold).

The options with the least time in shadow (2 September for Ariane 6 and 18 March for Falcon 9) do not meet the second criterion, since their transfer eclipses reach 83.3 and 77.8 min, and some of the options in Table 5.2 also arrive at GEO-P during an eclipse season. A slightly larger total time in shadow has therefore been accepted in exchange for a shorter maximum eclipse, since the battery capacity and the cold thermal case are driven by the longest eclipse and not by the accumulated shadow time. Among the options whose transfer eclipses stay below 72 min, the selected dates give the longest eclipse-free period after arrival.

Specifically, the consolidated nominal launch window baseline for the Lazarus mission is established as follows:

	Ariane 6 (Kourou)	Falcon 9 (Cape Canaveral)
GTO inclination	6°	27°
Launch (UTC)	29 May 2033, 23:00	2 Jun 2033, 11:00
Electric transfer duration	370 days	505 days
Arrival at GEO-P	3 Jun 2034	20 Oct 2034
Eclipse-free days after arrival	88	129
Eclipses during the transfer	328	196
Longest transfer eclipse	67.5 min	71.0 min
Time in shadow during the transfer	218.3 h	143.4 h
Eclipses in GEO-P	645	600
Total eclipses / time in shadow	973 / 822.0 h	796 / 703.3 h
Longest eclipse (sizing case)	**72.0 min (GEO-P)**	**72.0 min (GEO-P)**

Table 5.3: Selected launch options (8-year mission).

With both options the sizing eclipse is the 72.0 min equinox eclipse in GEO-P, and the spacecraft reaches GEO-P with 88 (Ariane 6) and 129 (Falcon 9) eclipse-free days before the next eclipse season. Specifically, the two consolidated reference launch baselines are justified as follows (A detailed analysis of the eclipses for both cases is given in Section 7.3.):

- **Nominal Baseline – Ariane 62/64 ($i_0 = 6°$, CSG Kourou):** Lift-off is scheduled on **29 May 2033 at** 23:00 **UTC**, at the onset of the northern summer solstice window and within the standard late-evening Arianespace GTO injection slot. Following a 370-day electric orbit-raising transfer (328 shadow transits totaling 218.3 h), the maximum transfer eclipse is capped at 67.5 **min**, providing a 4.5 min safety margin below the 72.0 min EPS sizing threshold. LAZARUS arrives at GEO-P on **3 June 2034**, securing 88 **consecutive eclipse-free days** throughout the northern summer solstice period (until the onset of the autumn eclipse season on 30 August) to complete in-orbit commissioning at full solar power.
- **Option – Falcon 9 ($i_0 = 27°$, CCSFS Cape Canaveral):** Lift-off is scheduled on **2 June 2033 at** 11:00 **UTC**, well inside the 06:00–15:00 UTC low-weather-risk window recommended for the Eastern Range. Due to the higher initial inclination (27°), the out-of-plane electric transfer requires 505 days, but benefits from fewer transfer eclipses (196 events, 143.4 h total) and maintains a maximum transfer eclipse of 71.1 **min** (< 72.0 min). Arrival at GEO-P occurs on **20 October 2034**—immediately after the conclusion of the autumn geostationary eclipse season (15 October)—providing 129 **uninterrupted eclipse-free days** prior to the spring eclipse season in late February 2035.

It is important to emphasize that the launch epochs and UTC hours reported in Table 5.3 represent the **ideal reference planning baseline** established by preliminary mission analysis; in actual flight implementation, the definitive Launch Period, calendar Launch Day (D_0), and exact lift-off instant (H_0) will be finalized jointly with the launch service authority accounting for manifest slot allocation, range tracking availability, local meteorological conditions on launch day, and final Collision Avoidance (COLA) conjunction screening.

6 Launch and Early Orbit Phase

6.1 Launch

As stated in previous chapters, two different launch vehicles are being considered for launching Lazarus into space: Ariane 6 and Falcon 9. Both operators state that it would take just 25min to 30min to inject Lazarus in its special GTO after take-off.

It must be taken into consideration that Ariane 6 starts from Kourou, while Falcon 9 mostly start from Kennedy Space Center or Cape Cañaveral. Therefore, depending on the selected launch vehicle, the inclination of the GTO will be different. For Ariane 6 launches an inclination of 6° is estimated for the GTO, while for Falcon 9 launches an inclination of 27° is estimated for the GTO. This considerable inclination difference must be taken into consideration for the transfer orbit, since the Delta-V budget will be very different.

Because of the dimensions and the mass of Lazarus, at first, it will not be assumed that the launch can be shared with other satellites.

6.2 Transfer Orbit

The launcher injects Lazarus directly into the GTO, at its perigee, about 30 min after lift-off. This orbit is the starting point of the low-thrust transfer to GEO-P.

6.2.1 Transfer Orbit Parameters

The GTO has its perigee at 250 km and its apogee at the altitude of the client orbit, 36 000 km. Its shape is the same for both launchers; only the inclination changes with the launch site. The right ascension of the ascending node depends on the launch date and time, so the values in Table 6.1 correspond to the two analysed launch dates.

Table 6.1: Parameters of the GTO for each launcher.

Ariane 6 (Kourou)

Parameter	Value
h_p / h_a [km]	250 / 36000
a [km]	24503.1
e [–]	0.7295
Period [h]	10.60
v_p / v_a [km s^{-1}]	10.198 / 1.595
ω [°]	178
i [°]	6

Falcon 9 (Florida)

Parameter	Value
h_p / h_a [km]	250 / 36000
a [km]	24503.1
e [–]	0.7295
Period [h]	10.60
v_p / v_a [km s^{-1}]	10.198 / 1.595
ω [°]	178
i [°]	27

The 27° orbit needs a much larger plane change during the transfer to GEO-P, which increases its Δv, propellant and duration (Section 7).

6.2.2 Coast in Transfer Orbit

After separation, Lazarus coasts in the GTO while the LEOP activities are carried out: acquisition of signal, deployment of the solar arrays, attitude acquisition, platform check-out and initial orbit determination. The Electric Propulsion (EP) system is then checked and the low-thrust transfer starts, which ends the LEOP.

The duration of this coast is To Be Determined (TBD). In the mission simulation the thrust starts at the injection, so any coasting time in GTO delays the arrival in GEO-P by the same amount. The 220 d or 81 d between the arrival and the first refuelling window easily absorb a coast of a few days.

7 Operational Orbit

7.1 Initial Parameters

The operations start in the GTO (Table 6.1) with the low-thrust transfer to the operational orbit, GEO-P, where Lazarus waits between services. After the last service it moves to the graveyard orbit. The three orbits are given in Table 7.1.

Table 7.1: Orbits of the operational phase (all circular and equatorial).

Parameter	**Client orbit**	**GEO-P**	**Graveyard**
Altitude [km]	36000	36050	36300
Radius [km]	42378.1	42428.1	42678.1
Period [h]	24.117	24.160	–
Drift of the clients relative to Lazarus [$^\circ\,\mathrm{d}^{-1}$]		0.6331	
Synodic period [d]		568.6	

GEO-P is slightly higher, and therefore slightly slower, than the client orbit. The clients overtake Lazarus slowly, and the same relative geometry repeats every synodic period. Lazarus uses this natural drift to reach each client without manoeuvring.

Table 7.2: State of Lazarus at the start of the operations and at the arrival in GEO-P.

Parameter	$i_0 = 6^\circ$	$i_0 = 27^\circ$
Mass at the start of the transfer [kg]	3000	
Initial inclination [°]	6	27
Arrival in GEO-P	3 Jun 2034	20 Oct 2034
Mass at arrival [kg]	2747.6	2695.6
Insertion angle θ_{ins} (EMC) [°]	113.3	243.1
Time in GEO-P until the disposal [d]	1957.1	1818.4

7.2 History of Orbital Elements

7.2.1 Transfer from GTO to GEO-P

The transfer raises the orbit from the GTO to GEO-P and removes the inclination. It is propagated including Solar Radiation Pressure (SRP), the Sun, the Moon and the Earth's shadow, and it is steered with a Q-law.

The thruster is not operated continuously. At every control step the Q-law gives the best achievable rate of decrease of its objective, $|\dot{Q}|_{\min}$, and the same quantity is evaluated along the whole osculating orbit. The relative effectivity

$$\eta_{\text{rel}} = \frac{|\dot{Q}|_{\min} - \min_\nu |\dot{Q}|_{\min}}{\max_\nu |\dot{Q}|_{\min} - \min_\nu |\dot{Q}|_{\min}} \in [0, 1] \tag{16}$$

compares the current point with the best and worst points of the orbit. The thruster is on only when $\eta_{\text{rel}} \geq 0.6$ and never in eclipse: this concentrates the thrust on the most efficient arcs and saves propellant at the cost of a longer transfer.

For Ariane 6 the threshold is $\eta_{\text{cut}} = 0.6$. For Falcon 9 the transfer has deliberately not been made as efficient as possible. With the shortest transfer, the refuelling of client E could coincide with a lunar eclipse of about 237 min predicted for 2041, which would require a much larger power system. The arrival in GEO-P has therefore been planned to avoid that eclipse, which gives a transfer of 505 d ($\eta_{\text{cut}} = 0.645$).

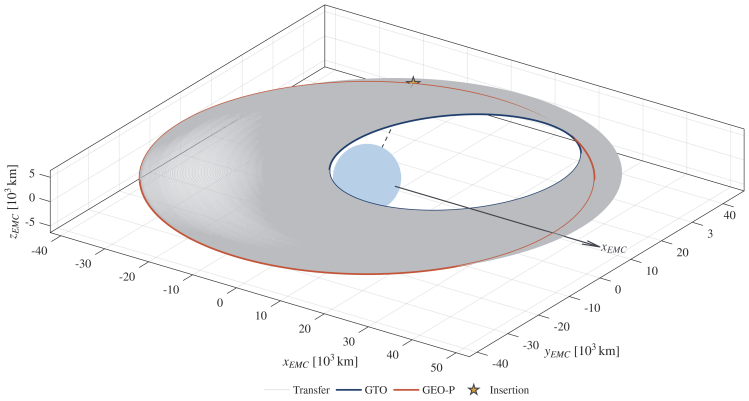
Table 7.3: GTO-to-GEO-P low-thrust transfer and impulsive reference.

Parameter	$i_0 = 6°$	$i_0 = 27°$
Coasting threshold η_{cut} [–]	0.600	0.645
Flight time [d]	370.0	505.1
Engine on [% TOF]	51.6	45.6
Revolutions [–]	568	821
Δv in plane [km s^{-1}]	1.690	1.679
Δv out of plane [km s^{-1}]	0.276	0.899
Δv total* [km s^{-1}]	1.724	2.099
Propellant [kg]	252.4	304.4
Chemical Δv (impulsive) [km s^{-1}]	1.491	1.798
Chemical propellant (I_{sp} 320 s) [kg]	1134.1	1308.2

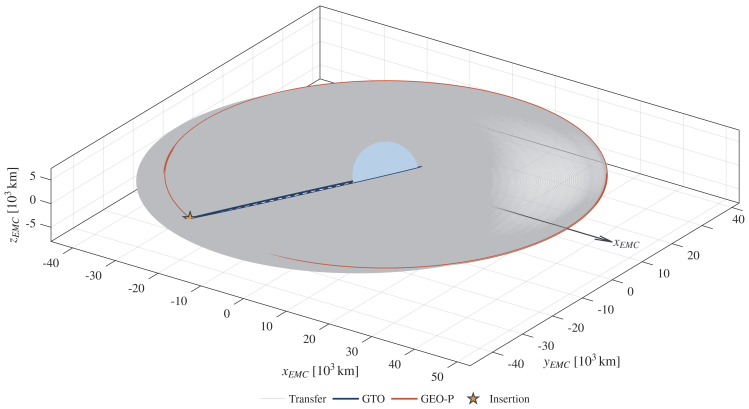
*The thrust has in-plane and out-of-plane parts at the same time, so the total is less than the sum of both components.

Raising the orbit costs practically the same with both launchers; the difference is the plane change, about three times larger from 27°. This is why Ariane 6 needs 52 kg less xenon. Compared with a chemical transfer, the electric transfer needs a little more Δv but about four times less propellant, which is the reason for using electric propulsion.

Figure 7.1 shows the spiral from the GTO to GEO-P. The Falcon 9 spiral is denser because it needs many more revolutions, and both transfers end at different points of the orbit, which sets the initial geometry with respect to the clients. The projections of Figure 7.2 show the plane change: in the xz plane the 27° transfer starts much more tilted and is flattened progressively onto the equator.

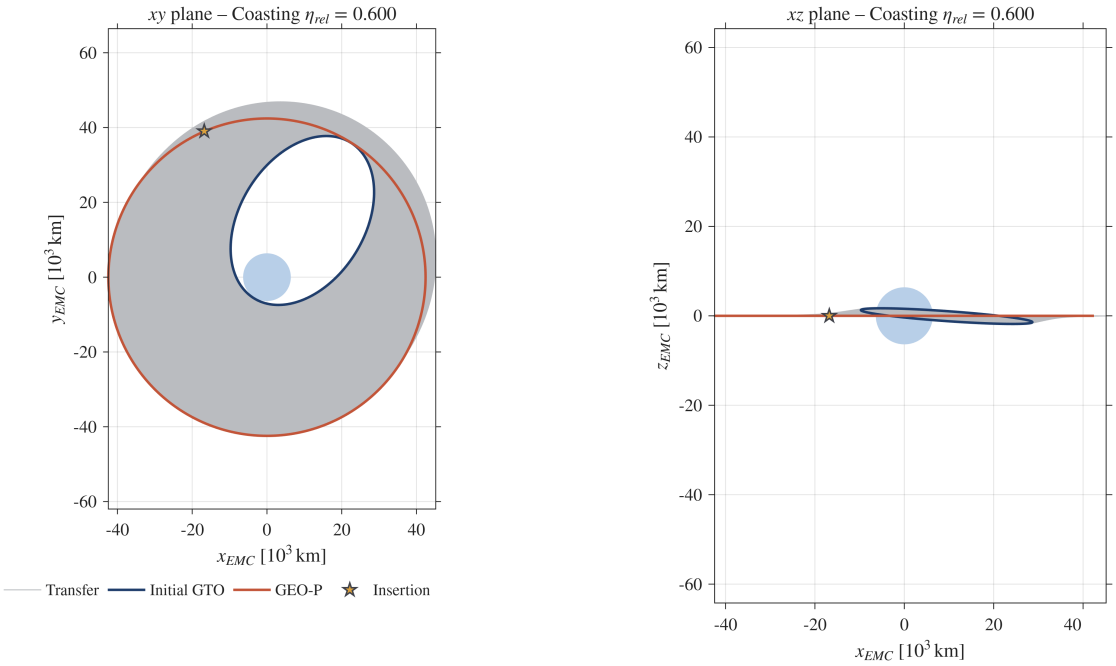


(a) $i_0 = 6°$

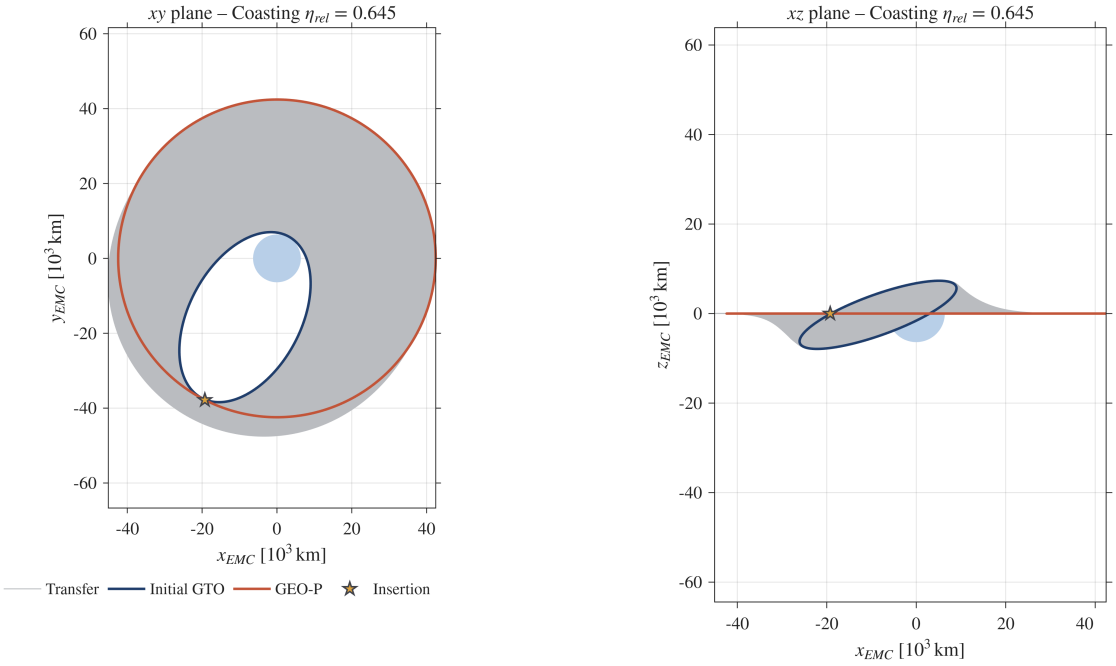


(b) $i_0 = 27°$

Figure 7.1: Low-thrust transfer in EMC axes and insertion point into GEO-P.

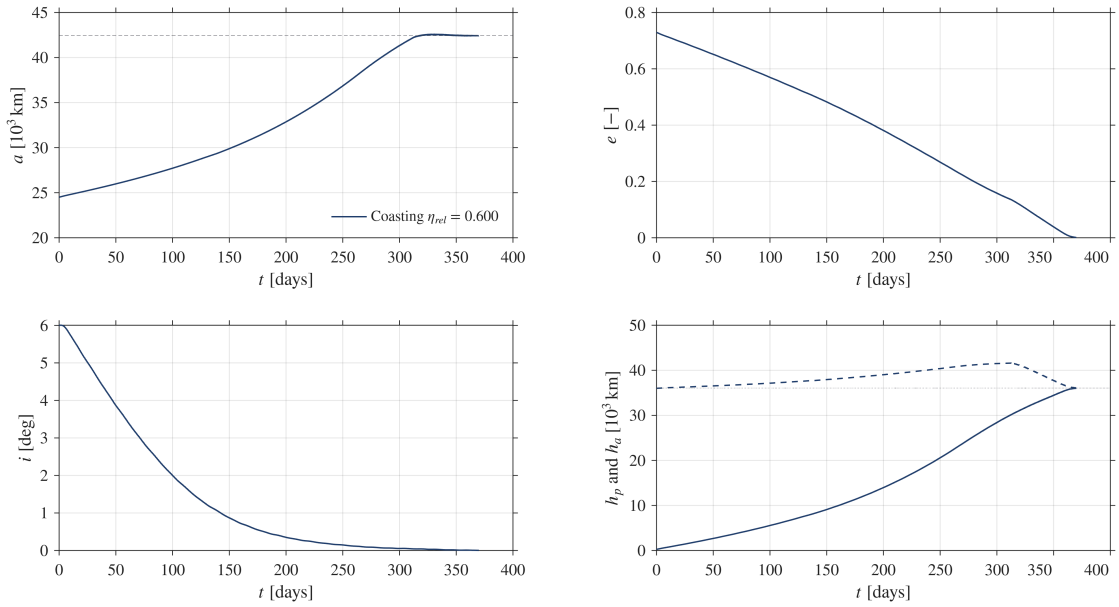


(a) $i_0 = 6°$

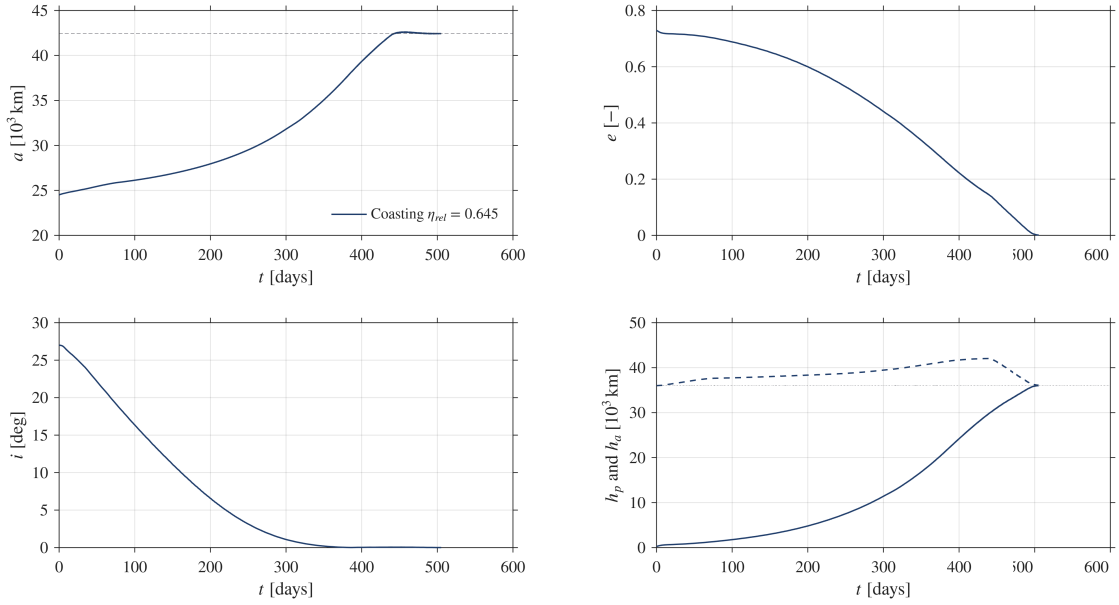


(b) $i_0 = 27°$

Figure 7.2: Projections of the low-thrust transfer on the EMC coordinate planes.



(a) $i_0 = 6°$



(b) $i_0 = 27°$

Figure 7.3: Orbital elements during the transfer.

Figure 7.3 shows the evolution of the elements: the semimajor axis grows up to GEO-P, while the eccentricity and the inclination are reduced to practically zero. The small residual eccentricity is removed by the first station-keeping manoeuvre (See Figure 7.4).

7.2.2 Arrival in GEO-P and Eclipse Seasons

In GEO-P the Earth's shadow is only reached around the equinoxes, in two seasons per year, with one eclipse per day of at most about 70 min. The first service, to client A, is the in-orbit demonstration, so it must take place outside a season (Table 7.4).

Table 7.4: Arrival in GEO-P and demonstration with client A with respect to the eclipse seasons.

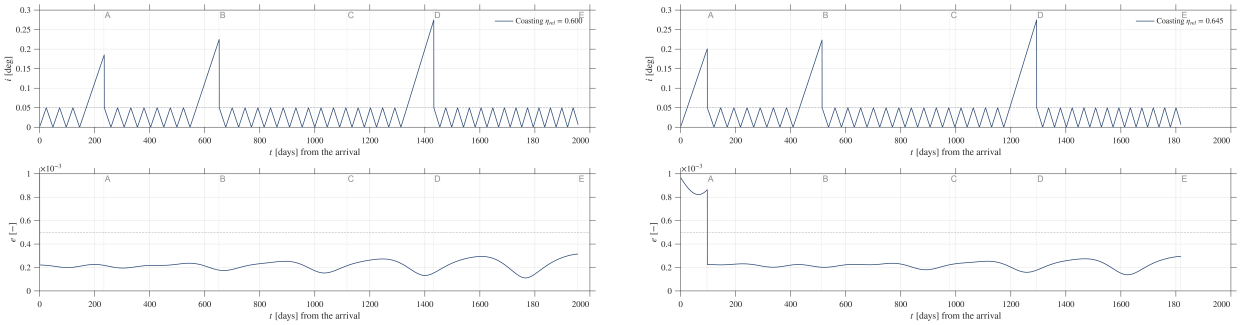
Parameter	$i_0 = 6°$	$i_0 = 27°$
Arrival in GEO-P	3 Jun 2034	20 Oct 2034
Arrival inside an eclipse season	No	No
Eclipse-free days after the arrival [d]	88	129
Time until the window of client A [d]	220	81
Eclipse-free period of the demonstration	15 Oct 2034 – 26 Feb 2035	
Docking with client A	24 Jan 2035	
Margin to the next season [d]	33.6	32.8

The Falcon 9 transfer ends just after the autumn season of 2034, so Lazarus arrives directly in the eclipse-free period of the demonstration. With Ariane 6 it arrives earlier and crosses that season while waiting in GEO-P, before any proximity operation. In both cases the demonstration takes place more than a month before the next season. The price of the longer Falcon 9 transfer is a shorter commissioning before the first window.

7.2.3 Evolution in GEO-P

In GEO-P the Sun and the Moon make the inclination drift, and the SRP makes the eccentricity oscillate. The station keeping keeps both within their limits (Section 7.6.1).

Figure 7.4 shows both elements from the arrival to the disposal. The inclination follows a saw-tooth between the limits of its band, one N–S correction every few weeks. The tall peaks appear after each 100 d phasing (clients A, B and D), during which no correction is made. The eccentricity starts at the residual value of the transfer, is corrected once and then stays below its limit for the rest of the mission.



(a) $i_0 = 6°$ (b) $i_0 = 27°$

Figure 7.4: Maintenance manoeuvres in the GEO-P.

7.3 Eclipses

This section presents a detailed analysis of the eclipses that the spacecraft will encounter throughout the mission, from injection into GTO until the end of life. The two launch window options

presented in Chapter 5 are analysed in detail.

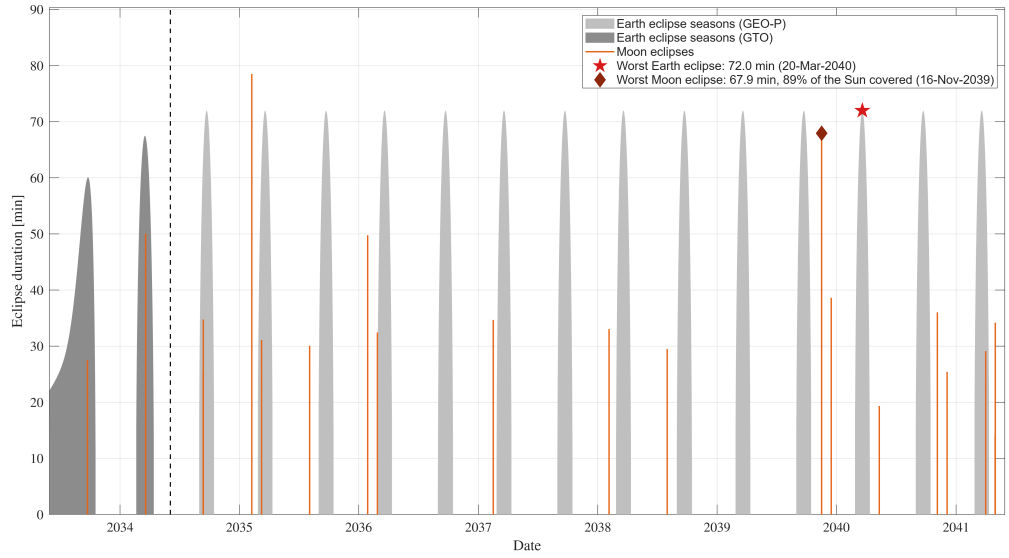
	Ariane 6	Falcon 9
Launch (UTC)	29 May 2033, 23:00	2 Jun 2033, 11:00
Transfer duration	370 days (568 orbits)	505 days (821 orbits)
Estimated arrival at GEO-P	3 Jun 2034	20 Oct 2034
First eclipse in GEO-P	31 Aug 2034	26 Feb 2035
Eclipse-free days after arrival	88	129
End of life	29 May 2041	2 Jun 2041

Table 7.5: Analysed launch options.

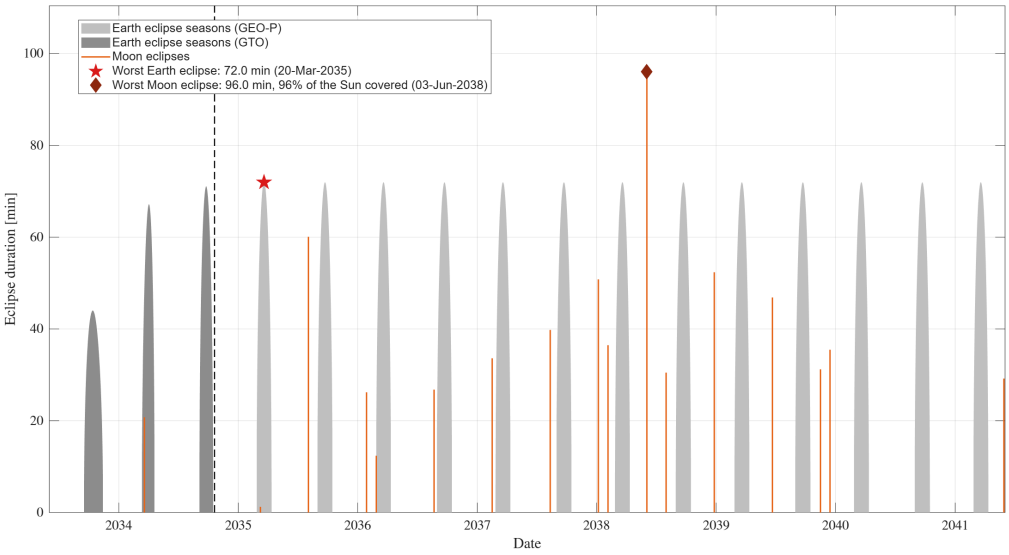
	Ariane 6		Falcon 9	
Year	Eclipses	Shadow [h]	Eclipses	Shadow [h]
1	328	218.3	152	103.0
2	91	86.3	90	83.3
3	92	86.4	91	86.3
4	93	86.5	92	86.4
5	93	86.5	92	86.4
6	93	86.5	93	86.5
7	92	86.4	93	86.5
8	91	86.3	93	86.4
Total	973	823.2	796	704.9

Table 7.6: Earth eclipses per mission year. Year 1 (and year 2 for Falcon 9) include the transfer.

Eclipses are computed with a conical shadow model (umbra/penumbra) and analytical Sun and Moon ephemerides, with time steps of 30 s during the transfer and 60 s in GEO-P. The expected eclipses are shown in Figure 7.5. Each eclipse is plotted at its date with its duration: the light grey areas are the eclipse seasons in GEO-P, the dark grey area the eclipses of the transfer, and the orange lines the eclipses by the Moon. The worst Earth and Moon eclipses are marked.



(a) Ariane 6

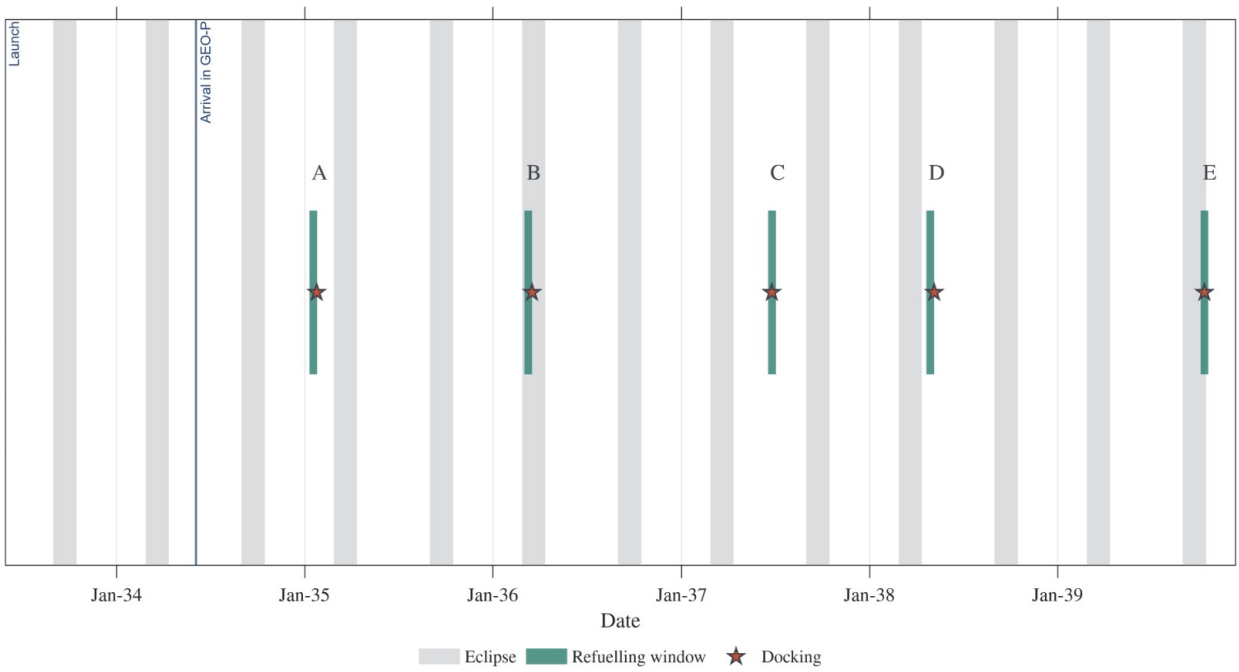


(b) Falcon 9

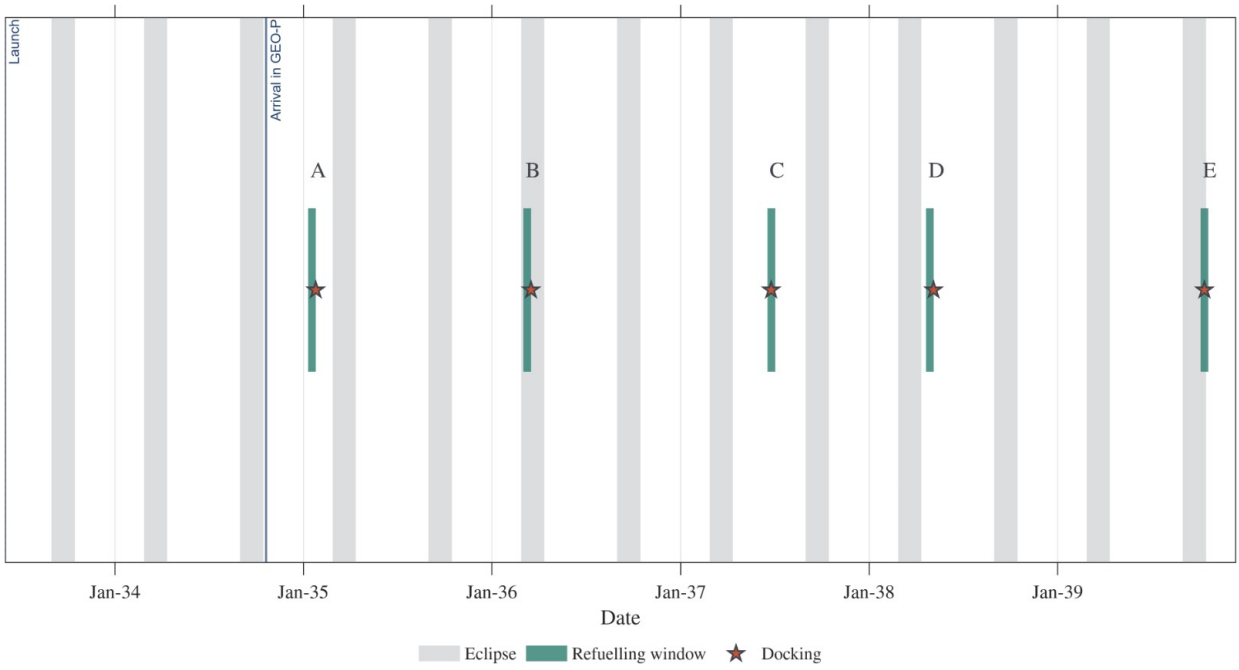
Figure 7.5: Eclipses during the whole mission

This analysis covers the transfer and the GEO-P orbit, without including the refuelling operations with the clients. This simplification has little effect on the Earth eclipses, since in GEO-P they depend only on the season and are almost the same at any longitude. In Moon eclipses, however, the result depends on the exact position of the spacecraft along the orbit, so they are analysed only to estimate how often they occur and to define a worst case (see Section 7.3.2).

Nevertheless, although the transfers to and from the client satellites have not been included in the eclipse computation, Figure 7.6 shows the refuelling windows of the five client operations (A to E) in order to estimate which manoeuvres could take place in eclipse.



(a) Ariane 6



(b) Falcon 9

Figure 7.6: GEO-P eclipse seasons, refuelling windows and docking dates.

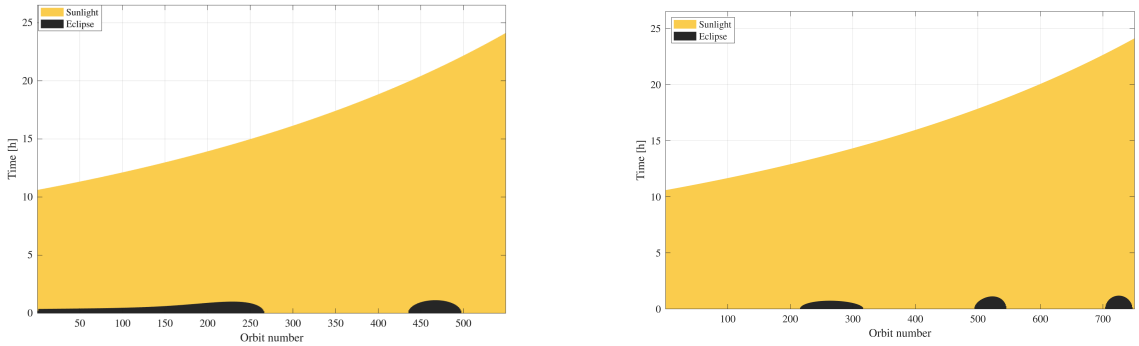
7.3.1 Eclipses by the Earth

Table 7.7 summarises the Earth eclipses of each phase. In both cases the longest eclipse of the mission (worst case) is the 72.0 min equinox eclipse in GEO-P (Figure 7.5), which will be the sizing case for the power subsystem.

Phase		Ariane 6	Falcon 9
Transfer	Orbital period	634–1451 min	633–1451 min
	Number of eclipses	328	196
	Longest eclipse (umbra)	67.5 (63.9) min	71.1 (66.9) min
	Date of the longest eclipse	18 Mar 2034	24 Sep 2034
	Max. fraction of the orbit in shadow	6.9 %	6.1 %
	Time in shadow	218.3 h	143.4 h
GEO-P	Orbital period	1449.6 min	1449.6 min
	Eclipse seasons	14	13
	Number of eclipses	645	600
	Longest eclipse (umbra)	72.0 (67.7) min	72.0 (67.7) min
	Max. fraction of the orbit in shadow	5.0 %	5.0 %
	Time in shadow	604.9 h	561.4 h
Total mission	**Number of eclipses**	**973**	**796**
	Time in shadow	**823.2 h**	**704.9 h**
	Worst case	**72.0 min (GEO-P)**	**72.0 min (GEO-P)**

Table 7.7: Earth eclipses per phase.

Transfer. At the beginning of the transfer the perigee is very low (185–250 km). Seen from such a low altitude the Earth covers a large part of the sky, so its shadow is crossed on almost every orbit, regardless of the season. These eclipses are short (about 10–30 min), because they occur near the perigee, where the spacecraft moves fastest. As the S/C raises the perigee, the orbit moves away from the Earth and its shadow becomes a narrow cylinder that the orbit only crosses when the Sun is close to the orbital plane, i.e. around the equinoxes. The periods with eclipses therefore become progressively shorter and turn into seasons, while each individual eclipse becomes longer, since the spacecraft moves more slowly at higher altitude and spends more time in the shadow. With the larger inclination of Falcon 9 (27°) the Sun is out of the orbital plane for longer, which explains why only 196 of its 750 orbits have an eclipse, compared with 328 of 549 for Ariane 6.e.



(a) Ariane 6

(b) Falcon 9

Figure 7.7: Orbital period and eclipse time per orbit during the transfer.

GEO-P. Once in GEO-P, the Earth's shadow only reaches the orbit when the solar declination is smaller than about 9°. This results in two eclipse seasons per year, centred on the equinoxes: from about 26 February to 12 April and from about 30 August to 15 October, each lasting about 45 days. During a season there is one eclipse per orbit, i.e. one per day, close to local midnight; its duration grows from a few minutes at the edges of the season to a maximum of 72.0 min at the equinox and decreases again. Each eclipse consists of a central umbra phase, in which the Earth completely hides the Sun and no solar power is available, and two short penumbra phases at entry and exit, in which the Sun is only partially covered. In GEO-P the penumbra lasts only about 2 min on each side, so 67.7 of the 72.0 min of the longest eclipse are spent in umbra (Figure 7.8). Outside the seasons there are no Earth eclipses at all. This pattern repeats every year of the mission and is common to both launch windows.

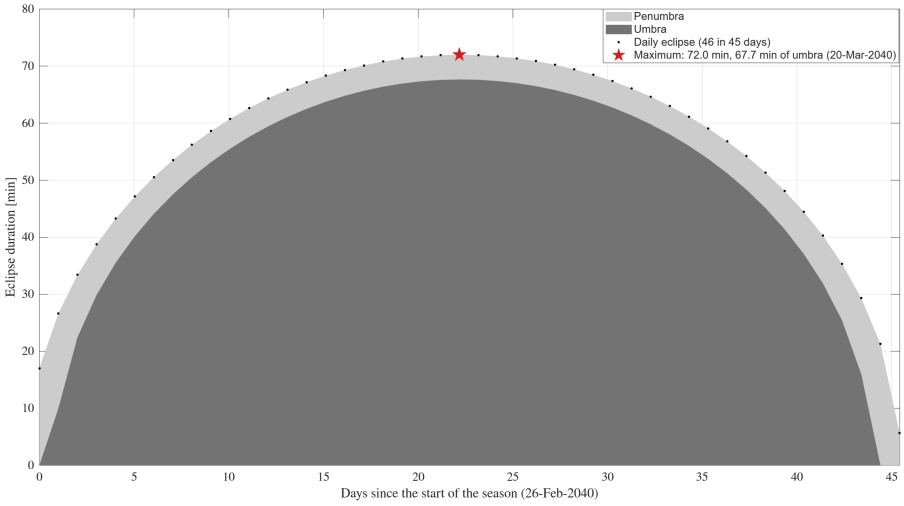


Figure 7.8: Detail of the GEO-P eclipse season containing the worst eclipse.

7.3.2 Eclipses by the Moon

As indicated at the beginning of Section 7.3, the refuelling operations are not considered here, and unlike Earth eclipses, Moon eclipses depend on the exact position of the spacecraft along the orbit, so these manoeuvres would change the dates and durations obtained here. For this reason, the results in Table 7.8 should be taken as an estimate of how often Moon eclipses occur and how long they last, not as a prediction of specific events. The worst cases below are used only to define a conservative design case for the power subsystem.

The eclipses by the Moon are shown as orange lines in Figure 7.5. They are few, irregular and do not follow any seasonal pattern, because they depend on the relative position of the Moon and on the longitude of the spacecraft at each instant. All of them are partial (penumbral): the spacecraft never enters the lunar umbra, so part of the Sun always remains visible. Their main figures are listed in Table 7.8.

	Ariane 6	Falcon 9
Number of events (transfer / GEO-P)	20 (2 / 18)	17 (1 / 16)
Time in partial shadow	12.0 h	10.5 h
Equivalent total-eclipse time	2.9 h	2.6 h
Worst case	16 Nov 2039, 16:14	3 Jun 2038, 09:13
Duration	67.9 min	96.0 min
Maximum fraction of the Sun covered	89 %	96 %
Equivalent total eclipse	27.6 min	40.3 min
Double eclipses (Earth + Moon)	2	2
Worst: time in shadow	63.1 min (9 Mar 2035)	64.0 min (9 Mar 2035)

Table 7.8: Eclipses by the Moon.

Since lunar eclipses are partial, their duration alone does not measure their impact: an event that covers a small part of the Sun for a long time removes less energy than a shorter but deeper one. The worst case is therefore defined as the length of a total eclipse that would cause the same loss of solar energy.

- **Ariane 6:** the worst case is the eclipse of 16 November 2039. It is not the longest event (that of 8 February 2035 lasts 78.5 min but covers at most 68 % of the Sun, equivalent to 26.7 min), but it combines a long duration (67.9 min) with the deepest coverage of the mission (89 %), which gives the largest energy loss, equivalent to 27.6 min of total eclipse.
- **Falcon 9:** the worst case is the eclipse of 3 June 2038, which is at the same time the longest (96.0 min) and the deepest (96 % of the Sun covered) event of the mission, equivalent to 40.3 min of total eclipse.

In both cases the worst lunar eclipse, and also the worst double Earth + Moon eclipse, remain below the 72.0 min Earth eclipse in energy, so the Earth eclipse is still the sizing case.

7.4 Sun-Spacecraft-Earth Angle

The Sun-Spacecraft-Earth (SSE) angle is the angle at the spacecraft between the Sun and the Earth directions, i.e. the angular separation between the centres of both bodies as seen from the spacecraft ($\approx 180°$: local noon, $\approx 0°$: local midnight). In GEO-P it varies each day between about $|\delta_\odot|$ and $180° - |\delta_\odot|$, so the daily minimum ranges from 23.4° (solstices) to almost 0° (equinoxes). An Earth eclipse occurs when the discs of the Earth and the Sun overlap, that is, when the daily minimum falls below the thresholds of Table 7.9. As shown in Figure 7.9, the eclipse seasons start exactly when the daily minimum crosses the penumbra threshold.

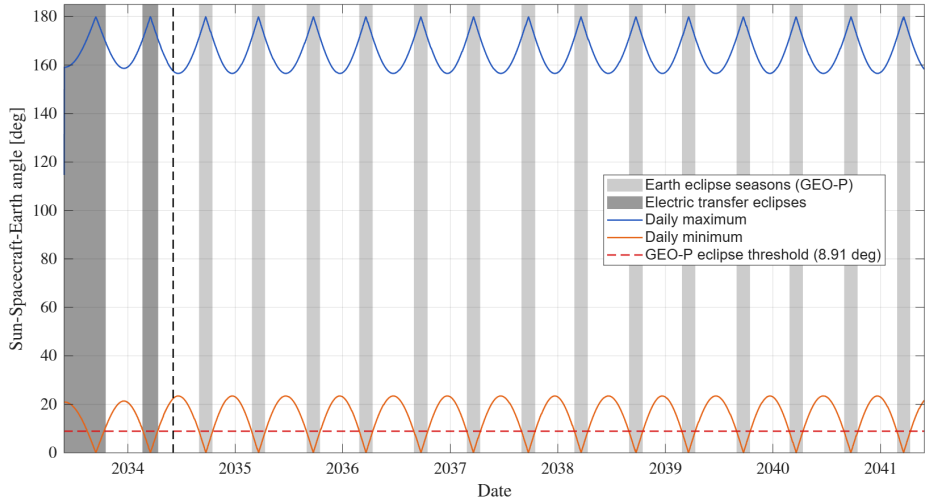
Condition	SSE angle
Entry into penumbra	$< 8.91°$
Sun centre at Earth limb	$< 8.65°$
Entry into umbra	$< 8.38°$

Table 7.9: SSE eclipse thresholds in GEO-P.

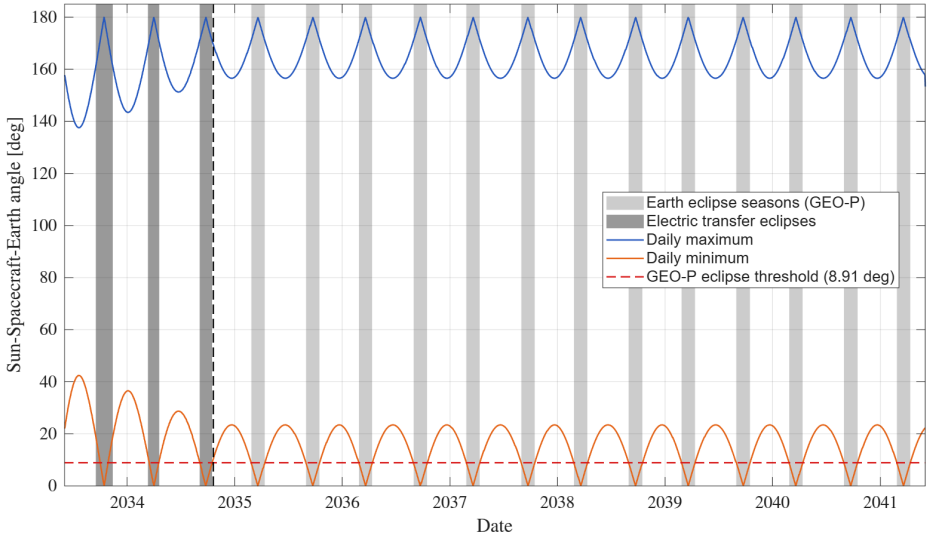
Table 7.10 compares the GEO-P Earth eclipses predicted with the SSE angle and the theoretical duration formula with those obtained with the conical shadow model. The number of eclipses agrees in both cases, except for one Ariane 6 and two Falcon 9 grazing eclipses at the edges of the seasons, which last less than the 1 min sampling step used for the SSE angle. The mean duration error is about 0.02 min, and the maximum error, below 1 min, also corresponds to short eclipses at the beginning or end of a season. The two methods therefore agree, which validates the eclipse durations that will be used for the sizing of the power subsystem.

	Ariane 6	Falcon 9
Eclipses: SSE angle / shadow model	644 / 645*	598 / 600*
Duration error, penumbra + umbra (mean / max)	0.02 / 0.87 min	0.02 / 0.60 min
Duration error, umbra (mean / max)	0.01 / 0.45 min	0.01 / 0.45 min

Table 7.10: Validation of the GEO-P Earth eclipses with the SSE angle.



(a) Ariane 6



(b) Falcon 9

Figure 7.9: Sun-Spacecraft-Earth angle: daily maximum and minimum.

7.5 Ground Station Coverage

7.5.1 Global Coverage

To provide coverage, various ground stations will be used. We will use ESA's Cebreros and Kourou stations as main stations for coverage during the standby period in the GEOP orbit.

In situations where we perform an orbital maneuver to position ourselves in a location outside the coverage of these stations, we will use other ESA stations and temporarily subcontracted NASA stations to carry out this type of operation outside the coverage range of our main stations. For this purpose, the following tables define the coverage ranges of the stations we will use in our mission.

Ground Station (K-band)	Coverage [-180º,180º]
Cebreros	-69.643, 60.908
Malargüe	-136.299, -2.496
New Norcia	47.998, -175.614
Goldstone (DSN)	176.168, -49.726

Table 7.11: Coverage range for K-band ground stations

Ground Station (S-band)	Coverage [-180º,180º]
Kourou	-129.0889, 23.480
Malindi	-36.134, 116.514
Dongara	40.598, -170.738
South Point	128.802, -80.121

Table 7.12: Coverage range for S-band ground stations

This allows for global coverage in both situations.

7.5.2 Detail Coverage

To obtain these values, the first thing we calculated is the maximum link distance of our stations, being approximately 40,636 km for high-gain antennas and approximately 41,177 km for emergency antennas. Therefore, we have verified that we can successfully establish communication links with our satellite.

Subsequently, we obtained the coverage in relation to the longitude of the ground stations to be used in our mission. To do this, we used the position where the satellite will be in its standby orbit until called for refueling; using this value, we calculated the angles formed by the station with respect to the satellite. In this way, we were able to obtain the different longitudinal coverages with respect to the station. This process has been carried out for all the ground stations we will use in the mission, observing that they cover the full 360°, enabling global coverage.

7.6 Orbit Manoeuvres and Corresponding Propellant Budget

The propellant of every manoeuvre is computed with the rocket equation

$$\Delta v = I_{sp}\, g_0 \ln\left(\frac{m_i}{m_f}\right), \qquad g_0 = 9.81\,\mathrm{m\,s^{-2}}, \tag{17}$$

applied manoeuvre by manoeuvre with the current mass of Lazarus. With 0.3 N every manoeuvre is a burn of several hours, centred on its optimal point and split into several passes when it is too long. Table 7.13 lists the manoeuvres of the operational phase.

Table 7.13: Manoeuvres of the operational phase.

Manoeuvre	**When**	Δv [$\mathrm{m\,s^{-1}}$]
Service	Each client	4.15
Phasing	Case 2 clients only	0.8 – 8.7
N–S (inclination)	About every 48 d while waiting	5.7
Eccentricity	Once	1.6 – 1.9
Disposal	End of the mission	9.0

Service manoeuvres. Every service is a transfer of 11 h to 0.5 km behind the client, the approach and docking, the retreat to 0.5 km and a symmetric return to GEO-P. It costs 4.15 $\mathrm{m\,s^{-1}}$: four impulses of 0.94 $\mathrm{m\,s^{-1}}$ for the outbound and return transfers and four small impulses of 0.10 $\mathrm{m\,s^{-1}}$ for the approach and the retreat (Figure 7.11).

The 11 h transfer time comes from the porkchop of Figure 7.10. For every transfer time there is an optimal relative angle at departure. The minimum Δv that does not cross the client orbit before the arrival is reached at 11 h, slightly shorter than the Hohmann transfer. The corresponding departure angle, with the client just behind Lazarus, is the one used for every service.

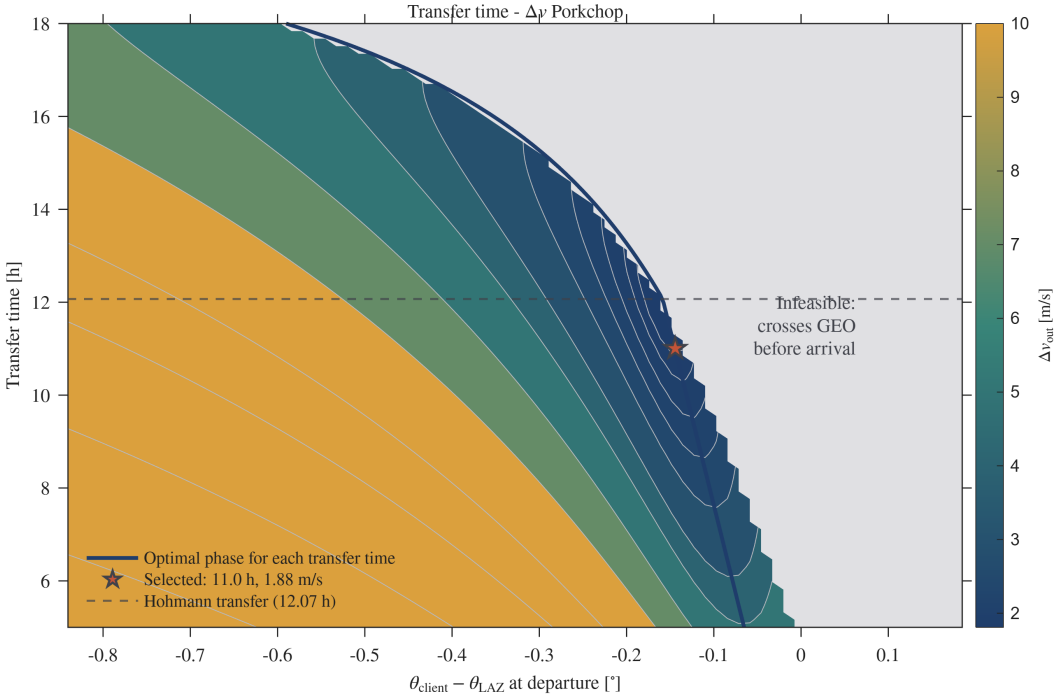


Figure 7.10: Outbound transfer GEO-P→client orbit: Δv versus relative angle at departure and transfer time. Solid line: optimal departure angle; star: selected 11 h solution; dashed: Hohmann transfer time; grey: infeasible.

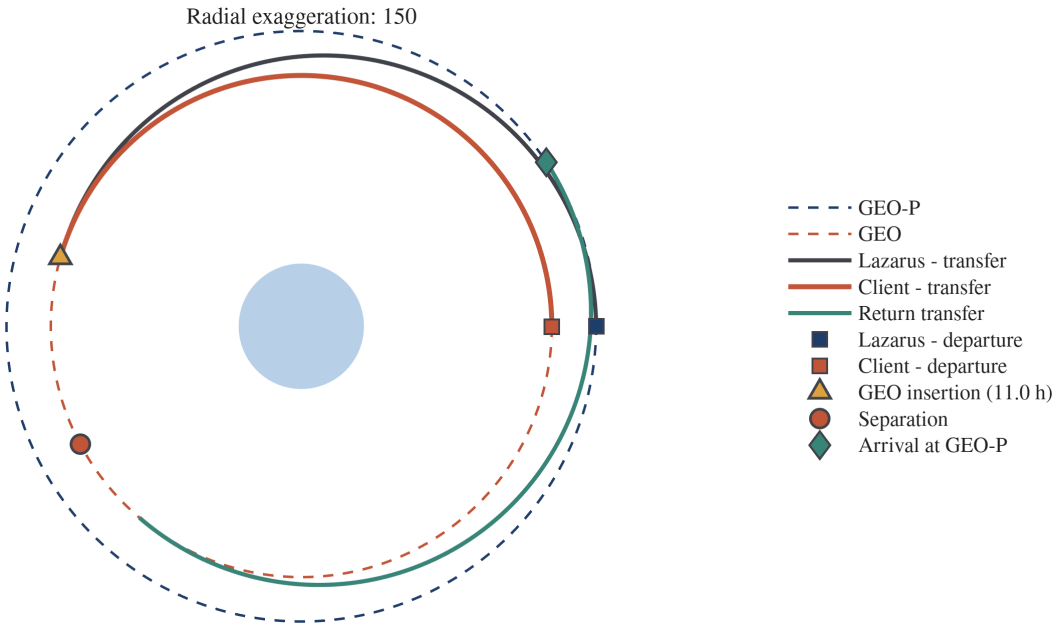


Figure 7.11: Complete service in the orbital plane (radial distance exaggerated 150 times): outbound transfer, insertion behind the client after 11 h, separation and return transfer to GEO-P.

Case 1 and Case 2. The operational case decides whether a service needs anything more than the service manoeuvres (Table 7.14).

- **Case 1** (Clients C and E). The natural drift brings the client to the optimal angle inside the window. Lazarus only waits in GEO-P doing station keeping, and the service costs just its $4.15\,\mathrm{m\,s^{-1}}$.
- **Case 2** (Clients A, B and D). Lazarus flies a phasing orbit, slightly higher than GEO-P, during the last 100 d before the departure. This limit keeps the service as close as possible to the end of the client's propellant: refuelling much earlier would give the client the same propellant but fewer extra years of life, which no client would accept. This has three consequences:
  - An extra Δv, proportional to the angle that the natural drift cannot cover in those 100 d.
  - No station keeping is done during the phasing, so the accumulated inclination is removed afterwards with a single larger N–S burn. The total cost does not change.
  - The docking takes place on the last day of the window, with no time margin. If margin is wanted, the phasing can be planned to end a few days earlier.

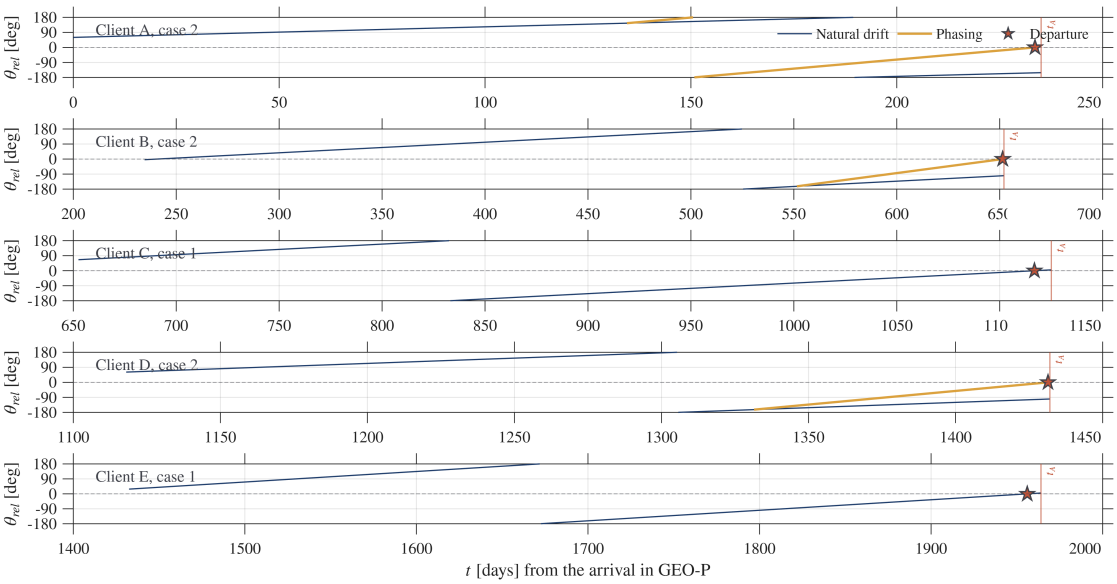
Table 7.14: Results of the five services. $\Delta\theta_0$: relative angle when Lazarus becomes available; natural wait: time the drift alone would need to reach the optimal angle.

Client	$\Delta\theta_0$ [°]	**Natural wait** [d]	**Case**	**Wait in GEO-P** [d]	**Phasing** [d]	Δv **phasing** [$\mathrm{m\,s^{-1}}$]	Δv **total** [$\mathrm{m\,s^{-1}}$]	**Docking**
A ($i_0 = 6°$)	60.0	473.7	2	134.5	99.1	8.74	12.88	24 Jan 2035
A ($i_0 = 27°$)	-74.2	117.0	2	0.0	95.7	0.80	4.94	24 Jan 2035
B	-3.9	5.9	2	316.0	99.9	5.70	9.85	16 Mar 2036
C	65.9	464.4	1	464.4	–	0	4.15	25 Jun 2037
D	61.4	471.4	2	213.5	99.9	5.69	9.84	5 May 2038
E	28.5	523.5	1	523.5	–	0	4.15	12 Oct 2039

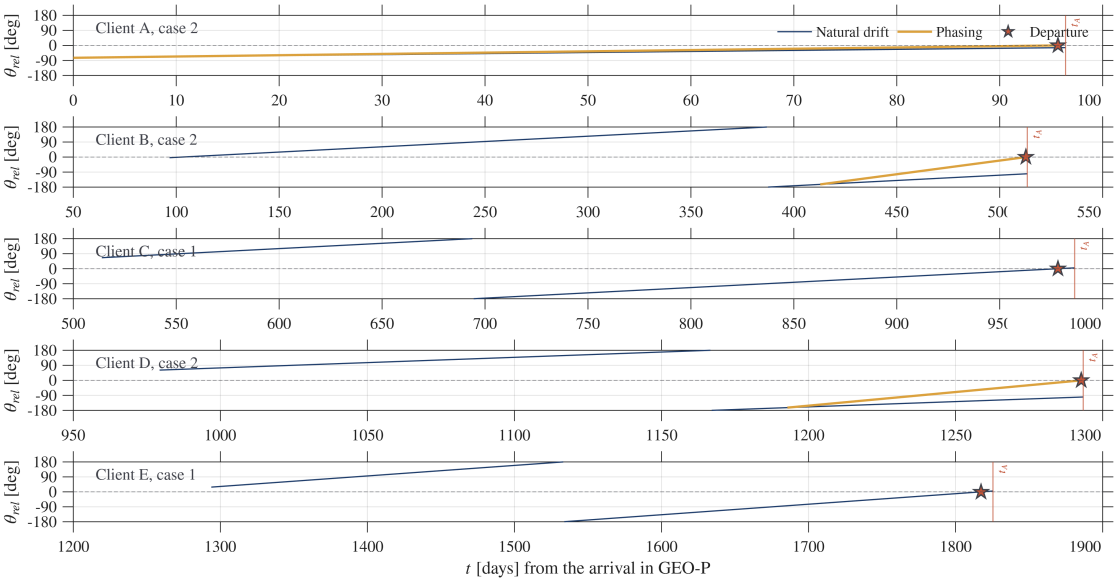
The only service that changes with the launcher is the first one. With Falcon 9 the natural drift almost covers the angle to client A by itself, so its phasing is ten times cheaper than with Ariane 6. This saving is small compared with the extra xenon of the Falcon 9 transfer.

Figure 7.12 shows the relative angle with each client from the moment Lazarus is available until the end of its window. The blue line is the natural drift. In **Case 1** (clients C and E) it crosses the optimal departure angle inside the window, and the departure (star) is placed on that crossing. In **Case 2** (clients A, B and D) the crossing falls outside the window, and the phasing (yellow) bends the line so that it reaches the optimal angle at the end of the window. Client B shows that the case is decided by the window and not by the geometry: its optimal angle arrives only a few days after Lazarus is free, but more than a year before its window opens.

The only difference between both launches is client A, because each transfer ends at a different point of the orbit and at a different date. From client B on, both campaigns are the same.



(a) $i_0 = 6°$



(b) $i_0 = 27°$

Figure 7.12: Relative angle with every client until the end of its window: natural drift (blue), phasing (yellow) and departure (star).

7.6.1 Natural Orbit Perturbations

Three perturbations act on GEO-P:

- **Inclination** (Sun and Moon). It drifts by about 0.8° per year and is kept within ±0.05°: when a limit is reached, an N–S manoeuvre takes it to the opposite limit. This is the dominant cost of the operational phase.
- **Eccentricity** (SRP). It oscillates yearly below the admissible 5×10^{-4}, so it only needs one correction to remove the residual eccentricity of the transfer.
- **Semimajor axis** (Earth's ellipticity, J_{22}). Its oscillation stays inside the ±5 km tolerance,

so it needs no correction.

Corrections are only made while Lazarus waits in GEO-P. The results interval by interval are given in Table 7.15.

Table 7.15: Station keeping in GEO-P, interval by interval.

	Duration [d]		**N–S manoeuvres**		Δv **[m s^{-1}]**	
Interval	6°	27°	6°	27°	6°	27°
Arrival → A	134.5	0.0	3	0	19.1*	0.0
A → B	316.8	316.1	7	7	48.4	50.5*
B → C	464.4	464.4	11	11	73.6	73.3
C → D	213.5	213.5	4	4	22.9	22.9
D → E	523.5	523.5	13	13	87.6	87.8
Total	1652.6	1517.4	38	35	251.6	234.5
Propellant [kg]					34.96	31.99
Δv per year [m s^{-1}]					55.6	56.4

*Includes the eccentricity correction.

The cost per year is the same in both cases, because it only depends on the luni-solar drift. The total depends on how long Lazarus waits in GEO-P: with Ariane 6 it arrives earlier, waits longer and spends a little more on station keeping. With Falcon 9 the phasing towards A starts on arrival, so the first correction is made after that service.

7.6.2 Perturbations due to Attitude Manoeuvres

The attitude of Lazarus changes the area exposed to the Sun, and therefore the SRP. Its main effect is on the eccentricity, whose yearly oscillation is proportional to the area-to-mass ratio:

$$e_n \propto C_R \frac{A}{m}. \tag{18}$$

With the nominal area the oscillation is less than half of the admissible eccentricity. Periodic eccentricity control would only be needed if the mean exposed area were more than twice the nominal value. The attitude changes for thrusting, rendezvous and docking are short compared with the yearly SRP cycle, so they are not expected to change the station-keeping strategy. The area also enters the Inter-Agency Space Debris Coordination Committee (IADC) disposal rule, and the graveyard orbit keeps a margin of about 45 km over its minimum perigee.

The attitude manoeuvres themselves are performed by the Attitude and Orbit Control System (AOCS); their propellant and their small effect on the orbit are TBD.

7.7 Reference Orbit for Radiation Analysis

The radiation environment of the mission has been obtained with SPENVIS (ESA's Space Environment Information System) for a reference orbit that represents the whole mission, from the injection into GTO to the end of the 8-year lifetime. For the thermal control subsystem, the radiation environment is needed to estimate the degradation of the thermal coatings (increase of the solar absorptivity α), which defines the end-of-life hot case.

7.7.1 Definition of the reference orbit

The mission cannot be represented by GEO-P alone. During the electric transfer the perigee rises slowly from the GTO to GEO-P, so the spacecraft crosses the inner proton belt and the outer electron belt twice per orbit for several months. GEO-P, at the outer edge of the electron belt, has a lower flux but lasts about 7 years. The reference orbit has therefore been defined in SPENVIS as a multi-segment mission.

SPENVIS allows a maximum of five segments, so the transfer has been divided into four segments followed by GEO-P. The segments do not have the same duration: the first two are shorter (1/8 of the transfer each) because during them the perigee crosses the proton belt, where the flux changes quickly with altitude; the third one covers 1/4 of the transfer and the fourth one, already outside the belts, the remaining half. Each segment is represented by the mean perigee altitude, apogee altitude and inclination of the approximate transfer (Section 7.3), in which these parameters vary linearly from the GTO to GEO-P. Tables 7.16a and 7.16b give the segments for both launch options.

Segment	Start (UTC)	Duration	h_p [km]	h_a [km]	i [deg]
1	29 May 2033, 23:30	46.25 days	2 487	36 003	5.62
2	15 Jul 2033, 05:30	46.25 days	6 962	36 009	4.88
3	30 Aug 2033, 11:30	92.50 days	13 675	36 019	3.75
4	30 Nov 2033, 23:30	185.0 days	27 100	36 038	1.50
GEO-P	3 Jun 2034, 23:30	6.99 years	36 050	36 050	0

(a) Ariane 6.

Segment	Start (UTC)	Duration	h_p [km]	h_a [km]	i [deg]
1	2 Jun 2033, 11:26	63.12 days	2 427	36 003	25.31
2	4 Aug 2033, 14:26	63.12 days	6 910	36 009	21.94
3	6 Oct 2033, 17:26	126.2 days	13 634	36 019	16.88
4	9 Feb 2034, 23:26	252.5 days	27 084	36 038	6.75
GEO-P	20 Oct 2034, 11:26	6.62 years	36 050	36 050	0

(b) Falcon 9.

Table 7.16: Reference orbit segments for the radiation analysis.

For each segment SPENVIS propagates a representative trajectory of 30 days and scales the orbit-averaged fluxes to the duration of the segment. Atmospheric drag and solar radiation pressure are not included, since each segment already represents the mean orbit of that part of the transfer and the lowest mean perigee (about 2 400 km) is well above the atmosphere.

The following models have been used:

- Trapped protons: AP-8 MIN, the worst case of the AP-8 model.
- Trapped electrons: AE-8 MAX, the worst case of the AE-8 model.
- Solar protons: ESP-PSYCHIC, with a 95 % probability of the fluence not being exceeded, for the whole mission (2033–2041).

7.7.2 Results

Table 7.17 summarises the integral fluences of the whole mission for both launch options, together with the fraction received during the electric transfer, and Figure 7.13 compares the total fluence spectra.

Table 7.17: Total mission integral fluences and contribution of the electric transfer.

	Ariane 6		Falcon 9	
	Fluence [cm^{-2}]	Transfer	Fluence [cm^{-2}]	Transfer
Trapped protons > 1 MeV	5.8×10^{13}	99.9 %	4.6×10^{13}	99.8 %
Trapped protons > 10 MeV	1.1×10^{11}	100 %	7.6×10^{10}	100 %
Trapped protons > 100 MeV	2.3×10^{9}	100 %	2.2×10^{9}	100 %
Trapped electrons > 0.1 MeV	7.2×10^{15}	19 %	7.3×10^{15}	23 %
Trapped electrons > 1 MeV	1.1×10^{14}	34 %	1.2×10^{14}	39 %
Trapped electrons > 2 MeV	1.0×10^{13}	48 %	1.1×10^{13}	52 %
Solar protons > 1 MeV	9.7×10^{11}	–	9.6×10^{11}	–
Solar protons > 10 MeV	1.8×10^{11}	–	1.8×10^{11}	–
Solar protons > 100 MeV	5.3×10^{9}	–	5.3×10^{9}	–

The trapped protons are received almost entirely during the electric transfer. Above 10 MeV the whole fluence comes from the first two segments, when the perigee crosses the inner proton belt, while in GEO-P the trapped protons are negligible above 1 MeV. The Ariane 6 option receives about 45 % more protons above 10 MeV than the Falcon 9 one: with an inclination of 6° the perigee stays close to the magnetic equator, where the proton belt is most intense, whereas the 27° orbit of Falcon 9 partly avoids its core.

The trapped electrons, on the other hand, are dominated by GEO-P, which is located in the outer electron belt: it accounts for about 80 % of the fluence above 0.1 MeV and 60–65 % above 1 MeV. Both options give almost the same electron fluence, since the longer transfer of Falcon 9 is compensated by its shorter stay in GEO-P.

The solar proton fluence is the same for both options, since it depends only on the mission dates.

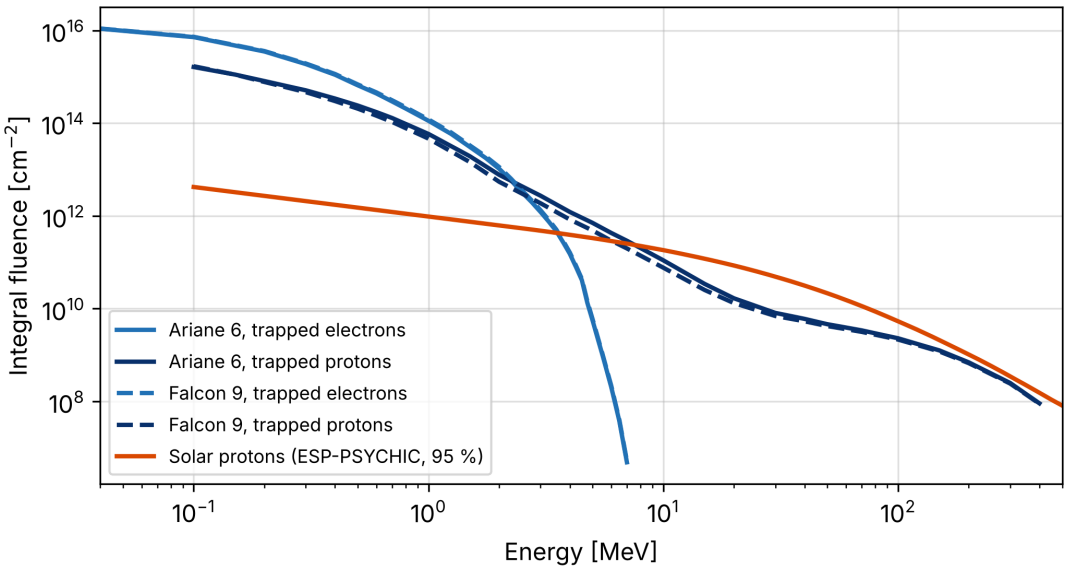


Figure 7.13: Total mission integral fluence of trapped electrons, protons and solar protons.

7.8 Orbital Lifetime

Unlike traditional geostationary assets whose service life is typically constrained by a fixed multi-year station-keeping schedule, the operational lifetime of the Lazarus technology demonstrator is primarily dictated by the depletion of its onboard Xenon propellant reserves rather than by structural or electronic degradation. Nevertheless, from a programmatic and systems engineering standpoint, the mission is sized for a maximum orbital lifetime of 8 years starting from launcher separation, considered as a worst-case-scenario useful for subsystems sizing. the mission itself will almost surely be completed in less than 8 years.

Following injection into a GTO and launcher separation, the spacecraft executes a low-thrust electric propulsion orbit-raising transfer lasting between 370 and 505 days (depending on the selected launch vehicle inclination and non-continuous thrust-coast strategy, as consolidated in Section 3.2.1). Upon completion of this transfer, Lazarus is injected into its nominal drift parking orbit, the GEO-P, located 50 km above the geostationary ring.

Once the in-orbit verification is completed Lazarus transitions into its routine utilization phase (Phase E). Spanning the remainder of the 8-year maximum operational envelope, Phase E sees the spacecraft loitering in GEO-P and executing phased orbital transfers down to the nominal GEO belt to deliver its 500 kg supercritical Xenon payload across 5 nominal cooperative client servicing cycles (100 kg transferred per mission).

Because Lazarus houses two independent Xenon storage systems—one dedicated to the 500 kg client refuelling payload and one reserved for its own 100% electric propulsion architecture (comprising the main thruster and the auxiliary thrusters for RVM/DKM and attitude control) the effective duration of Phase E is strictly coupled to the propellant mass fraction consolidated in the manoeuvre and propellant budget (Section 7.6) and the overall ΔV budget (Chapter 9).

Specifically, the onboard propulsion tank must accommodate the initial GTO-to-GEO-P orbit

raising and inclination correction, secular orbit maintenance within the GEO-P operational envelope, reaction-wheel momentum desaturation via the distributed electric thrusters, 5 complete phasing, rendezvous, docking, and return cycles to GEO-P, potential Collision Avoidance and Abort (CAM) manoeuvres, and a mandatory, inviolable propellant reserve allocated for End-of-Life (EOL) disposal.

Throughout the mission, onboard telemetry continuously tracks

- the remaining propellant mass
- the remaining payload mass
- subsystems degradation
- the completion of the 5 nominal refuelling missions
- satellite lifetime against the 8 years limit

The reaching of any of these limits, following predetermined levels, formally triggers the End-of-Life Review (ELR), verifying that the mission has completed its useful service and that all orbital elements are configured for safe disposal.

7.8.1 Graveyard Orbit and End-of-Life Disposal

Upon completion of the operational phase and formal approval at the ELR, the mission transitions into Phase F (Disposal, mapped to WBS work package WP-1350 and Section 4.5.4) to ensure full compliance with European and international space debris mitigation regulations.

Because deorbiting a spacecraft from the geostationary region back into the Earth's atmosphere requires an impractical ΔV budget, the Inter-Agency Space Debris Coordination Committee (IADC) and ESA space debris mitigation guidelines mandate that geostationary satellites be transferred at EOL into a supersynchronous graveyard orbit located well above the protected geostationary zone (which extends up to GEO + 200 km).

To prevent long-term lunisolar gravitational perturbations and solar radiation pressure from causing the retired spacecraft to drift back into active GEO slots, the disposal manoeuvre must achieve a minimum perigee altitude increase ΔH (in km) above the nominal geostationary radius, defined by the IADC formulation as:

$$\Delta H = 235 + 1000 \cdot C_R \cdot \frac{A}{m_{\text{EOL}}} \tag{19}$$

where C_R is the solar radiation pressure coefficient, A is the average cross-sectional area of the spacecraft, and m_{EOL} is the spacecraft dry mass at the end of life. Based on the preliminary spacecraft sizing (Chapter 10), Lazarus features a reference launch wet mass of $m_0 \approx 3,000$ kg (strictly bounded by the $4,500$ kg launcher limit), an exposed solar array and bus cross-sectional area of $A = 41.7$ m^2, and a radiation pressure coefficient of $C_R = 1.3$.

While the exact value of m_{EOL} is subject to final consolidation alongside the dry mass and propellant depletion budgets in Section 7.6 and Chapter 9, assuming a representative EOL mass between

1,500 kg and 2,500 kg (after offloading the 500 kg Xenon payload across the 5 servicing missions and expending operational fuel) yields a required minimum perigee raise ΔH between 257 km and 271 km above the nominal GEO ring.

Consequently, although early mission concepts considered a preliminary disposal altitude of GEO+ 200 km (150 km above GEO-P), the baseline disposal trajectory is established at a circularized graveyard orbit of GEO+300 km (i.e., 250 km above the nominal GEO-P), ensuring a conservative compliance margin over the IADC requirement.

Once the graveyard orbit injection is completed, the spacecraft executes a mandatory passivation sequence in accordance with ECSS standards to eliminate all stored onboard energy that could lead to accidental in-orbit fragmentation or explosion: all residual Xenon propellant and pressurant lines in both the payload and propulsion tanks are completely vented, the reaction wheels are spun down, the battery modules are permanently discharged and disconnected from the deployable solar arrays, and the communication transponders are deactivated, concluding with the formal End-of-Mission (EOM) declaration and the Mission Close-out Review (MCR).

8 Subject (Earth, Sky, Sun) Coverage

8.1 Subject Maps Presentation

The geographic locations of the Lazarus mission ground stations are represented below, demonstrating the system's global coverage capability from the GEOP orbit. For its analysis, the telemetry and telecommand network (Low Gain) has been separated from the network used for video transmission (High Gain).

Low-Gain Network (S-Band): Given the wide field of view of the antennas, the satellite illuminates practically the entire visible Earth disk. Coverage is restricted only by the curvature of the planet, leaving the polar caps hidden at very high latitudes. This geometric limitation does not affect the mission, as the required stations are located at mid and low latitudes. Taking advantage of the extensive overlapping windows of the four S-band stations, signal handovers have been established at the longitudinal midpoints, defining the operational ranges listed in the following table and delimited in the attached map.

Active Station	**Start of Communication (Handover received)**	**End of Communication (Handover passed to next)**
Malindi	78.56° (*Receives from Dongara*)	-6.33° (*Hands over to Kourou*)
Kourou	-6.33° (*Receives from Malindi*)	-104.61° (*Hands over to South Point*)
South Point	-104.61° (*Receives from Kourou*)	159.03° (*Hands over to Dongara*)
Dongara	159.03° (*Receives from South Point*)	78.56° (*Hands over to Malindi*)

Table 8.1: Handovers and coverage of the S-band station network.

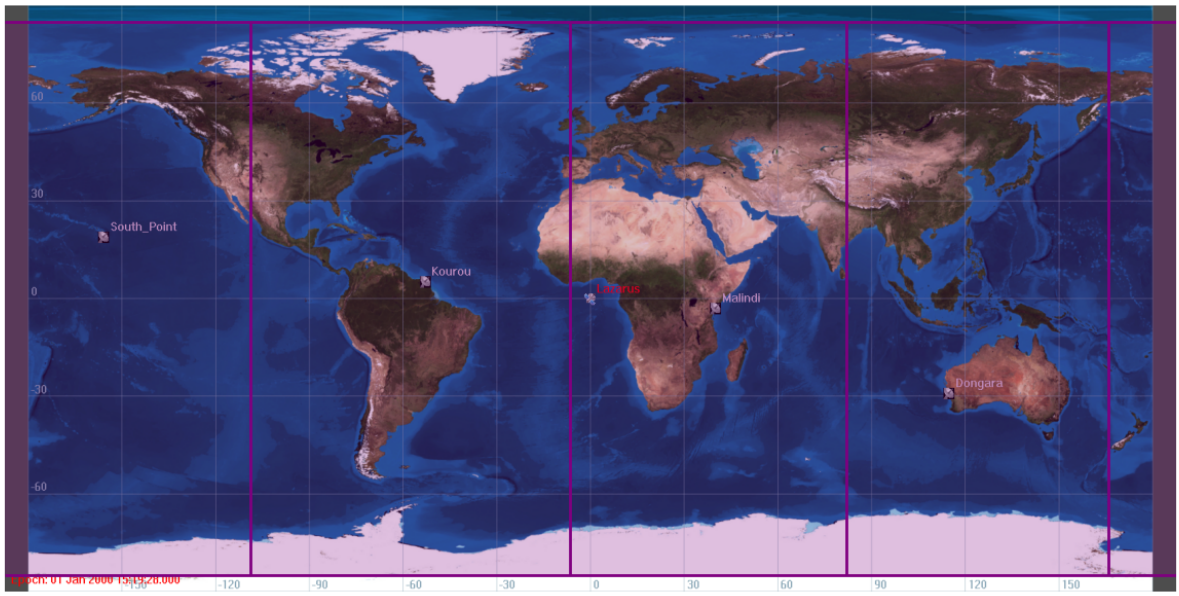


Figure 8.1: Global coverage map for the low-gain station network (S-Band).

High-Gain Network (K-Band): Unlike the previous network, the K-band link requires an active pointing mechanism due to its narrow beam. The satellite will maintain continuous pointing towards the station corresponding to its active region. The operational windows and switching points between the four high-gain stations are shown in the following table and its geographic representation.

Active Station (K-Band)	**Start of Communication (Handover received)**	**End of Communication (Handover passed to next)**
New Norcia	-179.72° (*Receives from Goldstone*)	54.45° (*Hands over to Cebreros*)
Cebreros	54.45° (*Receives from New Norcia*)	-36.07° (*Hands over to Malargüe*)
Malargüe	-36.07° (*Receives from Cebreros*)	-93.01° (*Hands over to Goldstone*)
Goldstone	-93.01° (*Receives from Malargüe*)	-179.72° (*Hands over to New Norcia*)

Table 8.2: Handovers and coverage of the K-band station network.

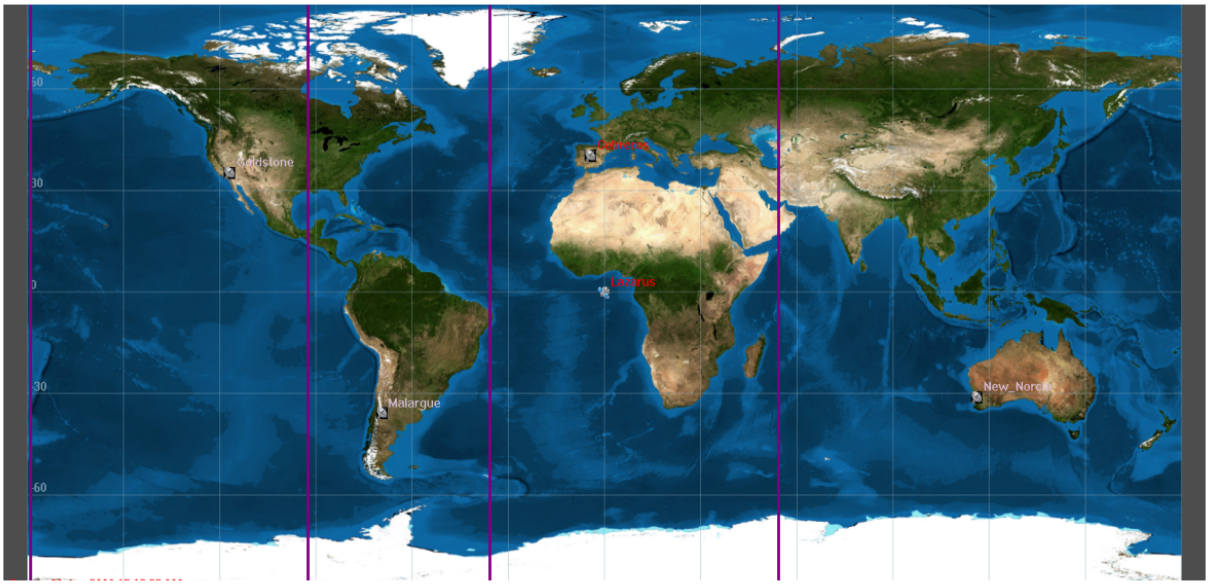


Figure 8.2: Global coverage map for the high-gain station network (K-Band).

To illustrate the directionality of this link, the following figure details the geometric projection of the directional beam over the Cebreros station. At the moment of maximum elevation, the projected footprint forms a closed ellipse (986.81 × 1347.97 km) with minimal distortion over the Iberian Peninsula. As the satellite drifts and the elevation angle decreases, the footprint progressively elongates. However, to prevent the beam from spilling over the Earth's horizon (*Earth limb spillover*), the active pointing will be handed over to the next ground station well before reaching this geometric limit.

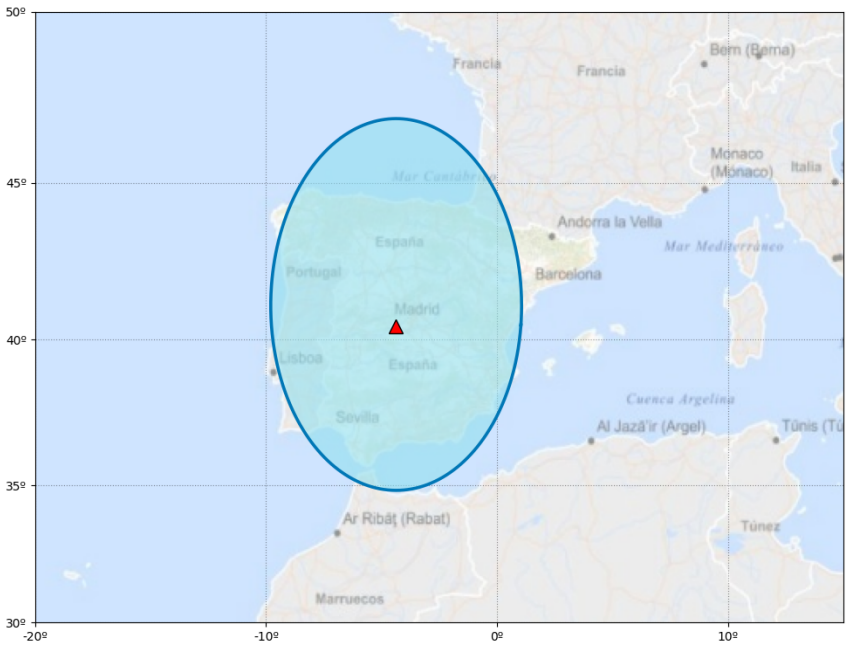


Figure 8.3: Cebreros coverage map for the high-gain station network (K-Band).

8.2 Communication Constraints

- **Sun Outages:** There will be situations where, due to the Sun's declination, microwave radiation enters the receiving beam of the ground stations, increasing the system noise temperature.
- **Structural Blockage:** The antennas' field of view must not be obstructed by any satellite component, and line-of-sight blockage due to the client satellite's position must be avoided as much as possible.
- **Energy Constraints During Eclipses:** During transit through the Earth's shadow, the solar panels do not generate power, and the communications subsystem must rely exclusively on secondary batteries. To preserve the energy balance and prioritize critical thermal control and attitude systems, continuous high-power video transmission in K-band will be restricted during peak eclipse periods, except for critical operations.
- **Pointing Constraints:** The K-band downlink demands strict pointing accuracy due to its narrow beam, requiring continuous operation of the pointing mechanism. There will be instances where, due to the characteristics of the client satellite or docking maneuvers, maintaining constant communication with the high-gain antenna will be challenging. During these periods, communications will be routed through the S-band.
- **Atmospheric Attenuation:** K-band operations suffer from significant attenuation due to atmospheric conditions such as rain. For this reason, emergency telemetry and telecommands will be transmitted in S-band, which is highly resilient to weather conditions.

9 Delta-V Budget

The estimation of the total Δv budget for the Lazarus mission relies on several assumptions regarding propellant management and operational phases (see Section 4.5). One of the (revised) requisites of Lazarus is that the mission profile spans an operational lifetime of 8 years ($a = 8$), during which 5 refueling operations ($X = 5$) are planned. Additionally it has to be considered that in microgravity environments, the extraction of the entirety of the propellant carried in the tank is in reality usually unfeasible. The fraction of the fuel trapped in the tanks, often refered to as the "residual" propellant, is sometimes given by the manufacturer of the tank, if not so, it is generally estimated at 3% of the total propellant mass carried in the tanks. This fraction of the propellant remains in the tanks and is therefore unavailable for maneuvers, but has to be considered.

The relevant mission phases for the estimation of Lazarus' Delta-V budget are the inclination correction of the orbital plane of the satellite in GEOP, maintaining GEOP, the desaturation of the reaction wheels, the several transfers from GEOP to GEO (and back to GEOP), the redenz-vous maneuveurs, the redenz-vous aborts and the decommisioning of the satellite into a graveyard orbit. For all other mission phases $\Delta v = 0$ is assumed. To estimate a conservative Delta-v for the whole mission, the worst case is being considered for every mission phase.

9.1 Phase 1: Orbit Injection

The initial mission phase involves transferring the spacecraft from GTO to the operational GEO-P. The chosen launch vehicle will inject the spacecraft into GTO with its apogee specifically at the GEOP altitude OF 36,050km. As previously stated, Lazarus can been launched with Ariane 6 or with Falcon 9. Since the inclination of the space ports, from which Falcon 9's are launched, is in average much higher than the inclination from the space port in Kourou (aprox. $i_{Falcon9} = 27°$ vs. $i_{Ariane6} = 5°$), it is taken into consideration as the worst case for estimating the Delta-V budget for the correction of the orbital plane inclination. taking this into consideration, the injection maneuver requires $\Delta v_{GTO->GEOP} = \mathbf{2139.8 m/s}$. As said, this value accounts for a worst-case scenario, which is a Falcon 9 launch. It includes both the circularization of the orbit at GEOP and the correction of the inclination of the orbital plane from 27° to 0°, while also factoring in solar radiation pressure perturbations.

9.2 Phase 2: Station Keeping and Attitude Control

Once in GEOP, the satellite must perform regular maintenance maneuvers to counteract orbital perturbations and desaturate its momentum exchange devices. The station-keeping budget is estimated at $\Delta v_{maintain} = \mathbf{52\ m/s\ per\ year}$. This calculation assumes a predefined worst-case scenario of two consecutive years in orbit without any rendezvous operations. The primary control objectives during this phase are maintaining the orbital inclination i $< 0.1°$ and keeping the eccentricity e $< 3 \times 10^{-4}$.

Simultaneously, the Attitude and Orbital Control System (AOCS) manages the spacecraft's constant NADIR-pointing orientation. Because the local magnetic field in GEO is too weak for magnetorquers to be effective, the desaturation of the 12 Nms reaction wheels, which become saturated approximately 60 times annually, must be performed using secondary actuators while the wheels are powered down. This desaturation process requires an additional $\Delta v_{desat} = \mathbf{1.3\ m/s\ per\ year}$.

9.3 Phase 3: Rendezvous and Refueling Operations

For Lazarus' lifespan of 8 years, 5 commercial refueling services are planned. Since the target clients are Xenon-propulsed satellites in GEO, each refueling involves necessarily transferring from GEO-P to GEO and returning back to Lazarus' nominal orbit GEO-P. For being able to be profitable, these transfers must be performed as efficient as possible. Nonetheless, due to the continuous low-thrust nature of the electric propulsion system (for the main propulsion system: 0.3N of thrust and an I_{sp} of 2,000s), traditional impulsive Hohmann transfers are not viable. The two-way transfer maneuvers with electric propulsion require each a $\Delta v_{GEOP-GEO} = \mathbf{13.8\ m/s\ per\ operation}$.

The orbital transfer from GEO-P to GEO finishes, when Lazarus is at a relative distance of 500m directly behind the target spacecraft. At this point, the final rendezvous approach begins. As already described in previous chapters, the approach is divided by 5 hold points at 200m, 100m, 20m, 5m, and 0m from the client's spacecraft. The distances between the hold points are all flown at different speeds, which can also be consulted in previous chapters.

After refueling the client's spacecraft, Lazarus returns revisiting all 5 hold points again, finishing the redenzvous again 500m behind the client's spacecraft. For this phase $\Delta v_{rendezvous}$ = **1.4 m/s per operation** is needed. Furthermore, a highly conservative abort scenario is factored into the total budget, assuming one failed approach sequence per refueling mission (X) occurring immediately before contact, adding a Δv_{abort} = **0.7 m/s per maneuver**.

9.4 Phase 4: End-of-Life Disposal

At the conclusion of its operational life, the worst-case scenario assumes Lazarus is located in GEOP and must be transferred to a compliant graveyard orbit. This orbit is defined as GEO + 300 km, at approximately 36,300km. Performing the most efficient transfer possible, this decommissioning maneuver requires a final $\Delta v_{deorbit}$ = **10.8 m/s**.

9.5 Total Delta-V Budget Summary

The Δv budget integrates all operational phases over the 8-year mission lifespan ($a = 8$) and 5 refueling missions ($X = 5$) and is calculated as follows:

$$\begin{aligned}\Delta v_{total} =& \Delta v_{GTO->GEOP} + (\Delta v_{maintain} + \Delta v_{desat}) \cdot a \\ &+ (\Delta v_{GEOP-GEO} + \Delta v_{rendezvous} + \Delta v_{abort}) \cdot X + \Delta v_{deorbit}\end{aligned}$$

Substituting the above mentioned values for each term:

$$\begin{aligned}\Delta v_{total} &= 2139.8 + (52.0 + 1.3) \cdot 8 + (13.8 + 1.4 + 0.7) \cdot 5 + 10.8 \\ &= 2139.8 + 426.4 + 79.5 + 10.8 \\ &= \mathbf{2656.5 m/s}\end{aligned}$$

It is important to keep in mind that the above estimated value $\Delta v_{total} = 2656.5 m/s$ is the estimated Delta-V budget for the worst case scenario. For comparison, a exemplary mission has been defined and summarized in Table 9.1.

Table 9.1: Phase-by-phase budget of the mission. Each service includes its phasing.

	Duration		**Δv [m s^{-1}]**		**Propellant [kg]**	
Phase	6°	27°	6°	27°	6°	27°
GTO → GEO-P	370.0 d	505.1 d	1724.4	2099.3	252.41	304.41
Station keeping before A	134.5 d	–	19.1	–	2.67	–
Service A	100.2 d	96.8 d	12.9	4.9	1.80	0.68
Station keeping A → B	316.8 d	316.1 d	48.4	50.5	6.75	6.92
Service B	101.0 d	101.0 d	9.8	9.9	1.37	1.35
Station keeping B → C	464.4 d	464.4 d	73.6	73.3	10.24	10.01
Service C	1.1 d	1.1 d	4.1	4.1	0.58	0.57
Station keeping C → D	213.5 d	213.5 d	22.9	22.9	3.19	3.12
Service D	101.0 d	101.0 d	9.8	9.8	1.36	1.34
Station keeping D → E	523.5 d	523.5 d	87.6	87.8	12.12	11.94
Service E	1.1 d	1.1 d	4.1	4.1	0.57	0.56
GEO-P → graveyard orbit	35.5 h	35.3 h	9.0	9.0	1.24	1.22
Total	6.38 years	6.37 years	**2025.8**	**2375.6**	**294.30**	**342.12**
Final mass [kg]					**2705.70**	**2657.88**

Some important notes about the above table are that the total Delta-V is smaller than the worst case scenario estimated above. The GTO -> GEO-P transfer takes between 86 % and 89 % of the used xenon in the whole of the mission. It is striking that in the operational phase, station keeping costs six to seven times more than the five refueling services together. Having a closer look at both launch vehicles, Ariane 6 would end the mission about 48 kg heavier than a Falcon 9. Also, with a 500 kg xenon tank, about 41 % (Ariane 6) and 32 % (Falcon 9) of the xenon would be left at the end of the mission, enough for several additional refueling services.

10 Preliminary S/C sizing

The preliminary spacecraft sizing is based on the study of structural configurations commonly used in space platforms, taking as references both satellites with similar characteristics and functions and different structural architecture solutions. This analysis allows the most suitable configurations to be identified to meet the mission requirements and establish an initial layout of the onboard subsystems.

As an initial configuration, a spacecraft with a rectangular prismatic geometry is proposed. Its architecture will be mainly influenced by the propellant storage and transfer requirements, the integration of the electric propulsion system, and the requirements associated with approach and docking operations with other satellites.

10.1 External Configuration

Due to the high electrical power demand of the propulsion system, the spacecraft will be equipped with deployable solar arrays, which will remain folded during launch and will be deployed once the appropriate orbital conditions have been reached.

Communication antennas, sensors required for approach operations, and the docking and xenon transfer mechanism will also be integrated into the external surfaces of the spacecraft.

10.2 Primary Structure

The primary structure will consist of an external rectangular framework formed by longitudinal and transverse beams, together with a central cylindrical tube acting as one of the main load-bearing elements of the spacecraft.

The external framework will be made of aluminium 7075-T6, using L-shaped and T-shaped beams to provide structural support and connect the external panels. The panels will consist of CFRP facesheets with an aluminium honeycomb core, allowing a reduction in structural mass while maintaining adequate stiffness.

The central tube will be made of CFRP, with a preliminary diameter of 1,200 mm, a wall thickness of 4 mm and a total height of 5,000 mm. This element will accommodate the main propellant tanks and provide a structural load path between the docking interface, the spacecraft body and the launcher attachment region.

To improve the stiffness of the central structure, five aluminium stiffening rings will be distributed along the tube. Additionally, twelve sandwich shear walls will connect different structural elements, contributing to the transmission of loads and the overall stiffness of the spacecraft.

The preliminary structural mass, including the external beams, sandwich panels, central tube, stiffening rings and shear walls, is estimated at 541.4 kg. This value will be updated as the structural configuration and the required load cases are further analysed.

10.3 Subsystem Layout

10.3.1 Attitude and Orbit Control System (AOCS)

The Attitude and Orbit Control System (AOCS) will include four reaction wheels arranged in a pyramidal configuration around the central structural tube, close to the spacecraft centre-of-mass plane. This arrangement will provide three-axis attitude control and redundancy in the event of a single reaction wheel failure.

Attitude determination will be performed using two star trackers, two fine Sun sensors, four coarse Sun sensors and one Inertial Measurement Unit (IMU). The star trackers and Sun sensors will be distributed over the external surfaces of the spacecraft, ensuring suitable fields of view while avoiding obstructions caused by the structure, solar arrays and other external equipment. The IMU will be installed on a rigid internal mounting surface, preferably close to the spacecraft centre of mass.

For rendezvous and proximity operations, two cameras, one 3D LiDAR and one laser rangefinder will be integrated into the upper region of the spacecraft, close to the docking interface. Their orientation will be selected to provide the required visibility of the client satellite during approach and docking operations.

Additionally, twelve SITAEL HT400 electric thrusters will be distributed into four assemblies of three thrusters each, located near the longitudinal corners of the spacecraft. Their orientation will be defined to generate the forces and torques required for attitude control, reaction-wheel desaturation and proximity operations.

The auxiliary thrusters are included in the Propulsion System mass budget to avoid double counting. The preliminary AOCS equipment selection, including the relative-navigation sensors, is summarised in Table 10.1.

Table 10.1: Preliminary AOCS equipment sizing.

Component	Model	Nº.	Unit mass [kg]	Dimensions [mm]
Reaction wheel	Rocket Lab 12.0 Nms [33]	4	5.000	⌀264 × 70
Star tracker	Sodern HORUS [34]	2	1.600	141 × 141 × 250
Fine Sun sensor	Tensor Tech FSS-15M [35]	2	0.040	22 × 15 × 5.26
Coarse Sun sensor	Bradford CoSS-R [36]	4	0.015	48 × 37 × 6
IMU	Safran STIM300 [37]	1	0.055	44.8 × 38.6 × 21.5
RPO camera (far/medium)	Infinity Avionics Lynx4MP-10 [38]	1	0.440	111 × 95 × 95
RPO camera (close)	Infinity Avionics Lynx4MP-10 [38]	1	0.440	90 × 100 × 100
3D LiDAR	Jena-Optronik RVS 3000-3D [39]	1	14.000	218 × 342 × 265
Laser rangefinder	Jenoptik DLEM-SR [40]	1	0.040	54 × 22 × 34
	Total mass		**38.315**	—

10.3.2 Propulsion System

The spacecraft propulsion system will be based entirely on electric propulsion using xenon, with four main thrusters dedicated to orbital transfers and twelve auxiliary thrusters for attitude control and proximity operations.

The four main electric thrusters will be installed on the lower face of the spacecraft, distributed around the central structural tube. Each thruster will incorporate a pointing mechanism to adjust its thrust direction and reduce unwanted moments during orbital manoeuvres. Their arrangement will be defined considering the spacecraft centre of mass, structural attachment points and plume clearance requirements.

The twelve auxiliary electric thrusters will be distributed into four assemblies of three thrusters each, located near the longitudinal corners of the spacecraft. Their orientation will be selected to provide the forces and torques required during attitude-control manoeuvres, reaction-wheel desaturation, rendezvous, docking and separation operations.

The spacecraft propulsion system will incorporate three MT Aerospace S-XTA 120 L xenon storage tanks, installed within the central structural tube and physically separated from the three tanks dedicated to the refuelling payload. Their arrangement will be selected to minimise centre-of-mass displacement as propellant is consumed throughout the mission.

According to the manufacturer's technical catalogue [56], each S-XTA 120 L tank has a stated

maximum xenon storage capacity of 220 kg. Therefore, the three selected tanks provide a combined nominal storage capacity of 660 kg, exceeding the 500 kg required for spacecraft propulsion.

This preliminary selection provides sufficient nominal storage capacity for the mission. The final usable capacity, operating conditions and qualification status will be confirmed during subsequent design phases.

Additional space and mass allowances will be reserved for the propulsion feed system, power-processing electronics, valves, regulators and associated equipment.

The preliminary selection of propulsion equipment, including the corresponding mass and dimensional characteristics, is summarised in Table 10.2.

Table 10.2: Preliminary propulsion system equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
Main thruster	Safran PPS5000 [41]	4	12.30	$270.5 \times 270.5 \times 119.4$
Auxiliary thruster	SITAEL HT400 [58]	12	2.77	$163 \times 102 \times 80$
Xenon storage tank	MT Aerospace S-XTA 120L [56]	3	16.80	$\varnothing 440 \times 1072$
Xenon propellant	XeHPV	1	500.00	—
Other equipment	TBD	1	78.00	TBD
	Total mass [kg]		**710.84**	—

10.3.3 Electrical Power System

The Electrical Power System (EPS) will consist of two deployable solar array wings, each containing five solar panels, together with the batteries and power management and distribution equipment.

The solar arrays will be attached to opposite lateral faces of the spacecraft and will remain folded during launch. Each panel has preliminary dimensions of 1.5 m × 2.0 m, providing a total geometric area of 30 m^2. The configuration will incorporate deployment and Sun-tracking mechanisms, whose masses are included in the current solar array mass allocation.

Two battery units will be accommodated in the lower internal region of the spacecraft, distributed symmetrically around the central structural tube to maintain a balanced mass distribution. The remaining electrical equipment, including the EPS modules, Power Control and Distribution Unit (PCDU), Peak Power Unit (PPU) and Shunt Unit, will be installed on internal mounting surfaces, considering their electrical connections, accessibility and thermal dissipation requirements.

The preliminary mass and dimensional characteristics of the selected EPS components are summarised in Table 10.3.

10.3.4 Payload and Docking System

The Lazarus payload will be responsible for the storage and transfer of xenon propellant to client spacecraft, as well as providing the mechanical and fluid interfaces required for docking and refuelling operations.

The refuelling interface will be installed at the centre of the spacecraft upper face, aligned with the central structural tube to facilitate the transmission of docking loads through the primary

Table 10.3: Preliminary Electrical Power System equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
Solar panel	TBD	10	9.50	$1500 \times 2000 \times 30$
EPS module	TBD	2	5.50	$\varnothing 200 \times 250$
Battery	TBD	2	21.75	$280 \times 220 \times 220$
Power Control and Distribution Unit (PCDU)	TBD	1	32.00	$450 \times 350 \times 200$
Peak Power Unit (PPU)	TBD	1	25.00	$420 \times 350 \times 180$
Shunt Unit	TBD	1	14.00	$400 \times 250 \times 80$
Cabling	TBD	1	20.00	—
	Total mass [kg]		**240.50**	—

structure.

The payload xenon storage system will consist of three MT Aerospace S-XTA 120 L tanks, installed within the central structural tube and physically separated from the three tanks dedicated to spacecraft propulsion.

According to the manufacturer's technical catalogue [56], each S-XTA 120 L tank has a stated maximum xenon storage capacity of 220 kg. Therefore, the three selected tanks provide a combined nominal capacity of 660 kg, exceeding the 500 kg required for the five planned refuelling operations.

The fluid transfer system will also incorporate a custom-designed astroFUEL transfer tank with an internal capacity of 1.5 L. Its preliminary mass and external dimensions remain to be determined.

The docking and refuelling interface will be based on the Orbit Fab RAFTI system. Additional mechanical capture and locking equipment will be required to establish a rigid connection with the client spacecraft. The corresponding equipment mass and mechanical interfaces remain subject to further definition.

The preliminary equipment selection is summarised in Table 10.4.

Table 10.4: Preliminary Payload and Docking System equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
Refuelling interface	Orbit Fab RAFTI [30]	1	1.10	$96.012 \times 96.012 \times 99.873$
Xenon storage tank	MT Aerospace S-XTA 120L [56]	3	16.80	$\varnothing 440 \times 1072$
Xenon transfer tank	astroFUEL (1.5 L, custom design)	1	TBD	TBD
Xenon propellant	XeHPV	1	500.00	—
	Subtotal [kg] (excluding TBD)		**551.50**	—

The identified payload mass subtotal is 551.50 kg, including 500 kg of xenon propellant. This value excludes the custom transfer tank, the mechanical capture and locking mechanism, and any additional fluid transfer equipment not yet included in the subsystem mass allocation.

10.3.5 Communications System

The Communications System will consist of one high-gain antenna (HGA), two low-gain antennas (LGA) and one GNSS receiving antenna, together with the corresponding radio-frequency and data-handling equipment.

The HGA will be installed on the lateral face of the spacecraft, supported by an antenna point-

ing mechanism to maintain the required orientation towards the ground stations. The two LGA antennas will be located on opposite lateral faces, providing complementary coverage during attitude manoeuvres and proximity operations. The GNSS antenna will be positioned considering the visibility requirements of navigation signals and possible obstructions caused by the spacecraft structure and deployed solar arrays.

The RF amplifiers, transponders, On-Board Computer (OBC) and Remote Terminal Unit (RTU) will be accommodated on internal mounting surfaces, mainly within the central equipment region. Their final distribution will consider accessibility, mechanical interfaces, electrical connections and thermal dissipation requirements.

The preliminary selection of communications and data-handling equipment, including the corresponding mass and dimensional characteristics, is summarised in Table 10.5.

Table 10.5: Preliminary Communications and Data Handling equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
High-Gain Antenna	Beyond Gravity K-band [?]	1	1.750	∅575 × 160
Low-Gain Antenna	Beyond Gravity S-band Helix [?]	2	0.235	∅65 × 289
GNSS Antenna	Beyond Gravity GEO GNSS PEC [?]	1	0.715	∅239 × 179
RF Amplifier	TBD	2	TBD	250 × 150 × 100
Transponder	TBD	TBD	TBD	TBD
On-Board Computer (OBC)	TBD	TBD	TBD	200 × 200 × 150
Remote Terminal Unit (RTU)	TBD	TBD	TBD	150 × 150 × 100

10.3.6 Thermal Control System

The Thermal Control System (TCS) will combine passive and active thermal control elements to maintain the spacecraft equipment within their allowable operating temperature ranges throughout the mission.

The preliminary configuration will consider thermal insulation, radiators and heating elements, with particular attention to the xenon storage tanks, batteries, electronic equipment and electric propulsion system. Radiators will be located on suitable external surfaces, considering their visibility to deep space and possible interference with the solar arrays, propulsion plumes and other external components.

The final arrangement of the thermal control equipment will depend on the thermal requirements and heat dissipation characteristics of each subsystem, as well as the temperature variations experienced during orbital transfers, eclipses and refuelling operations.

A preliminary mass allocation of 150 kg has been reserved for the Thermal Control System, as summarised in Table 10.6. This value will be refined once the thermal architecture and individual equipment requirements have been defined.

10.3.7 Structure and Mechanisms

The mechanical subsystem will include the mechanisms and structural interfaces required for the deployment, orientation and attachment of the main spacecraft components.

Table 10.6: Preliminary Thermal Control System equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
Thermal insulation (MLI)	TBD	TBD	TBD	TBD
Radiators	TBD	TBD	TBD	TBD
Heaters	TBD	TBD	TBD	TBD
Heat pipes / thermal straps	TBD	TBD	TBD	TBD
Thermal control equipment	TBD	TBD	TBD	TBD
Preliminary mass allocation			**150.00**	—

The solar array deployment mechanisms will be integrated into opposite lateral faces of the spacecraft, while the main electric thruster pointing mechanisms will be installed in the lower region. Additionally, an antenna pointing mechanism will support the high-gain antenna, allowing its orientation towards the ground stations.

The mechanical integration will also consider the docking capture and locking assembly, the xenon tank mounting supports and the structural interface with the launch vehicle.

The final configuration and mass allocation of these components will be defined considering the launch loads, deployment requirements, operational clearances and compatibility with the primary structure.

Table 10.7: Preliminary Structure and Mechanisms equipment sizing.

Component	Model	Qty.	Unit mass [kg]	Dimensions [mm]
Main thruster pointing mechanism	Beyond Gravity APPMAX2	4	13.80	TBD
Antenna pointing mechanism	TBD	1	15.00	TBD
Solar array deployment mechanism	TBD	2	TBD	TBD
Docking capture and locking mechanism	TBD	1	TBD	TBD
Launch vehicle adapter interface	TBD	1	TBD	TBD
Xenon tank mounting assemblies	Custom design	6	TBD	TBD
	Subtotal [kg] (excluding TBD)		**70.20**	—

10.4 Preliminary Dimensions

Based on the initial distribution of the onboard subsystems, a preliminary rectangular spacecraft configuration has been established.

The main spacecraft body has approximate dimensions of 2.9 m × 2.9 m × 5.0 m. When the folded solar array configuration is considered, the overall preliminary envelope increases to 3.4 m × 2.9 m × 5.0 m.

These dimensions have been selected to provide sufficient internal space for the propellant tanks, propulsion equipment, power subsystem, attitude-control components and the remaining onboard systems.

The central cylindrical tube, with a preliminary diameter of 1.2 m, will accommodate the main propellant tanks. The surrounding space will be used to distribute the remaining equipment, considering their individual dimensions, mounting requirements and the need to maintain a balanced mass distribution.

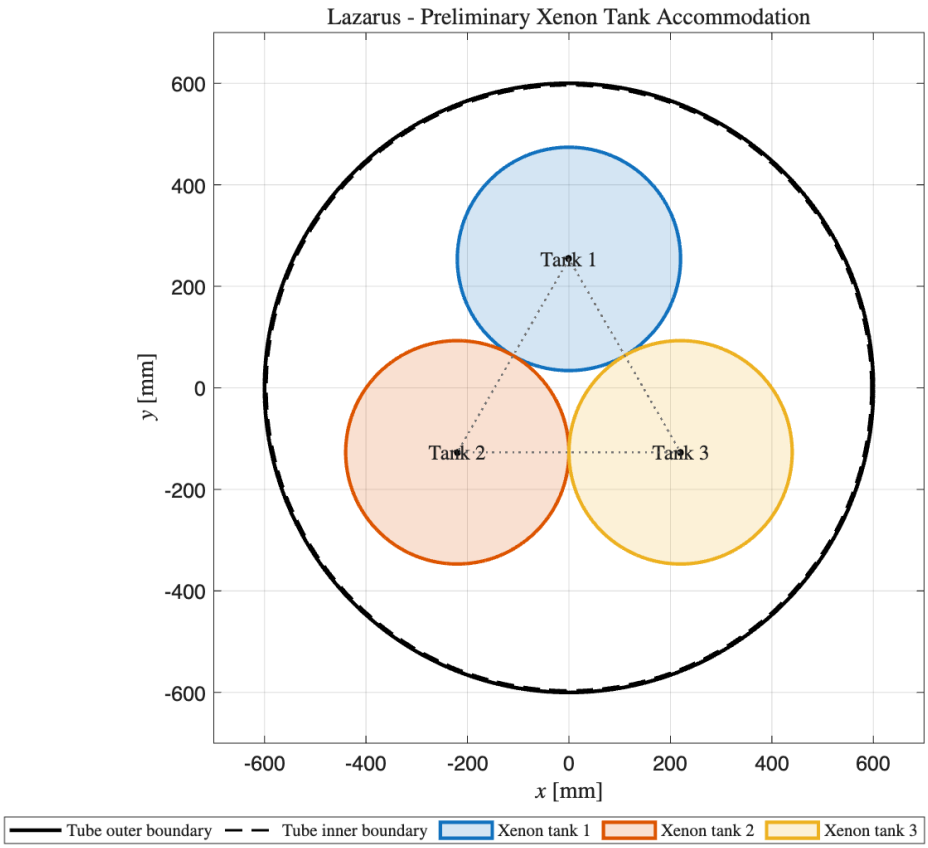


Figure 10.1: Preliminary cross-sectional arrangement of three xenon storage tanks within the central structural tube. The triangular configuration provides a theoretical radial clearance of approximately 122 mm between the tank assembly and the inner tube wall.

The electrical power system will include ten solar panels distributed into two deployable wings, each containing five panels. The preliminary dimensions of each panel are 1.5 m × 2.0 m, with a thickness of approximately 30 mm.

The final spacecraft dimensions will depend on the detailed positioning of the onboard equipment, the space required for structural supports and the compatibility with the selected launch vehicle fairing envelopes.

Table 10.8: Preliminary spacecraft structural dimensions.

Structural component	**Material**	**Qty.**	**Dimensions [mm]**
Spacecraft main body	—	1	2900 × 2900 × 5000
Stowed spacecraft envelope	—	1	3400 × 2900 × 5000
Central structural tube	CFRP	1	∅1200 × 5000
Central tube wall thickness	CFRP	—	4
External sandwich panels	CFRP / Al honeycomb	TBD	16 (thickness)
Longitudinal longerons	Al 7075-T6	TBD	TBD
Transverse frames	Al 7075-T6	TBD	TBD
Internal stiffening rings	Al 7075-T6	5	TBD
Internal shear walls	CFRP / Al honeycomb	12	TBD
Equipment mounting supports	TBD	TBD	TBD

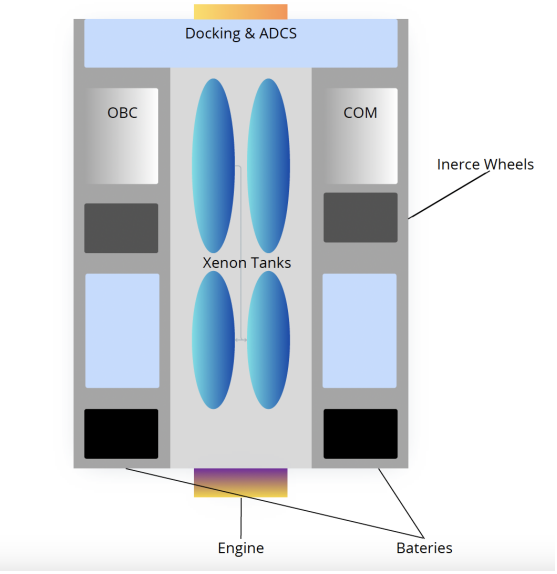


Figure 10.2: Preliminary spacecraft configuration showing the external arrangement of the main subsystems and deployable solar arrays.

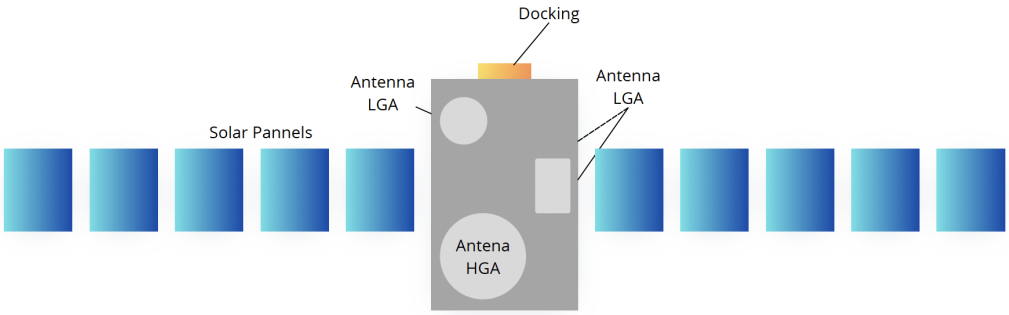


Figure 10.3: Preliminary internal spacecraft layout showing the approximate accommodation of the main subsystems, xenon storage tanks and structural elements.

The preliminary spacecraft configuration is illustrated in Figures 10.2 and 10.3, showing the approximate external and internal accommodation of the main subsystems, respectively. The layouts are conceptual and will be further refined during subsequent design phases.

10.5 Evolution of the centre of gravity

A preliminary analysis of the spacecraft centre of gravity (CG) was performed to assess its displacement as xenon is consumed and transferred to the clients. The model considers three 90 L tanks in the lower tier for spacecraft propulsion and three 120 L tanks in the upper tier for the refuelling payload. Within each tier, the tanks are equally spaced at angular intervals of 120° around the longitudinal axis.

The remaining spacecraft mass is assumed to be constant, with its CG located at the geometric centre of the tank assembly envelope. The empty tank masses are included separately, while the

xenon in each tank is represented by a variable mass located at its geometric centre. Consequently, internal fluid motion is neglected. The spacecraft CG is calculated as

$$\mathbf{r}_G(t) = \frac{m_{\text{fixed}}\mathbf{r}_{\text{fixed}} + \sum_{j=1}^{6} m_j(t)\mathbf{r}_j}{m_{\text{fixed}} + \sum_{j=1}^{6} m_j(t)}, \tag{20}$$

where m_{fixed} includes all non-consumable masses, and $m_j(t)$ denotes the xenon remaining in tank j.

The reference case assumes an initial spacecraft mass of 3000 kg, including 500 kg of xenon for propulsion and 500 kg for client refuelling. The consumption history follows the orbital analysis for an initial GTO inclination of 6°, with propellant consumption interpolated linearly within each mission phase. Five payload transfers of 100 kg are modelled, each lasting 1.5 h. Both tank groups are depleted symmetrically, with equal mass withdrawn from the three tanks in each tier.

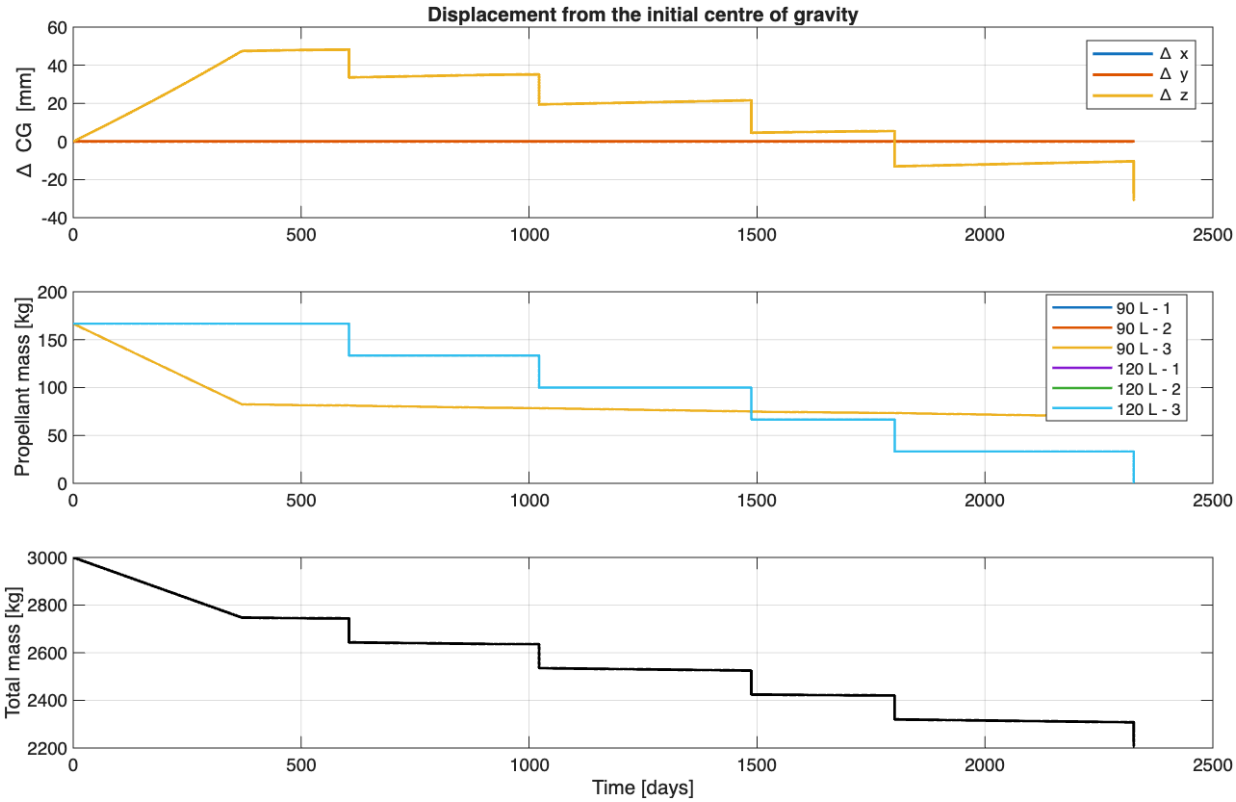


Figure 10.4: Evolution of the spacecraft CG displacement, xenon mass in each tank and total spacecraft mass for the 6° reference scenario. Displacements are measured relative to the initial CG.

As shown in Figure 10.4, the lateral displacements remain approximately zero owing to the symmetric arrangement and drawdown strategy. The CG therefore moves along the longitudinal axis. During the initial GTO-to-GEO-P transfer, consumption from the lower tanks shifts the CG upwards, reaching a maximum displacement of approximately +48 mm. Subsequently, each payload

transfer removes mass from the upper tier and produces a downward shift, partially counteracted by the continued consumption of propulsion xenon from the lower tier.

At the end of the mission, the vertical displacement is approximately −31 mm relative to the initial position. The resulting vertical excursion is therefore approximately 79 mm between the highest and lowest positions. The total spacecraft mass decreases to approximately 2206 kg after consuming 293.79 kg of propulsion xenon and delivering the 500 kg payload.

These results provide a preliminary CG envelope for the assumed geometry and mass distribution. The orbital propellant budgets are retained from the supplied analysis without recalculating the manoeuvres after each payload delivery. The predicted displacement should therefore be updated as the spacecraft mass budget, equipment layout and tank drawdown strategy are refined.

For the initial GTO inclination of 27°, the CG evolution follows the same general pattern as the 6° case. However, the larger propulsion xenon consumption during the GTO-to-GEO-P transfer, approximately 304.41 kg over 505 days, produces a greater upward displacement. The maximum vertical displacement reaches approximately +58 mm relative to the initial CG, while the lateral displacements remain approximately zero under the symmetric drawdown assumption.

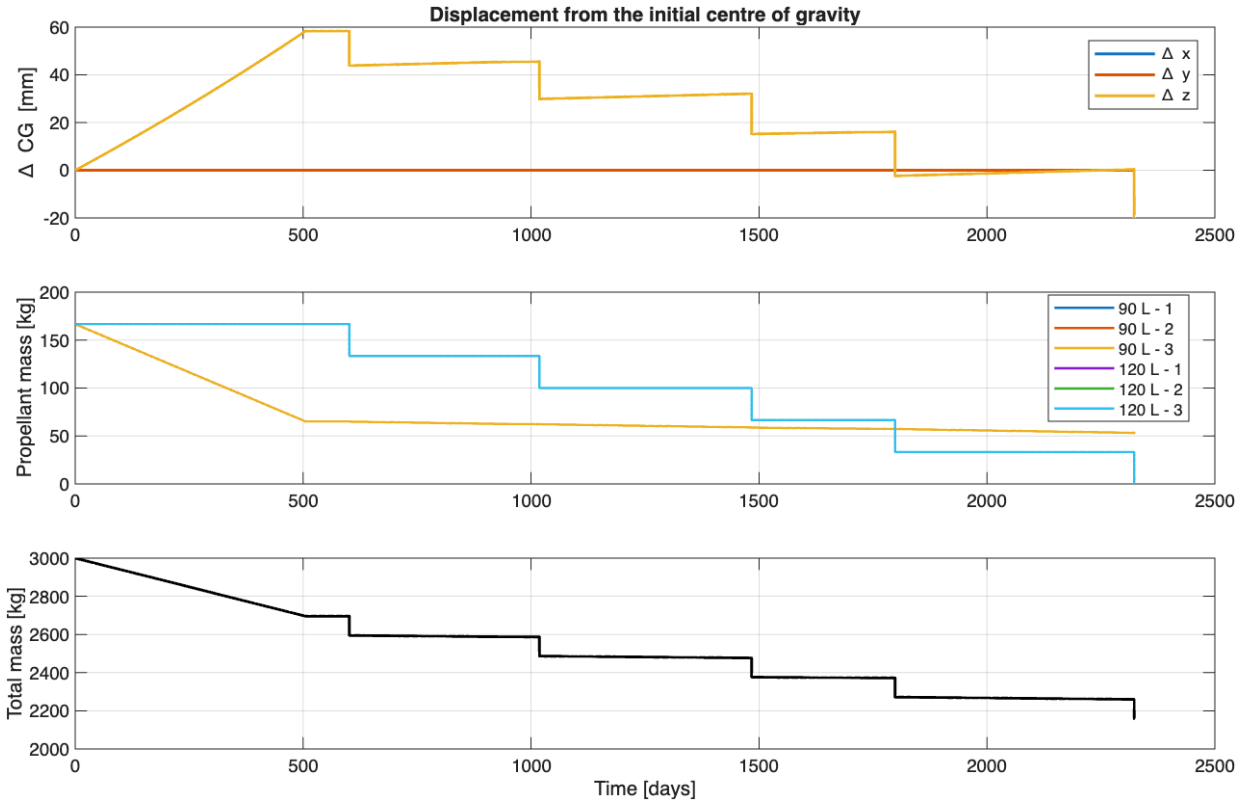


Figure 10.5: Evolution of the spacecraft CG displacement, xenon mass in each tank and total spacecraft mass for the 27° reference scenario. Displacements are measured relative to the initial CG.

The five payload transfers subsequently shift the CG downwards as xenon is removed from the upper tanks. Between services, consumption from the lower propulsion tanks produces a gradual upward drift. At the end of the mission, the CG lies approximately 20 mm below its initial position, giving an overall vertical excursion of approximately 78 mm.

The final spacecraft mass is approximately 2158 kg, accounting for 341.85 kg of propulsion xenon consumed and 500 kg delivered to the clients. Compared with the 6° scenario, the greater depletion of the lower tanks shifts the final CG upwards relative to that case, despite the lower final spacecraft mass.

Table 10.9: Preliminary spacecraft equipment mass and dimensions breakdown.

Component	Manufacturer	Model / Material	Qty.	Unit mass [kg]	Dimensions	Total mass [kg]
Communications						
High-gain antenna	Beyond Gravity	K-band all-metal reflector	1	1.750	∅575 × 160 mm	1.750
Low-gain antenna	Beyond Gravity	S-band helix TT&C	2	0.235	∅65 × 289 mm	0.470
GNSS antenna	Beyond Gravity	GEO GNSS PEC	1	0.715	∅239 × 179 mm	0.715
Amplifiers	TBD	TBD	2	TBD	250 × 150 × 100 mm	TBD
Transponders	TBD	TBD	TBD	10.000	TBD	TBD
Other communications equipment	TBD	Preliminary allocation	1	60.000	TBD	60.000
					Communications subtotal	**62.935**
Data Handling						
On-board computer (OBC)	TBD	TBD	TBD	TBD	200 × 200 × 150 mm	TBD
Remote terminal unit (RTU)	TBD	TBD	TBD	TBD	150 × 150 × 100 mm	TBD
Other data handling equipment	TBD	Preliminary allocation	1	35.000	TBD	35.000
					Data Handling subtotal	**35.000**
Payload and Docking						
Refuelling interface	Orbit Fab	RAFTI	1	1.100	96.012 × 96.012 × 99.873 mm	1.100
Xenon storage tanks	MT Aerospace	S-XTA 120 L family	3	16.800	∅440 × 1072 mm	50.400
Xenon transfer tank	astroFUEL	1.5 L custom design	1	TBD	TBD	TBD
Payload xenon	—	XeHPV	1	500.000	—	500.000
					Payload subtotal (excluding TBD)	**551.500**
Propulsion						
Main electric thrusters	Safran	PPS5000	4	12.300	270.5 × 270.5 × 119.4 mm	49.200
Auxiliary electric thrusters	SITAEL	HT400	12	2.770	163 × 102 × 80 mm	33.240
Xenon storage tanks	MT Aerospace	S-XTA 120 L family	3	16.800	∅440 × 1072 mm	50.400
Propulsion xenon	—	XeHPV	1	500.000	—	500.000
Other propulsion equipment	TBD	Preliminary allocation	1	78.000	TBD	78.000
					Propulsion subtotal	**710.840**

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Component	Manufacturer	Model / Material	Qty.	Unit mass [kg]	Dimensions	Total mass [kg]
Thermal Control						
Thermal control equipment	TBD	Preliminary allocation	1	150.000	TBD	150.000
					Thermal subtotal	**150.000**
AOCS and Relative Navigation						
Reaction wheels	Rocket Lab	12.0 Nms Reaction Wheel	4	5.000	∅264 × 70 mm	20.000
Star trackers	Sodern	HORUS	2	1.600	141 × 141 × 250 mm	3.200
Fine Sun sensors	Tensor Tech	FSS-15M	2	0.040	22 × 15 × 5.26 mm	0.080
Coarse Sun sensors	Bradford Space	CoSS-R	4	0.015	48 × 37 × 6 mm	0.060
Inertial measurement unit	Safran	STIM300	1	0.055	44.8 × 38.6 × 21.5 mm	0.055
Far/medium-range camera	Infinity Avionics	Lynx4MP-10	1	0.440	111 × 95 × 95 mm	0.440
Close-range camera	Infinity Avionics	Lynx4MP-10	1	0.440	90 × 100 × 100 mm	0.440
3D LiDAR	Jena-Optronik	RVS 3000-3D	1	14.000	218 × 342 × 265 mm	14.000
Laser rangefinder	Jenoptik	DLEM-SR	1	0.040	54 × 22 × 34 mm	0.040
					AOCS subtotal	**38.315**
Electrical Power System						
Solar array panels	TBD	Deployable solar arrays	10	9.500	1500 × 2000 × 30 mm	95.000
EPS modules	TBD	TBD	2	5.500	∅200 × 250 mm	11.000
Batteries	TBD	TBD	2	21.750	280 × 220 × 220 mm	43.500
Power Control and Distribution Unit	TBD	PCDU	1	32.000	450 × 350 × 200 mm	32.000
Peak Power Unit	TBD	PPU	1	25.000	420 × 350 × 180 mm	25.000
Shunt Unit	TBD	TBD	1	14.000	400 × 250 × 80 mm	14.000
Power cabling	TBD	Preliminary allocation	1	20.000	TBD	20.000
					EPS subtotal	**240.500**
Primary Structure						
L-section lateral longerons	—	Aluminium 7075-T6	4	8.000	$L = 5000$ mm; $t = 5$ mm	32.000
T-section lateral longerons	—	Aluminium 7075-T6	4	10.000	$L = 5000$ mm; $t = 5$ mm	40.000
Upper/lower L-section beams	—	Aluminium 7075-T6	8	4.700	$L = 2900$ mm; $t = 5$ mm	37.600

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Component	Manufacturer	Model / Material	Qty.	Unit mass [kg]	Dimensions	Total mass [kg]
Upper/lower T-section beams	—	Aluminium 7075-T6	4	5.500	$L = 2900$ mm; $t = 5$ mm	22.000
Upper stiffening ring	—	Aluminium 7075-T6	1	11.000	$D_i = 1200$, $D_o = 1400$, $t = 6$ mm	11.000
Intermediate stiffening rings	—	Aluminium 7075-T6	3	11.000	$D_i = 1200$, $D_o = 1400$, $t = 6$ mm	33.000
Lower stiffening ring	—	Aluminium 7075-T6	1	11.000	$D_i = 1200$, $D_o = 1400$, $t = 6$ mm	11.000
Lateral sandwich panels	—	CFRP / Al honeycomb	4	41.000	2900×5000 mm; $t = 16$ mm	164.000
Upper/lower sandwich panels	—	CFRP / Al honeycomb	2	24.000	2900×2900 mm; $t = 16$ mm	48.000
Central structural tube	—	CFRP	1	120.000	$\varnothing 1200 \times 5000$ mm; $t = 4$ mm	120.000
Internal shear walls	—	CFRP / Al honeycomb	12	1.900	850×800 mm; $t = 16$ mm	22.800
					Primary Structure subtotal	**541.400**
Mechanisms						
Solar array deployment	TBD	Included in solar array masses	TBD	—	—	—
Solar array hold-down/release	TBD	Pending definition	TBD	TBD	TBD	TBD
Solar array tracking	TBD	Included in solar array masses	TBD	—	—	—
Client capture and locking	TBD	Pending definition	TBD	TBD	TBD	TBD
Antenna pointing mechanism	Beyond Gravity	APM	1	15.000	$320 \times 375 \times 340$ mm	15.000
Thruster pointing mechanisms	Beyond Gravity	APPMAX2-Baseline	4	13.800	$486 \times 484 \times 281$ mm	55.200
					Mechanisms subtotal (excluding TBD)	**70.200**
					TOTAL IDENTIFIED WET MASS [kg]	**2400.690**

Table 10.10: Preliminary spacecraft dry mass budget.

Subsystem	**Dry mass [kg]**
Communications	62.935
Data Handling	35.000
Payload and Docking	51.500
Propulsion	210.840
Thermal Control	150.000
AOCS and Relative Navigation	38.315
Electrical Power System	240.500
Primary Structure	541.400
Mechanisms	70.200
Total identified dry mass	**1400.690**

Table 10.11: Preliminary spacecraft wet mass budget.

Mass contribution	**Mass [kg]**
Total identified dry mass	1400.690
Propulsion xenon	500.000
Payload xenon	500.000
Total identified wet mass	**2400.690**

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